

Supply & Demand — Must-Know

The essentials you cannot walk into the exam without

- › Solve for equilibrium · surplus · deadweight loss
- › Elasticity & the revenue rule
- › Taxes: the wedge & who really pays
- › Competitive firm ($P = MC$) vs market power (markup)

The 12 things you must know

1. **Demand slopes down** (= willingness to pay), **supply slopes up** (= marginal cost / willingness to accept). They cross at the **equilibrium** (P^* , Q^*).
2. **To find equilibrium: set quantity demanded = quantity supplied, solve for p, then plug back in for q.**
3. **Consumer surplus** = triangle **under demand, above price**. **Producer surplus** = triangle **above supply, below price**. Both are $\frac{1}{2} \cdot \text{base} \cdot \text{height}$.
4. **Elasticity** $\epsilon = (\% \Delta Q) / (\% \Delta P)$ in absolute value. $\epsilon > 1$ elastic, $\epsilon < 1$ inelastic. **The slope coefficient is NOT the elasticity.**
5. **Inelastic demand:** raising price **raises** revenue. **Elastic demand:** raising price **lowers** revenue.
6. **A shift's split into price vs quantity depends on elasticity.** More elastic demand $\rightarrow$ shift moves quantity a lot, price a little.
7. **A tax drives a wedge:** buyer price rises, seller price falls, quantity falls. **It doesn't matter who legally pays** — same outcome.
8. **Tax incidence: the more INELASTIC side bears the bigger share.** A tax creates a **deadweight loss**.
9. **Competitive (price-taking) firm: produce where $P = MC$.** Its supply curve is its MC curve (above AC). Exit if $P < AC$.
10. **Long run under competition: entry drives price to $P = MC = AC$** (zero economic profit).
11. **Firm with market power: $MR < P$.** For linear demand, **MR has the same intercept but is twice as steep.** Produce where **$MR = MC$** , then read the price up on the demand curve (so **$P > MC$**).
12. **Markup rule: $(P - MC)/P = 1/\epsilon$.** More elastic demand $\rightarrow$ smaller markup.
Market power $\rightarrow$ deadweight loss; competition removes it.

CORE SKILL 1

Solve for the Equilibrium

THE METHOD (WORKS EVERY TIME)

Step 1. Write demand and supply as quantities: $q_D(p)$ and $q_S(p)$.

Step 2. Set $q_D = q_S$ and solve for the price p^* .

Step 3. Put p^* back into either equation to get q^* .

WORKED EXAMPLE

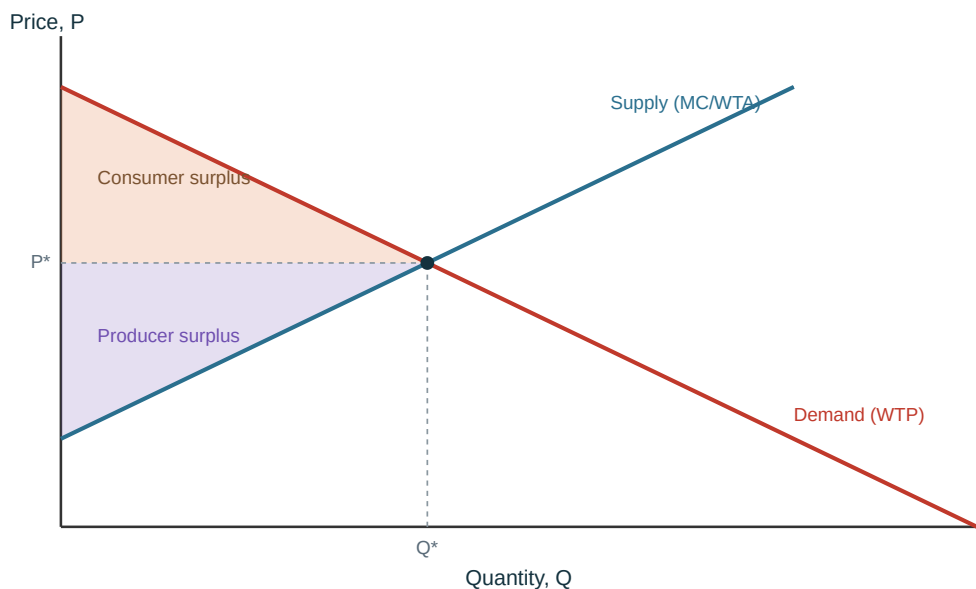
Demand $q = 200 - 5p$, supply $q = -50 + 20p$.

Set equal: $200 - 5p = -50 + 20p \rightarrow 250 = 25p \rightarrow p^* = 10$. Then $q^* = 200 - 5(10) = 150$.

EXAM TRAP

"**Normal slopes**" means demand down, supply up. If a question says supply is **perfectly inelastic** it's a **vertical** line (fixed quantity); **perfectly elastic** is a **horizontal** line (fixed price). Excess supply (price too high) pushes price **down**; excess demand (price too low) pushes it **up**.

Consumer & producer surplus — the master picture



The one diagram to memorise. Consumer surplus = area **under demand, above P^*** . Producer surplus = area **above supply, below P^*** . Total surplus (CS+PS) is **maximised** at the competitive equilibrium — no deadweight loss.

SURPLUS FORMULAS (STRAIGHT-LINE CURVES)

CS = $\frac{1}{2} \times Q^* \times (\text{demand choke price} - P^*)$ (choke = the price where $q = 0$)

PS = $\frac{1}{2} \times Q^* \times (P^* - \text{supply intercept price})$

WORKED EXAMPLE (SAME NUMBERS)

Invert: demand $p = 40 - q/5$ (choke 40); supply $p = 2.5 + q/20$ (intercept 2.5). At $q^*=150$, $p^*=10$:

CS = $\frac{1}{2} \cdot 150 \cdot (40 - 10) = 2250$. **PS** = $\frac{1}{2} \cdot 150 \cdot (10 - 2.5) = 562.5$.

CORE SKILL 2

Elasticity & Revenue

DEFINITION

$\epsilon = |\% \Delta Q / \% \Delta P| = |(dQ/dP) \cdot (P/Q)|$ — always take the absolute value for demand.

ϵ	Name	Quantity response	Raise price → revenue...
> 1	Elastic	very responsive	falls
$= 1$	Unit-elastic	—	unchanged (max)
< 1	Inelastic	barely responds	rises

THE #1 ELASTICITY TRAP

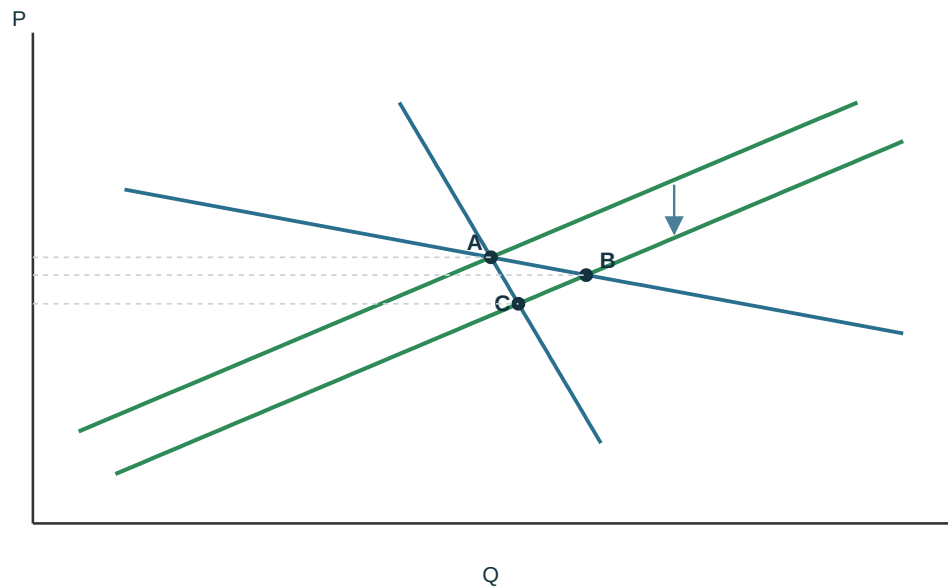
If demand is $q = 200 - 5p$, the **5 is NOT the elasticity** — it's the slope coefficient ($dq/dp = -5$). Elasticity also multiplies by **P/Q**: at $p=10$, $q=150$, $\epsilon = 5 \cdot (10/150) = 1/3$ (inelastic). Always compute P/Q .

WORKED EXAMPLE

Used cars have elasticity 0.6 (inelastic). If price rises 10%, quantity falls only $0.6 \times 10 = 6\%$. Since quantity falls *less* than price rose, **total spending on used cars rises**.

Shifts: elasticity decides price vs quantity

A cost fall (or good harvest) shifts **supply right/down**; a popularity change shifts **demand**. How the new equilibrium splits into a price change vs a quantity change depends on **elasticity**.



Same supply shift, two demands. With **elastic** (flat) demand → price barely falls, quantity jumps (point B). With **inelastic** (steep) demand → price drops a lot, quantity barely moves (point C).

CLASSIC MCQ

"Price will fall by the same amount as costs." **False** — the price change depends on the elasticities of BOTH curves, almost never one-for-one. Also: a price change is a **movement along** a curve, **not** a shift of it.

TWO-SHOCK QUESTIONS (BOTH CURVES MOVE)

If supply shifts left **and** demand shifts left (e.g. fewer shrimp caught + less popular): **quantity falls for certain**, but the **price change is ambiguous** — it depends on the relative size of the two shifts.

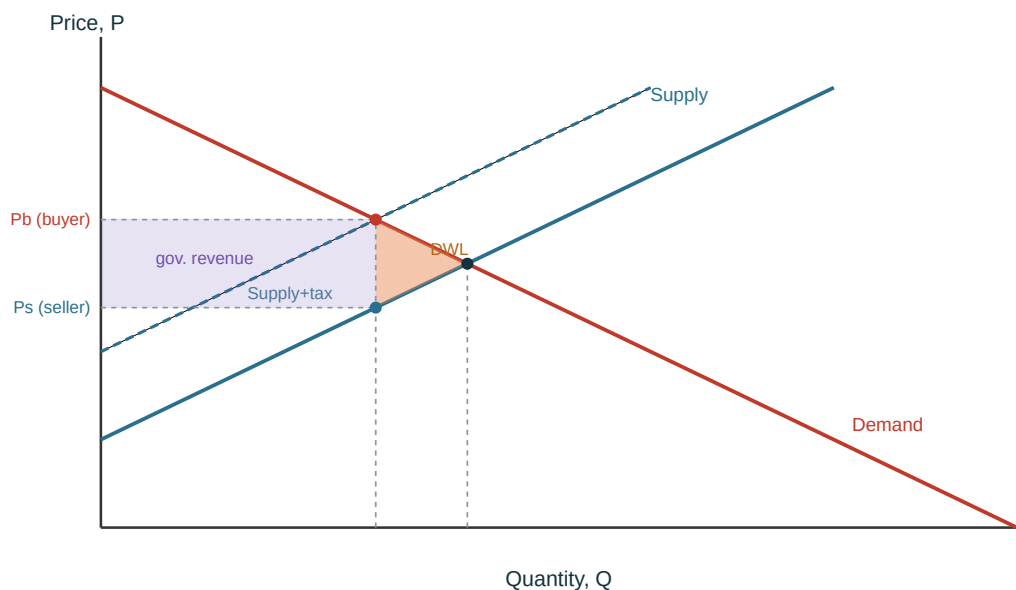
CORE SKILL 3

Taxes — the Wedge

HOW TO SOLVE A SPECIFIC (PER-UNIT) TAX T

Let p = market price. If the **seller** remits the tax: seller keeps $p - t$, buyer pays p . Rewrite supply using $(p - t)$, set new supply = demand, solve.

Buyer price – seller price = t always. Quantity falls.



The tax wedge. Buyer price P_b rises, seller price P_s falls, and $P_b - P_s = t$. Government collects the purple rectangle ($t \times Q$). The orange triangle is the **deadweight loss** — trades that no longer happen.

WORKED EXAMPLE

Supply $q = -80 + 20p$, demand $q = 130 - 10p$. Pre-tax: $p = 7$, $q = 60$. A €3 seller tax $\rightarrow$ supply becomes $q = -80 + 20(p-3) = -140 + 20p$. New: $-140 + 20p = 130 - 10p \rightarrow p$ (buyer) = 9, seller price = 6, $q = 40$. **Buyer price +2, seller price -1.**

INCIDENCE — WHO REALLY PAYS (MEMORISE)

The more INELASTIC side of the market bears the LARGER share of the tax.

- Perfectly inelastic supply $\rightarrow$ sellers bear it all, **consumer price unchanged**.
- Perfectly elastic supply $\rightarrow$ buyers bear it all, market price rises by the full t .
- **It makes no difference whether the buyer or the seller legally pays** — identical prices & quantities. Subsidies = negative taxes.

TRAP

"The impact on the producer price is larger if supply is more elastic." **False** — more elastic supply means sellers bear **less**, so the producer price moves **less**.

UNIT 8

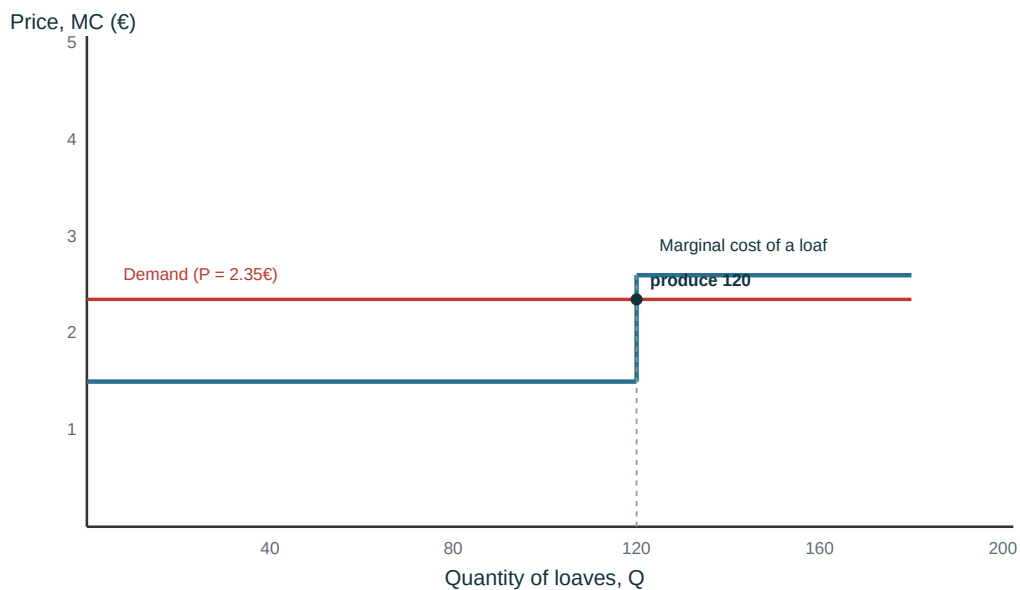
The Competitive (Price-Taking) Firm

PRICE-TAKER

Many firms, identical product → each firm **takes the market price as given** and faces a **horizontal** demand at that price. It only chooses **how much** to produce.

THE RULES (KNOW THESE COLD)

1. Produce where $P = MC$ (because for a price-taker, $MR = P$, and profit-max is $MR = MC$).
2. The firm's **supply curve = its MC curve**, for the part **above AC**.
3. **Shut down / exit** if $P < AC$. Producer surplus = area between price and MC.
4. Market supply = **horizontal sum** of all firms' MC curves.
5. **Long run**: entry drives price down to $P = MC = AC$ → zero economic profit.



Price-taker's choice. Demand is horizontal at the market price. Produce every unit whose price exceeds its marginal cost — here up to where P meets the step in MC.

EFFICIENCY (THE BIG UNIT 8 RESULT)

The competitive equilibrium **maximises total surplus with no deadweight loss** — provided price-taking, complete contracts and **no externalities**. This is what "perfect competition is efficient" means.

TRAPS

· "MR = price" is only true for a **price-taker**. · "Producer surplus is zero in perfect competition" is **false** in general (only if supply is perfectly elastic / constant MC). · Strong economies of scale → markets tend to be **more concentrated**.

UNIT 7

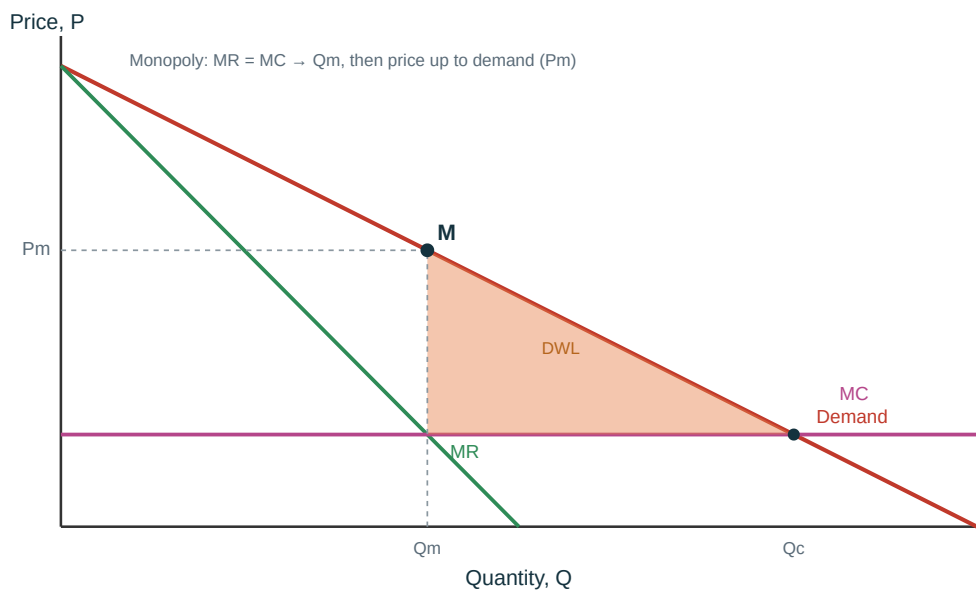
The Firm With Market Power

MARKET POWER

A firm selling a **differentiated** product faces a **downward-sloping** demand curve and can **set a price above marginal cost**. To sell one more unit it must lower the price on **all** units → $MR < P$.

THE TWO MUST-KNOW FACTS

Linear demand $P = a - bQ \Rightarrow MR = a - 2bQ$ — same intercept, twice as steep.
Profit max: $MR = MC$. Find Q there, then read the **price up on the demand curve** (so $P > MC$).



Monopoly/market-power picture. MR (twice as steep) meets MC at Q_m ; the price P_m is read up on demand. Because output is restricted below the competitive level Q_c , there is a **deadweight loss** (orange).

WORKED EXAMPLE (MONOPOLY)

Demand $q = 200 - 5p \rightarrow$ inverse $p = 40 - q/5$, so $MR = 40 - 2q/5$. $MC = 2$. Set $MR = MC$:
 $40 - 2q/5 = 2 \rightarrow q = 95$, $p = 40 - 95/5 = 21$. Consumer surplus = $\frac{1}{2} \cdot (40 - 21) \cdot 95 = 902.5$.

THE MARKUP RULE (LERNER)

$(P - MC)/P = 1/\epsilon$ — the price markup equals the **inverse of the demand elasticity**. More elastic demand (bigger ϵ , more competition) → **smaller markup, price closer to MC**. As $\epsilon \rightarrow \infty$ you get perfect competition ($P = MC$, no DWL).

MC & AC (COST SIDE, BOTH UNITS)

$AC = C(Q)/Q$ (slope of the ray from the origin); MC = cost of one more unit (slope of the total-cost curve). If $MC < AC \rightarrow AC$ falling; if $MC > AC \rightarrow AC$ rising. So **MC cuts AC at the minimum of AC**. **Economies of scale** = AC falls as output rises (spreading fixed costs; increasing returns).

Formula Cheat-Sheet & Top Traps

Thing	Formula / rule
Equilibrium	set $q_D = q_S$, solve for p^* , then q^*
Consumer surplus	$\frac{1}{2} \times Q^* \times (\text{demand choke} - P^*)$
Producer surplus	$\frac{1}{2} \times Q^* \times (P^* - \text{supply intercept})$
Elasticity of demand	$\epsilon = (dQ/dP) \cdot (P/Q) $; >1 elastic, <1 inelastic
Revenue & elasticity	inelastic: $P \uparrow \Rightarrow \text{revenue} \uparrow$; elastic: $P \uparrow \Rightarrow \text{revenue} \downarrow$
Specific tax	$P_b - P_s = t$; who pays is irrelevant; DWL created
Tax incidence	less-elastic side bears more of the tax
Competitive firm	$P = MC$; supply = MC above AC ; exit if $P < AC$
Long-run competition	$P = MC = AC$ (zero economic profit)
Market power output	$MR = MC$, then P from demand ; $P > MC$
Linear demand MR	$P = a - bQ \Rightarrow MR = a - 2bQ$ (twice as steep)
Markup (Lerner)	$(P - MC)/P = 1/\epsilon$
AC vs MC	$MC < AC \Rightarrow AC$ falling; they cross at min AC

TOP EXAM TRAPS (FROM PAST PAPERS)

- Slope coefficient $\neq$ elasticity (multiply by P/Q).
- Who legally pays a tax doesn't change the outcome.
- More elastic side of the market bears **less** of a tax.
- A cost cut does **not** cut price one-for-one.
- $MR = P$ only for a price-taker; for market power $MR < P$.
- Setting $P > MC$ (market power) is **not Pareto efficient** (deadweight loss).
- Economies of scale $\neq$ economies of scope (scale = one good, output $\uparrow$; scope = two goods, shared input).
- "Increasing returns / a $>$ proportional output rise" = economies of scale, **not** diseconomies.

TWO-MINUTE SELF-TEST

1. Demand $q = 100 - 2p$, supply $q = -20 + 2p$. Find p^* , q^* , CS, PS. ($p^*=30$, $q^*=40$; $CS=\frac{1}{2}\cdot 40\cdot(50-30)=400$; $PS=\frac{1}{2}\cdot 40\cdot(30-10)=400$.)
2. A monopolist faces $p = 60 - 2q$, $MC = 4$. Find q , p . ($MR = 60 - 4q = 4 \rightarrow q = 14$, $p = 32$.)
3. A €4 tax; demand more inelastic than supply. Who bears more? (*Buyers — the more inelastic side.*)

This is a condensed exam-revision sheet for Units 7 (the firm & market power) and 8 (supply & demand / competitive markets). The full, detailed unit guides follow this section — use them for the diagrams and derivations. Good luck tomorrow!

Unit 7 — The Firm and its Customers

Costs, demand, profit maximization & market power

A companion study guide and running lecture notes. Every cost curve, isoprofit diagram, marginal-revenue graph and worked example from the slides is redrawn here so you can follow along visually while the lecturer talks. Definitions, intuition boxes, worked solutions and exam tips are added throughout. Academic year 2025–2026.

HOW TO USE THESE NOTES

Skim the **definition boxes** before each lecture. During the lecture, follow the **redrawn diagrams** — they use the same axes and colours as the slides. Colour convention used throughout: **red = demand / marginal revenue**, **magenta = marginal cost**, **teal = average cost**. After the lecture, test yourself with the **self-test questions** at the end.

COVERAGE NOTE — PLEASE READ

These notes are built from the two uploaded slide decks: **Part 1** (slides 1–20: Introduction & Production/Costs) and **Part 2** (slides 40–51: Profit Maximization). The slides in between — **Section C “Demand, elasticity and revenue”** (≈ slides 21–39) and the detailed slides for **E “Gains from Trade”** and **F “Competition, differentiation and market power”** — were not in the uploaded files. Because Part 2 *relies* on those ideas (elasticity, the demand curve, marginal revenue), the sections marked “*bridging*” below reconstruct them in the standard CORE/KU Leuven treatment so your notes stay coherent. Cross-check them against the missing slides when you get them.

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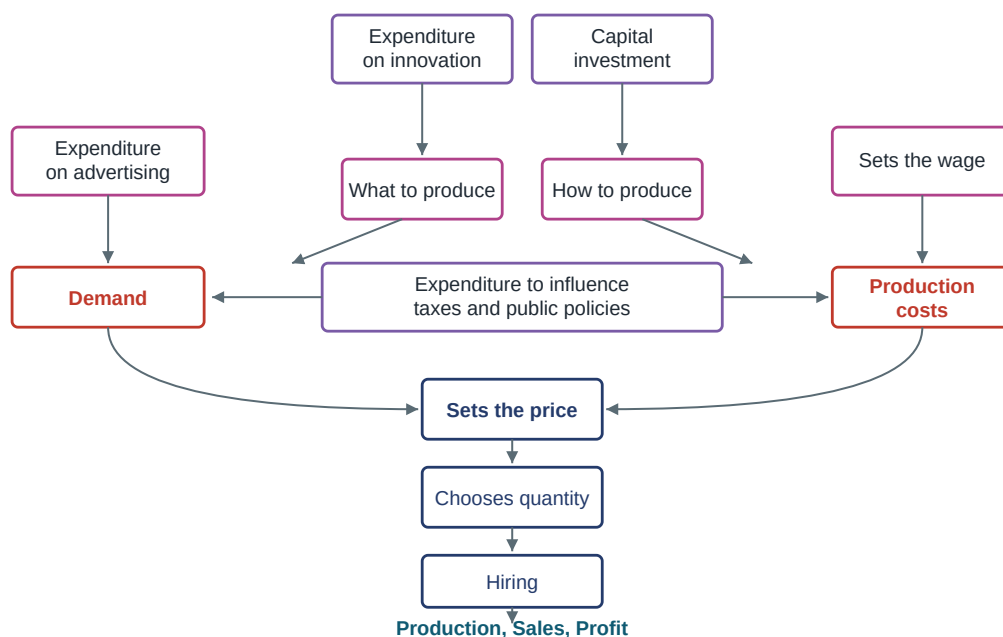
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|--|---|
| 1. A. Introduction — the firm's decisions | 7. D. Profit maximization |
| 2. B. Production & costs: key concepts | 8. Isoprofit curves & $MRS = MRT$ |
| 3. Fixed, variable & total cost | 9. Markup = $1 / \text{elasticity}$; $MR = MC$ |
| 4. Average cost, marginal cost & their relationship | 10. E. Gains from trade (<i>bridging</i>) |
| 5. Economies of scale, scope & returns to scale | 11. F. Competition & market power (<i>bridging</i>) |
| 6. C. Demand, elasticity & revenue (<i>bridging</i>) | 12. Summary, glossary & self-test |

A. Introduction — the firm and its decisions

A firm is a web of decisions. This unit zooms in on two of them — **what price to set** and **how much to produce** — and shows they depend on **demand** (how customers respond to price) and on **production costs**.

The firm's decisions

The firm's many choices feed into each other and ultimately into production, sales and profit:



Two inputs are decisive for pricing and output: **Demand** (left) and **Production costs** (right). Everything else — innovation, advertising, wages, lobbying — ultimately works through those two channels to “Sets the price → Chooses quantity → Hiring”.

What this unit is about

- Key firm decisions: **what price to set** and **what quantity to produce**.
- These depend on the **demand function** (how responsive quantity sold is to price — its **price elasticity**) and the **cost function** (how costs change with output).
- We model firms that produce **differentiated products** and therefore have some **market power** (they face a downward-sloping demand curve, unlike price-takers).
- We measure the **gains from trade** (surplus) and ask how **policy** can affect its division between firms and customers.

THE CORE TRADE-OFF IN ONE LINE

Setting a **lower** price raises quantity sold but lowers revenue per unit; a **higher** price does the reverse. Whether a price change raises or lowers *revenue* depends on the **elasticity of demand**. Whether it raises *profit* also depends on the **cost of producing** the extra unit (marginal cost). Profit maximization is about balancing these.

B. Production & costs — key concepts

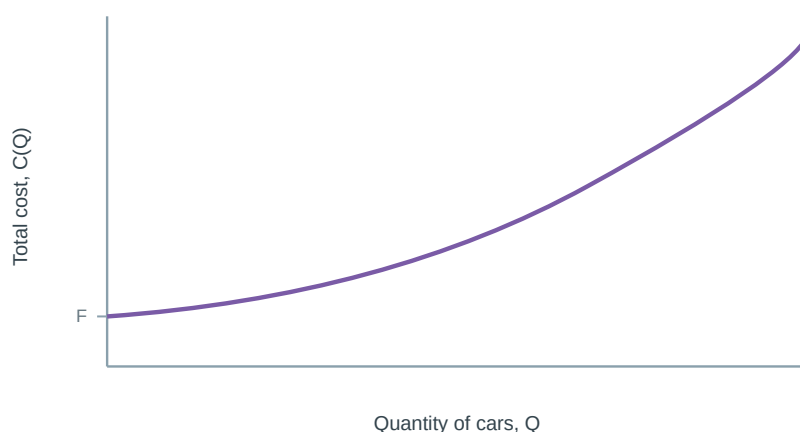
To choose price and output well, managers must know how **costs vary with the quantity produced**. That relationship is the **cost function**.

Fixed, variable and total cost

DEFINITIONS

- **Cost function $C(Q)$** — shows how *total* production cost varies with quantity Q produced.
- **Fixed costs** — do not vary with output (e.g. opportunity cost of capital, R&D). They can depend on the planning horizon (long run vs short run). “Fixed” means *not varying with the production rate*, not literally unchangeable.
- **Variable costs** — increase with output (e.g. wages, materials).

Total cost of production, $C(Q)$



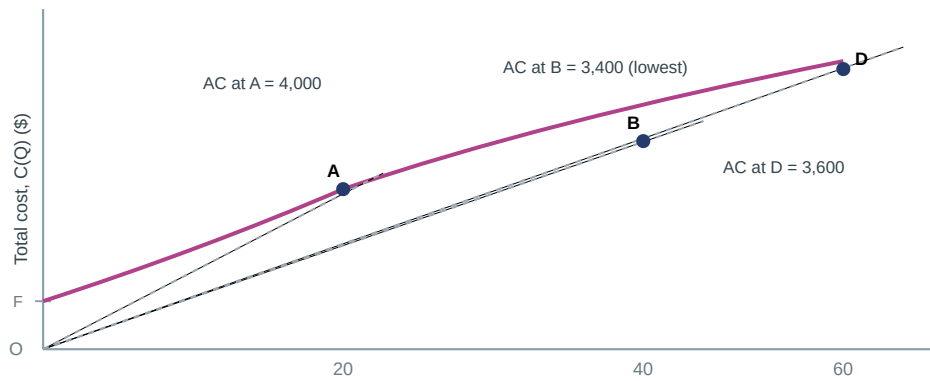
Total cost starts at the **fixed cost F** (the intercept, output = 0) and rises with output. Here it is *convex* (gets steeper) — each extra car costs a bit more than the last, i.e. rising marginal cost.

Average cost, marginal cost, and how they relate

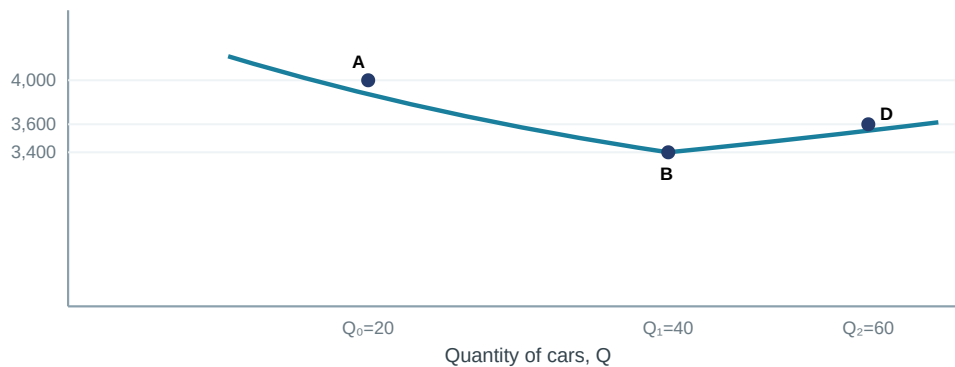
TWO COST MEASURES YOU MUST NOT CONFUSE

- **Average cost (AC)** = cost *per unit* produced = $C(Q)/Q$. Graphically it is the **slope of the ray from the origin** to a point on the total-cost curve.
- **Marginal cost (MC)** = the effect on total cost of producing *one additional* unit. Graphically it is the **slope of the total-cost curve** (its tangent) at a point.

Total cost $C(Q)$: AC = slope of ray from origin O



Average cost of production (\$)



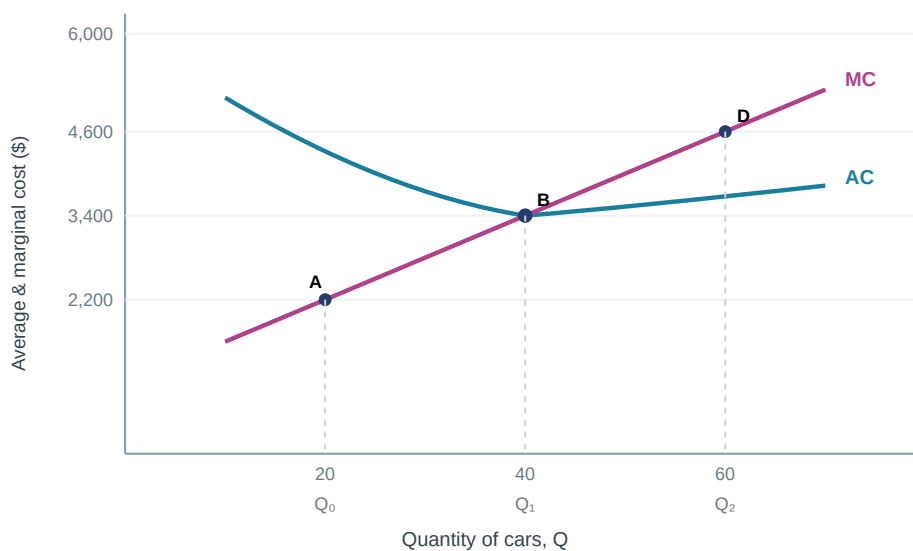
Top: average cost at a point = slope of the ray from the origin O. That slope is smallest at B, so **AC is lowest at B (3,400)**. Bottom: plotting those slopes gives the familiar **U-shaped average-cost curve** — AC falls, then rises (e.g. overtime, machine breakdowns at high output).

THE MC-AC RULE (LEARN THIS COLD)

- If $AC > MC \rightarrow$ average cost is **decreasing** (the cheap extra unit pulls the average down).
- If $AC < MC \rightarrow$ average cost is **increasing** (the expensive extra unit pulls the average up).

Consequence: the MC curve always cuts the AC curve at the AC curve's lowest point.

MC cuts AC at the minimum of AC



To the left of B, MC is below AC $\rightarrow$ AC falling. To the right, MC is above AC $\rightarrow$ AC rising. So MC must pass exactly through B, the **bottom of the AC curve** (here $Q_1 = 40$, cost = 3,400).

Advantages of being large: economies of scale

Many sectors are dominated by a few large firms. A key explanation: **average costs fall as output rises**.

DEFINITIONS — DON'T CONFUSE THESE TWO

- **Economies of scale** — average cost of production *decreases* as output increases. (Large firms get a cost advantage.) Newspaper industry (strong scale economies) vs the restaurant sector (weak).
- **Economies of scope** — cheaper to produce two products *jointly* in one firm than separately in two, usually because a **common input is shared** (e.g. internet + cable TV; insurance + banking).

Economies of scale come from two broad sources — **technological advantages** and **cost advantages**.

1. Technological advantages → returns to scale

Costs are fundamentally linked to the production process. Scaling inputs up can raise output more or less than proportionally:

Returns to scale	Output vs inputs	Effect on average cost
Increasing returns	Output rises <i>more</i> than proportionally to inputs (inputs rise less than proportionally to output)	AC decreasing
Constant returns	Output rises <i>proportionally</i> with inputs	AC constant
Decreasing returns	Output rises <i>less</i> than proportionally to inputs (inputs rise more than proportionally to output)	AC increasing

SOURCES OF INCREASING RETURNS

- Specialization & indivisibilities in the production process
- Engineering reasons — e.g. a gas pipeline's material cost scales with its *circumference*, but capacity with its *cross-sectional area*; container ships, tankers

SOURCES OF DECREASING RETURNS

- Coordination becomes harder at scale
- Supervision / management overheads grow

2. Cost advantages

- **Fixed costs** — when fixed costs are large, spreading them over more units lowers AC as scale rises. (Fixed = doesn't vary with the *production rate*: R&D, IT, advertising. A pharma firm's R&D can vary year to year, but not with how many pills it makes.)
- **Bargaining power** — large firms buy inputs on more favourable terms (e.g. the Ahold–Delhaize merger).

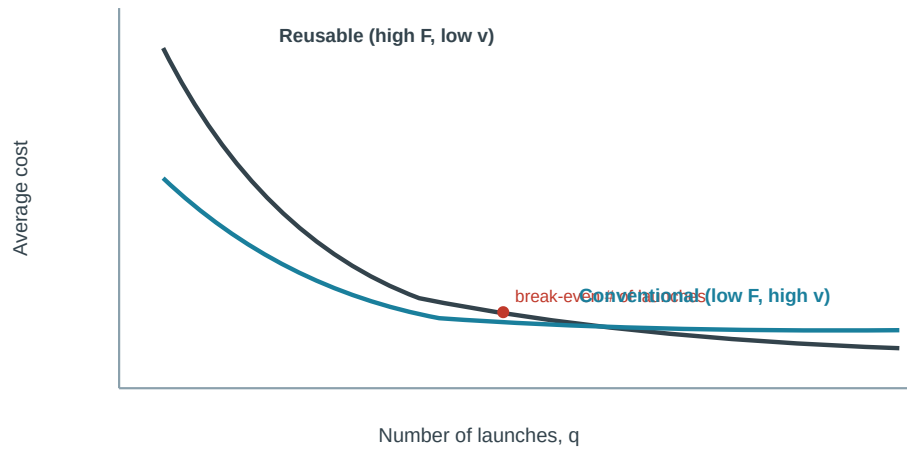
Worked example: how SpaceX makes space travel cheaper

Fixed development cost F and constant variable cost per launch v give a textbook economy of scale:

$$TC(q) = F + v \cdot q \Rightarrow AC(q) = v + \frac{F}{q} \quad (\text{decreases as } q \text{ rises})$$

SpaceX makes rocket boosters **reusable**: this *lowers* variable cost per launch v but *raises* fixed development cost F (engineering the boosters to return). Reusability wins **only if the number of launches is high enough**:

Average cost with and without reusable boosters



At **low** launch counts the reusable design's big fixed cost dominates (higher AC). Past the crossing point, its lower variable cost wins → **reusable is cheaper per launch when volume is high**. Same $v + F/q$ logic as any economy of scale.

C. Demand, elasticity & revenue (bridging — slides not in upload)

WHY THIS SECTION IS HERE

The profit-maximization material in Part 2 constantly uses the **demand curve**, **price elasticity** and **marginal revenue**. These were on the missing middle slides, so here is the standard treatment you need to make Section D make sense.

The demand curve & willingness to pay

DEFINITION — DEMAND CURVE

Shows, for each price, the quantity customers will buy (equivalently, for each quantity, the highest price at which it can all be sold). A firm with **market power** (a differentiated product) faces a **downward-sloping** demand curve: to sell more it must lower the price. Each point on the curve is a customer's **willingness to pay (WTP)** — its reservation price.

Price elasticity of demand

DEFINITION — PRICE ELASTICITY OF DEMAND (E)

The **percentage change in quantity demanded** divided by the **percentage change in price** (taken as a positive number):

$$\varepsilon = - \frac{\% \text{ change in quantity}}{\% \text{ change in price}} = - \frac{\Delta Q / Q}{\Delta P / P}$$

- $\varepsilon > 1$ → demand is **elastic** (quantity very responsive; cutting price *raises* revenue).
- $\varepsilon < 1$ → demand is **inelastic** (quantity barely responds; cutting price *lowers* revenue).
- $\varepsilon = 1$ → revenue is at a maximum.

Total & marginal revenue

DEFINITIONS

- **Total revenue** $R = P \cdot Q$.
- **Marginal revenue (MR)** = the change in total revenue from selling *one more* unit = dR/dQ .

WHY MR IS BELOW THE PRICE

To sell one extra unit a firm with market power must **lower the price on all units**. So $MR = (\text{price of the new unit}) - (\text{revenue lost on the units it could have sold dearer})$. That “lost revenue” term makes **$MR < P$** whenever demand slopes down. **For a linear (straight-line) demand curve, the MR curve has the same intercept and is exactly twice as steep.**

This last fact is used repeatedly in Section D, so keep it handy: *linear inverse demand* $P = a - bQ \Rightarrow R = aQ - bQ^2 \Rightarrow MR = a - 2bQ$.

D. Pricing & production decisions — profit maximization

Now bring **demand and cost together** to find the profit-maximizing price and quantity. There are **two equivalent methods**, and (obviously) they give the same answer:

1. **Isoprofit curves + demand curve** — the “constrained optimization under scarcity” picture.
2. **Marginal revenue = marginal cost** — the marginal picture.

Method 1 — Isoprofit curves

DEFINITIONS

- **(Economic) profit** = total revenue – total cost: $\pi = PQ - TC(Q) = Q(P - AC)$.
- **Isoprofit curve** — all price–quantity combinations that give the *same* profit (the firm's “indifference curve”).
- The **zero-profit curve is the AC curve** (where $P = AC$, so $\pi = 0$).
- Isoprofit curves **above the AC curve** = positive profits; curves **further from the origin** = higher profit.

SLOPE OF AN ISOPROFIT CURVE

$$\text{slope} = - \frac{P - MC}{Q}$$

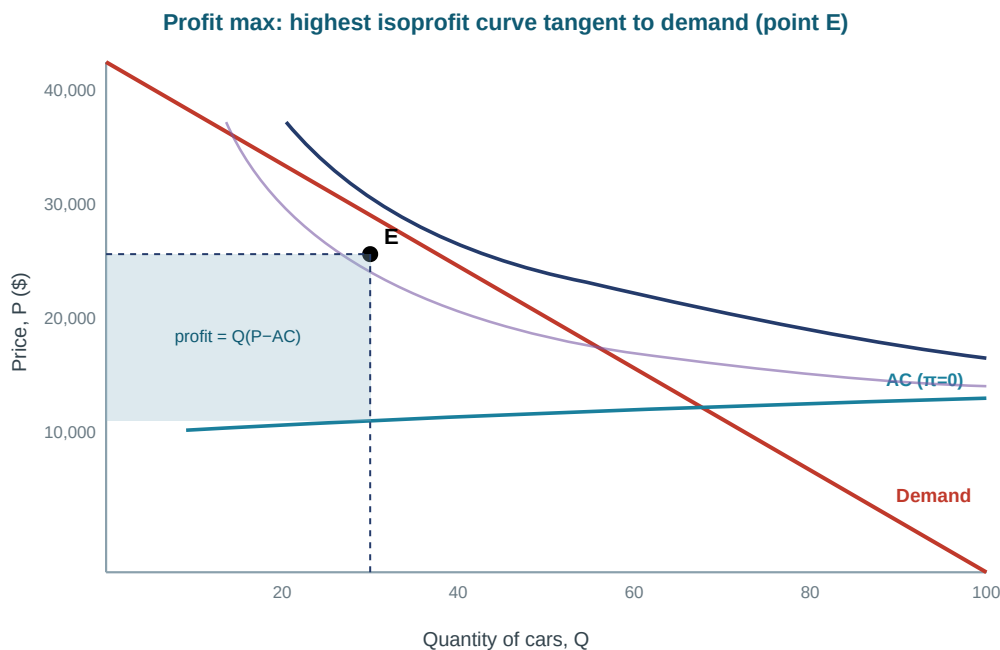
When Q rises by one unit, profit changes by $(P - MC)$; to hold profit constant, price must change by $-(P - MC)/Q$.

- If $MC > P$ (MC above the isoprofit curve) → isoprofit curve slopes **up** (need a higher price to offset the costly extra unit).
- If $MC < P$ (MC below) → isoprofit curve slopes **down**.

Method 1 as constrained optimization (MRS = MRT)

This is exactly the consumer's problem from Unit 3, with new labels:

Consumer (Unit 3)	Firm (Unit 7)
Indifference curves	Isoprofit curves — slope = MRS
Budget constraint / feasible frontier	Demand curve — the feasible frontier; slope = MRT
Optimum where $MRS = MRT$ (tangency)	Profit max where $MRS = MRT$ — highest isoprofit curve <i>tangent</i> to demand



The firm reaches the **highest isoprofit curve that still touches the demand curve** — the tangency **E**. There, the isoprofit slope (MRS) equals the demand slope (MRT). At E, $P > MC$ and $P > AC$, so profit (the shaded rectangle $Q(P-AC)$) is positive.

Profit-maximizing price: the markup rule

Setting $MRS = MRT$ and doing the algebra gives one of the most important results in the course — the **Lerner markup rule**:

$$\frac{P - MC}{P} = \frac{1}{\epsilon}$$

Slope of isoprofit (MRS)	Slope of demand (MRT)	Set $MRS = MRT \Rightarrow$
$-(P - c)/Q$	$-P/(\epsilon Q)$	$(P - c)/Q = P/(\epsilon Q) \Rightarrow (P - c)/P = 1/\epsilon$

READING THE MARKUP RULE (HERE $c = MC$)

The **price markup** (price-cost margin) equals the **inverse of the demand elasticity**. **More elastic demand (larger ϵ) $\Rightarrow$ smaller markup $\Rightarrow$ lower optimal price**. A firm whose customers are very price-sensitive cannot charge much above marginal cost. This single equation links *market power* (Section F) to *elasticity*.

Method 2 — Marginal revenue = marginal cost

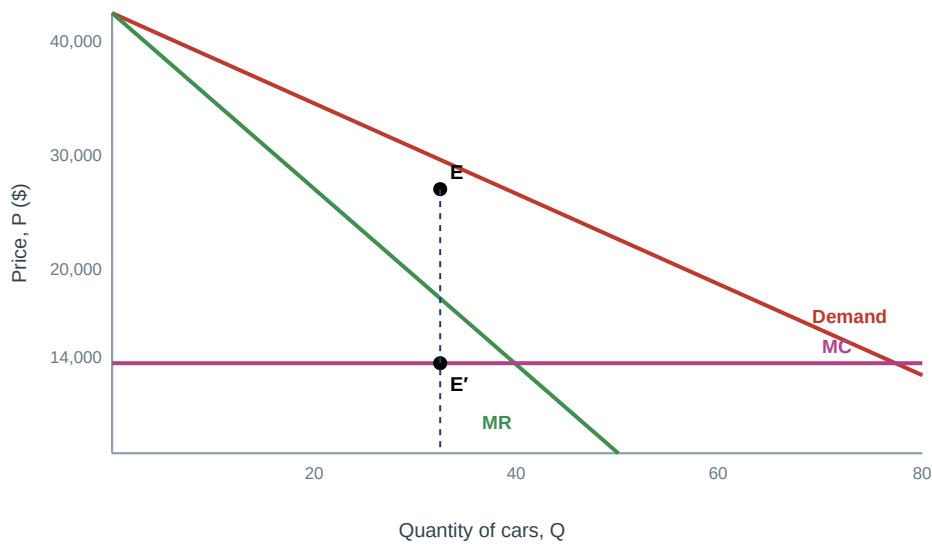
THE $MR = MC$ RULE

To maximize profit, produce at the scale where **marginal revenue = marginal cost**.

- If $MR > MC \rightarrow$ sell more (the extra unit adds more revenue than cost).
- If $MR < MC \rightarrow$ sell less.
- Algebra: $\pi = PQ - TC(Q)$; the first-order condition $d\pi/dQ = 0$ gives **$MR = MC$** .

Then read the **price off the demand curve** at that quantity. One can show this price also satisfies $(P - MC)/P = 1/\epsilon$ — the two methods agree.

Find Q where $MR = MC$, then go up to demand for P



For a linear demand curve, **MR has the same intercept but is twice as steep**. The profit-maximizing quantity is where **MR meets MC (E')**; the profit-maximizing price is found by going straight up to the demand curve (E). Note E sits above MC — **price exceeds marginal cost**.

Worked numerical example

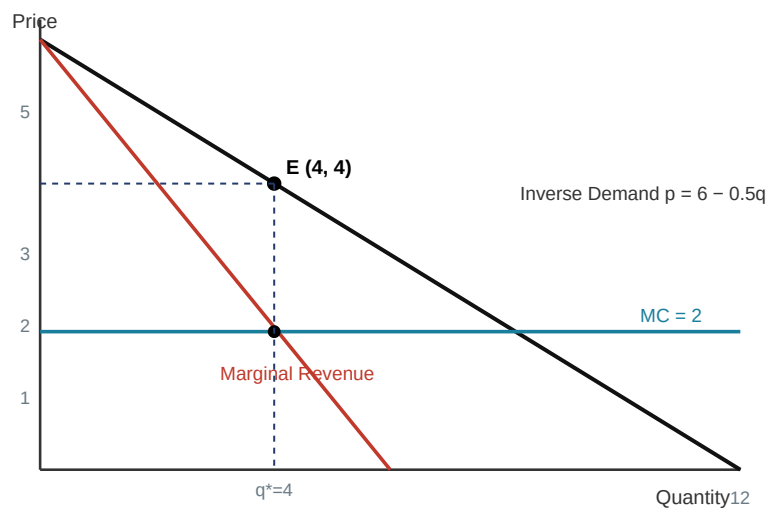
SOLVE IT STEP BY STEP

Demand: $Q = 12 - 2P$, so **inverse demand** $P = 6 - 0.5Q$. **Marginal cost** $MC = 2$.

- Total revenue:** $R = PQ = (6 - 0.5Q)Q = 6Q - 0.5Q^2$.
- Marginal revenue:** $MR = dR/dQ = 6 - Q$ (straight line, twice as steep as demand — same 6 intercept).
- Set $MR = MC$:** $6 - Q = 2 \Rightarrow Q^* = 4$.
- Price from demand:** $P^* = 6 - 0.5(4) = 4$.

Answer: $Q^* = 4$, $P^* = 4$. (Check: price 4 > MC 2, as market power implies.)

Example: $p = 6 - 0.5q$, $MR = 6 - q$, $MC = 2$



MR (red) hits MC (teal) at $q^* = 4$; going up to the demand curve gives $p^* = 4$. The gap between price and MC (4 vs 2) is the firm's markup, sustained by its market power.

E. Gains from trade (bridging — slides not in upload)

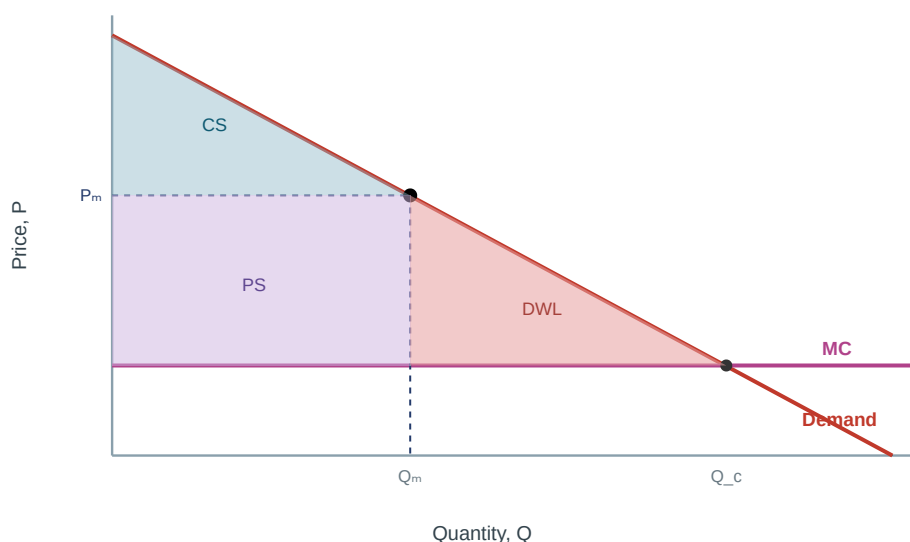
ABOUT THIS SECTION

Part 2 ends on the title slide “Gains from Trade”; its content slides weren't in the upload. Here is the standard CORE/KU Leuven treatment of **surplus** and the **welfare cost of market power**, which is where the unit heads next.

DEFINITIONS — MEASURING THE GAINS FROM TRADE

- **Consumer surplus (CS)** — total *willingness to pay* minus what customers actually pay. Area **below the demand curve**, above the price.
- **Producer surplus (PS)** — revenue minus variable cost of production. Area **above the MC curve**, below the price.
- **Total surplus** = CS + PS — the total gains from trade.
- **Deadweight loss (DWL)** — surplus that is *lost* because trades that would benefit both sides don't happen (e.g. when a firm restricts output to keep price high).

Surplus & the welfare cost of market power



A firm with market power sets price P_m above MC and sells only Q_m (less than the competitive Q_c where price = MC). Consumers keep CS (blue), the firm earns PS (purple). The red triangle is **deadweight loss** — mutually beneficial trades between Q_m and Q_c that never happen. This is the efficiency cost of market power that policy tries to reduce.

F. Competition, differentiation & market power (bridging)

DEFINITION — MARKET POWER

The ability of a firm to **set its price above marginal cost** without losing all its customers. It comes from producing a **differentiated product** (brand, quality, location) so the firm faces a **downward-sloping** demand curve rather than being a price-taker.

HOW IT ALL TIES TOGETHER

The markup rule $(P - MC)/P = 1/\epsilon$ is the bridge:

- **Strong differentiation** → **less elastic demand (small ϵ)** → **large markup, high price** (much market power).
- **More competition / closer substitutes** → **more elastic demand (large ϵ)** → **small markup, price near MC** (little market power).

In the limit of perfect competition, $\epsilon \rightarrow \infty$, the markup $\rightarrow 0$, and price = marginal cost (no deadweight loss).

Policy angle: because market power creates deadweight loss, competition policy, regulation and lowering barriers to entry aim to push ϵ up (more substitutes) and shrink the markup — redistributing surplus toward customers and improving efficiency.

Summary

1. **Costs.** The cost function splits into **fixed** and **variable** costs. $AC = C/Q$ (slope of ray from origin); MC = slope of the cost curve. **MC cuts AC at AC's minimum.**
2. **Economies of scale** (AC falls with output) come from **technological** advantages (increasing returns to scale) and **cost** advantages (spreading fixed costs, bargaining power). Don't confuse with **economies of scope**.
3. **Demand & revenue.** A firm with market power faces a downward-sloping demand curve; $MR < P$, and for linear demand MR is twice as steep.
4. **Profit maximization.** Two equivalent methods: highest **isoprofit curve tangent to demand** ($MRS = MRT$), or $MR = MC$. Both give price via the demand curve, with $P > MC$ and (if profitable) $P > AC$.
5. **Markup rule.** $(P - MC)/P = 1/\epsilon$: markup = inverse elasticity. More elastic demand → lower price.
6. **Gains from trade & market power.** Surplus = CS + PS; market power raises price above MC, creating **deadweight loss**. Competition and policy shrink the markup toward the efficient outcome.

Glossary — quick revision

Term	One-line meaning
Cost function $C(Q)$	How total cost varies with output.
Fixed / variable cost	Doesn't vary with output / increases with output.
Average cost (AC)	Cost per unit, C/Q ; slope of ray from origin.
Marginal cost (MC)	Cost of one more unit; slope of the cost curve.
MC–AC rule	MC below AC → AC falling; MC above → AC rising; they cross at min AC.
Economies of scale	AC decreases as output rises.
Economies of scope	Cheaper to produce two goods jointly (shared input).
Returns to scale	Output response to a proportional rise in all inputs (increasing / constant / decreasing).
Demand curve	Quantity customers buy at each price (downward-sloping under market power).
Price elasticity ϵ	$-(\% \Delta Q)/(\% \Delta P)$; >1 elastic, <1 inelastic.
Marginal revenue (MR)	Revenue from one more unit; $<$ price; twice as steep for linear demand.
Isoprofit curve	Price–quantity combinations giving equal profit; zero-profit curve = AC.
$MRS = MRT$	Profit max: isoprofit slope = demand slope (tangency).
Markup rule (Lerner)	$(P - MC)/P = 1/\epsilon$.
Market power	Ability to price above MC via product differentiation.
Consumer / producer surplus	Gains to buyers / sellers from trade.
Deadweight loss	Surplus lost when beneficial trades don't happen (e.g. under market power).

Self-test questions

1. A firm's AC is 3,600 and its MC is 4,600 at the current output. Is average cost rising or falling? What does this imply about where you are relative to minimum AC?

MC (4,600) > AC (3,600), so AC is rising → you are to the right of the minimum-AC point. Producing a bit less would lower average cost.

2. Demand is $Q = 20 - 2P$ and $MC = 3$. Find the profit-maximizing quantity and price.

Inverse demand: $P = 10 - 0.5Q$. $R = 10Q - 0.5Q^2 \Rightarrow MR = 10 - Q$. Set $MR = MC$: $10 - Q = 3 \Rightarrow Q^* = 7$. Price: $P^* = 10 - 0.5(7) = 6.5$. (Check $P > MC$.)

3. Two firms have the same MC. Firm A's customers are very price-sensitive ($\epsilon = 4$); Firm B's are loyal ($\epsilon = 1.5$). Which charges the higher markup, and why?

Markup = $1/\epsilon$. Firm A: $1/4 = 25\%$. Firm B: $1/1.5 \approx 67\%$. Firm B charges the higher markup — less elastic demand (more market power) lets it price further above MC.

4. Why does making rocket boosters reusable only lower average cost per launch if the number of launches is large?

Reusability raises fixed development cost F but lowers variable cost v , so $AC = v + F/q$. At small q the larger F/q dominates and AC is higher; only once q is large enough does the lower v outweigh the spread-out fixed cost, so the reusable AC curve dips below the conventional one.

5. At the profit-maximizing point, is price above or below marginal cost? Above or below average cost? What does each tell you?

Price > MC always for a firm with market power (that gap is the markup). Price vs AC determines profitability: $P > AC$ means positive economic profit; $P = AC$ means zero profit; $P < AC$ means a loss.

Supply & Demand

Unit 8 — Study Guide & Lecture Notes

- Price-taking firms: $MR = MC = P$ and the supply curve
- Competitive equilibrium, surplus & efficiency
- Shifts, elasticities, short-run vs long-run
- Taxes, incidence & deadweight loss

BEFORE YOU BEGIN

How to use these notes & where this unit fits

This guide follows the Unit 8 slides section-by-section and expands the reasoning so you can read it alone or annotate it in class. Every supply-and-demand diagram is redrawn cleanly, with equilibria, surplus areas and tax wedges computed from the actual functions so the geometry is exact.

DEFINITION BOX

The precise definition to reproduce in the exam.

WORKED EXAMPLE

Numbers carried through, matching the slide examples.

LISTEN FOR IN THE LECTURE

Points usually stressed verbally — leave room for your own notes.

CHECK YOURSELF

Quick questions to test recall.

Where this unit sits

The previous unit studied firms with **market power** — price **setters** who face a downward-sloping demand curve and whose outcomes are generally **not Pareto-efficient**. This unit studies the opposite polar case: **price-takers** in **competitive markets**. The two guiding questions:

- How does the behaviour of a **price-taking** firm differ from a price-setting one?
- **Can competition improve market outcomes?** (Spoiler: under strong assumptions, competitive equilibrium is Pareto-efficient — it exhausts all gains from trade.)

Outline

1. **A.** Introduction — market power vs price-taking
2. **B.** Demand and supply — WTP, WTA, the firm's supply curve, market supply, elasticity

3. **C.** Competitive equilibrium — market clearing, surplus and efficiency
4. **D.** Factors that affect equilibrium — shifts, elasticities, short vs long run
5. **E.** Applications — taxes, incidence and deadweight loss
6. **F.** Perfect competition — the assumptions behind the efficiency result

SECTION A

Introduction

Two kinds of firm sit at opposite ends of a spectrum:

PRICE SETTER (PREVIOUS UNIT)

A firm with **market power** can **set its own price**. It faces a downward-sloping demand curve, produces where $MR = MC$ with a **markup** over cost, and the outcome is **generally not Pareto-efficient** (some gains from trade go unrealised).

PRICE TAKER (THIS UNIT)

A firm in a market with **many firms selling identical products** cannot influence the price. It **takes the market price as given** and just decides **how much** to produce. Its supply depends purely on its **costs**.

The unit builds up the price-taking model, derives the market supply curve, finds the competitive equilibrium, and asks whether competition delivers an efficient outcome — then stress-tests that conclusion with shifts, elasticities and taxes.

LISTEN FOR IN THE LECTURE

Keep the contrast alive throughout: a price **setter** chooses a **point on a demand curve**; a price **taker** chooses a **quantity along its own cost curve** at a price it cannot change. Almost every result in this unit flows from that single difference.

SECTION B

Demand & Supply

THE TWO CURVES

- **Demand curve** — the total quantity all consumers together want to buy at each price. It represents the **willingness to pay (WTP)** of buyers.
- **Supply curve** — the total quantity all firms together would produce at each price. It represents the **willingness to accept (WTA)** of sellers.

Supply in a competitive market

When there are **many firms** selling **identical (homogeneous)** products:

- Firms are **price-takers**.
- The demand curve facing an **individual** firm is **horizontal** at the current market price (sell as much as you like at that price; charge a cent more and you sell nothing). Contrast this with the downward-sloping **market** demand curve.
- A price-taking firm **cannot benefit** from charging a different price and **cannot influence** the market price.
- Its supply therefore depends only on the **costs it incurs in production**.

The individual firm's supply — a bakery

A bakery can make **120 loaves at a constant marginal cost of €1.5**. It can produce more, but must pay **overtime**, so marginal cost jumps to **€2.6** beyond 120. The market price is **€2.35**.

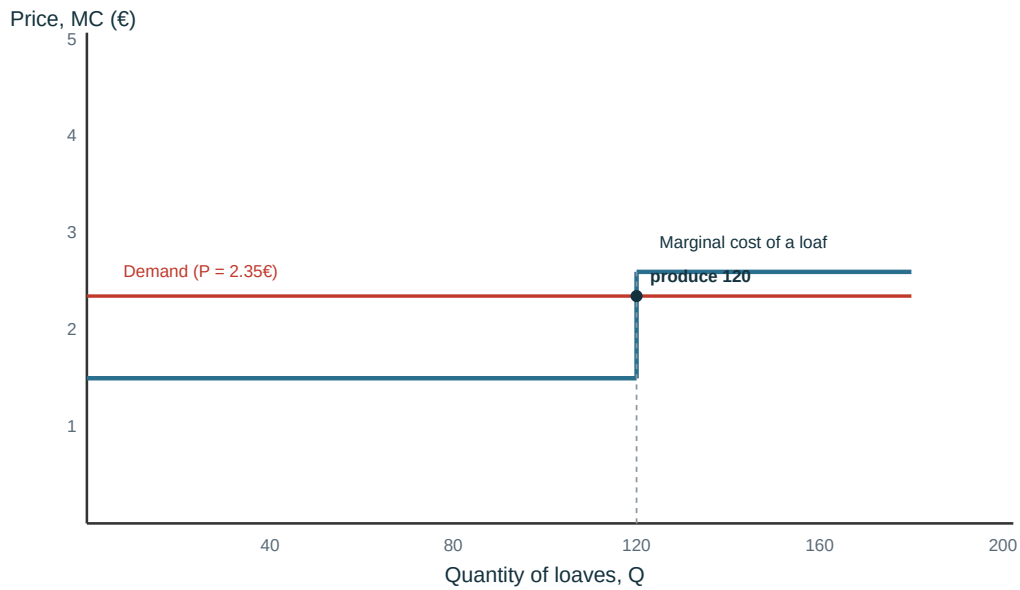


Figure 1. The firm's marginal cost is a **step**: €1.5 up to 120 loaves, then €2.6. The individual firm's demand is **horizontal** at the market price €2.35. **Rule: produce every unit whose price exceeds its marginal cost.** Here $2.35 > 1.5$ but < 2.6 , so the firm makes exactly **120 loaves**.

THE SUPPLY DECISION

- If **price > marginal cost** of a unit → **produce and sell it**.
- If **marginal cost > price** → **do not produce it**.
- Optimal production is where **price = marginal cost**, so the firm's **supply curve is its marginal cost curve**.

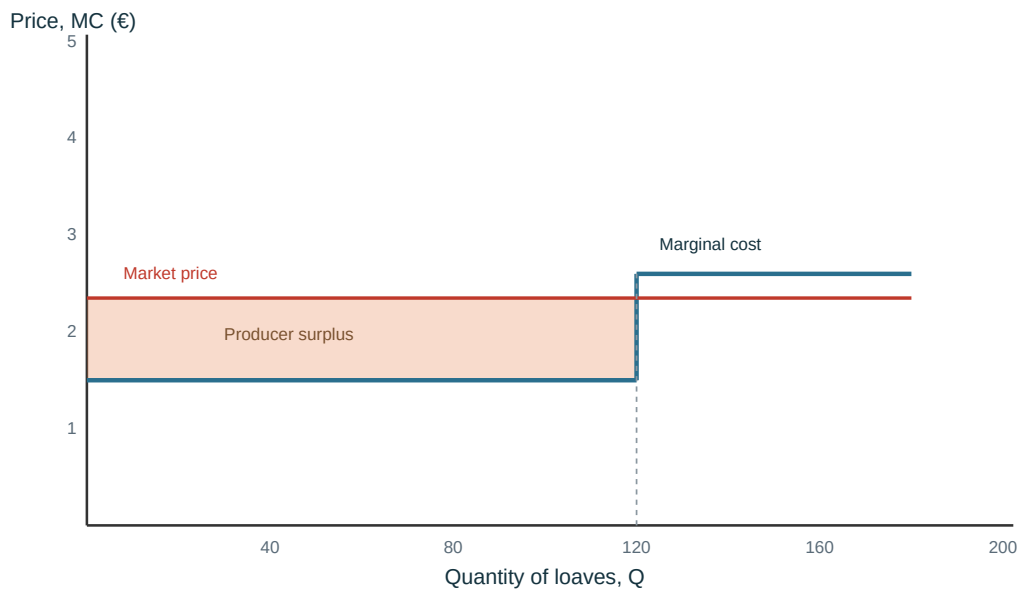


Figure 2. Producer surplus is the area between the market price and the marginal cost curve, over the units produced: here $(\text{€}2.35 - \text{€}1.5) \times 120 = \text{€}102$. It is the firm's gain from trading at the market price.

WHY $P = MC$ IS PROFIT-MAXIMISING ($MR = MC$)

Marginal revenue is the extra revenue from one more unit. The profit-maximising condition is $MR = MC$. For a **price-taker**, each unit sells at the market price, so $MR = P$. Hence the price-taker's rule $P = MC$ is exactly the general $MR = MC$ rule — perfectly consistent.

Should the firm produce at all? The shut-down check

Producing where $P = MC$ maximises profit *given* that you produce. But you must also check that revenue covers costs:

- Produce only if $P \geq AC$ (price at least covers **average cost**).
- Otherwise **exit** the market (unless it is a short-run phenomenon). So the supply curve is the part of the **MC curve that lies above the AC curve**.

From firm to market supply

The **market supply curve** is the **horizontal sum** of all firms' quantities at each price — add up how much every firm supplies at each price.

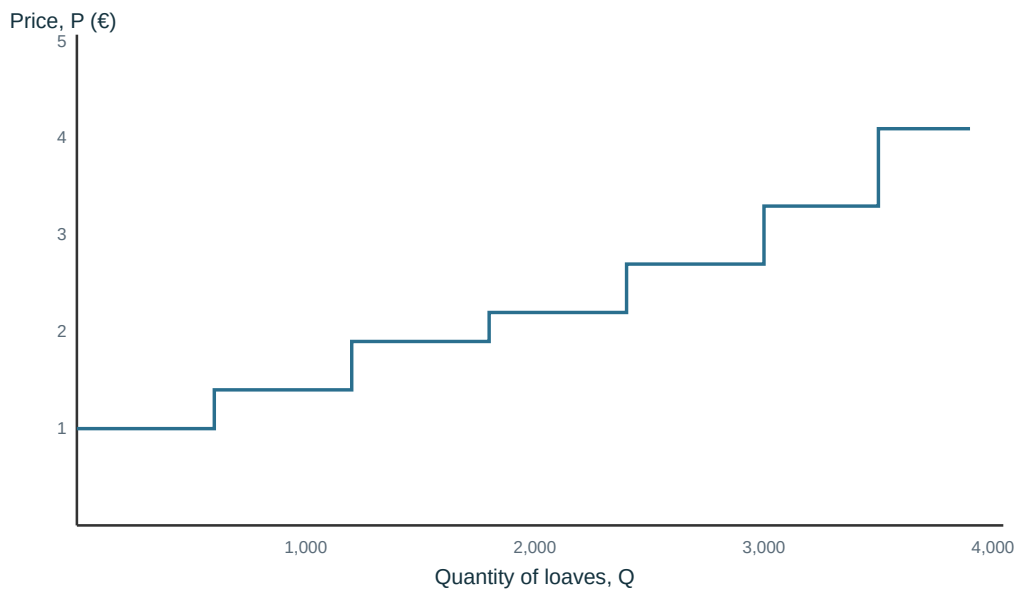


Figure 3. Summing many firms' step-shaped marginal costs produces a rising **staircase**: as price rises, more firms (and more of each firm's higher-cost units) become worth producing.

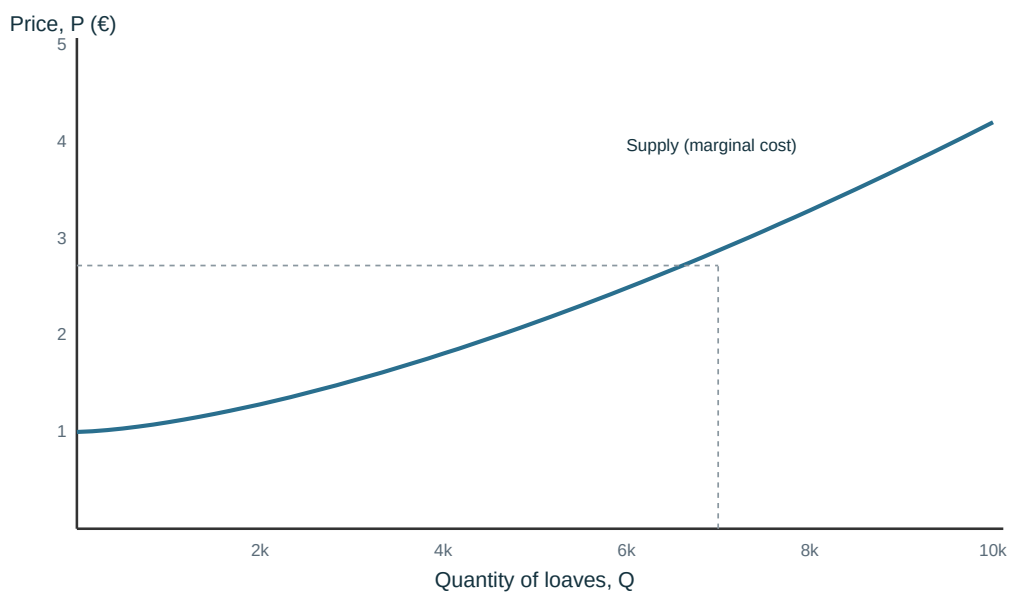


Figure 4. With many suppliers the staircase **smooths out** into an upward-sloping curve — the market's **marginal cost curve**. Higher prices call forth more output.

DEFINITION — PRICE ELASTICITY OF SUPPLY

How responsive quantity supplied is to price:

$$\epsilon_S = \frac{\% \Delta \text{ quantity supplied}}{\% \Delta \text{ price}}$$

Same characteristics as the elasticity of demand: a **flatter** supply curve is more **elastic** (very responsive), a **steeper** one is more **inelastic**. Elasticity will turn out to determine who bears a tax and how much prices move after a shock.

SECTION C

Competitive Equilibrium

Combine supply and demand to determine the equilibrium **price and quantity**.

DEFINITION — COMPETITIVE EQUILIBRIUM

A situation where **all buyers and sellers are price-takers** and, at the prevailing **market-clearing price P^*** , **quantity supplied = quantity demanded**.

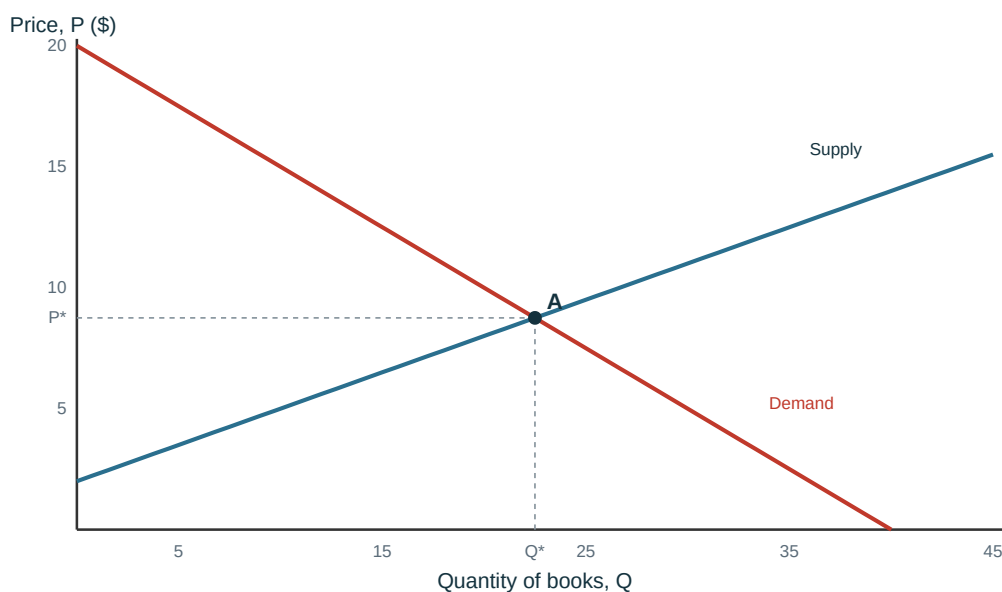


Figure 5. Equilibrium A sits where the supply and demand curves cross, pinning down the market-clearing price P^* and quantity Q^* .

Why the price is pushed to P^*

- If price is **above P^*** → **excess supply** (sellers want to sell more than buyers will buy) → **downward** pressure on price until $P = P^*$.
- If price is **below P^*** → **excess demand** → buyers **bid the price up** until $P = P^*$.
- Because products are **identical**, buyers will buy from any seller, and everyone **takes the prevailing price as given**.

Worked example — tanker (shipping) market

Demand and supply as functions of price p :

$$D = 40 - 0.15p \quad S = -10 + 0.10p$$

Market clears when $S = D$:

$$40 - 0.15p = -10 + 0.10p \implies 50 = 0.25p \implies p = 200, \quad D = S = 10$$

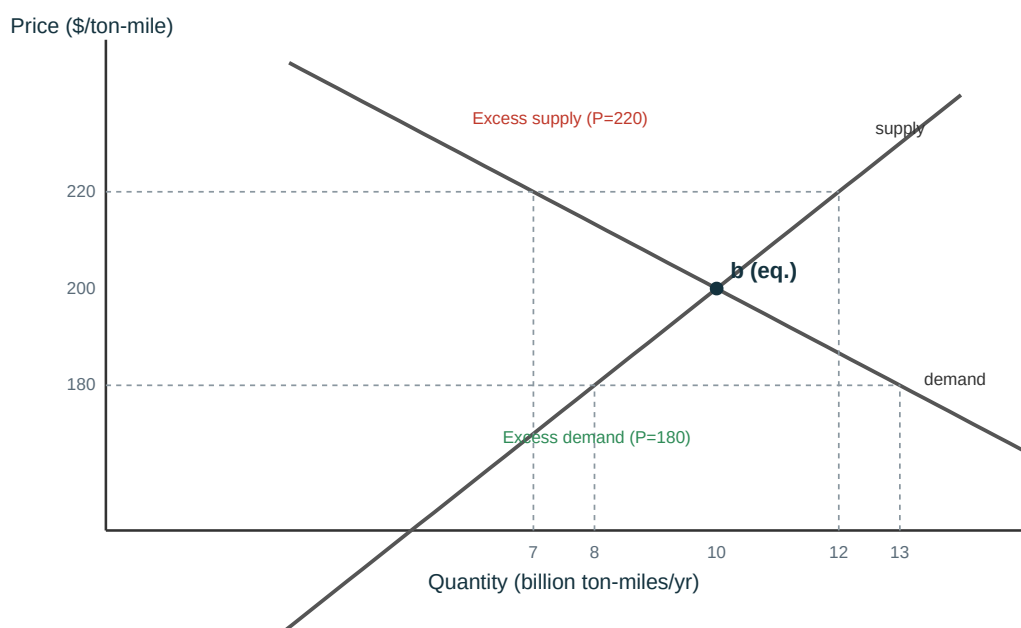


Figure 6. Equilibrium at $p = 200$, $q = 10$. At $p = 220$ quantity supplied (12) exceeds demanded (7) → **excess supply** → price pushed down. At $p = 180$ demanded (13) exceeds supplied (8) → **excess demand** → price pushed up. Both forces drive p back to 200.

Efficiency — all gains from trade are exploited

KEY EFFICIENCY RESULT

At the competitive equilibrium the **sum of consumer and producer surplus is maximised** — there is **no deadweight loss**. Every unit whose willingness to pay exceeds its marginal cost gets produced, and no unit beyond that. This is **not** the same as the point where *consumer* surplus alone is maximised (buyers would prefer a lower price, but that would shrink total surplus).

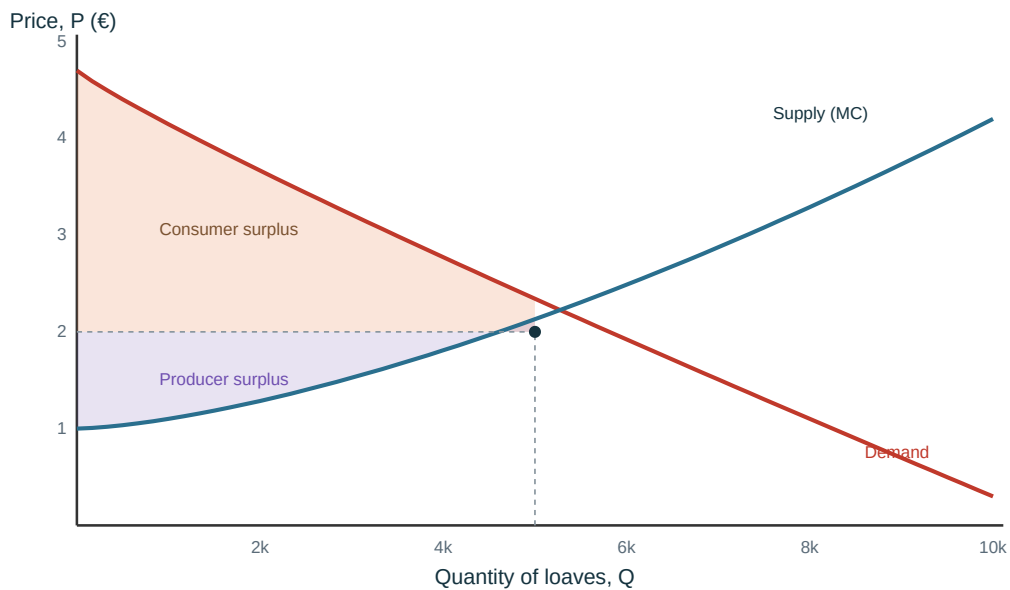


Figure 7. Consumer surplus (between demand and price) plus producer surplus (between price and supply) is as large as possible at equilibrium. Any other quantity would leave some mutually beneficial trades unmade — a deadweight loss.

When is competitive equilibrium Pareto-efficient? Assumptions & caveats

ASSUMPTIONS FOR EFFICIENCY

- Participants are **price-takers** (many firms, homogeneous products).
- Contracts are **complete**.
- Product quality/characteristics are **observed**.
- The transaction **only affects buyers and sellers** — **no external effects** (e.g. no pollution, cf. Unit 12).

CAVEATS

- If the assumptions fail, the allocation **may not be efficient**.
- **Fairness:** the split of total surplus depends on **elasticities** — the side that is **less elastic captures a smaller share** (share of surplus is inversely related to elasticity).
- Genuine price-takers are **hard to find** in real life.

CHECK YOURSELF

At the tanker equilibrium, what happens to price if a regulation fixes $p = 220$? Who is rationed — buyers or sellers? Why does the market-clearing price maximise *total* surplus but not *consumer* surplus?

SECTION D

Factors that Affect Equilibrium

The whole supply or demand curve can **shift** due to **exogenous shocks** — technological change, a change in popularity, input prices, entry/exit. Buyers and sellers then adjust until the market clears again at a **new equilibrium**.

A supply shift — improved baking technology

1. Better technology **lowers marginal costs** → the **supply curve shifts down/out**.
2. At the old price there is now **excess supply** → a **move along the demand curve**.
3. The price **falls** to a new, lower equilibrium (and quantity rises).

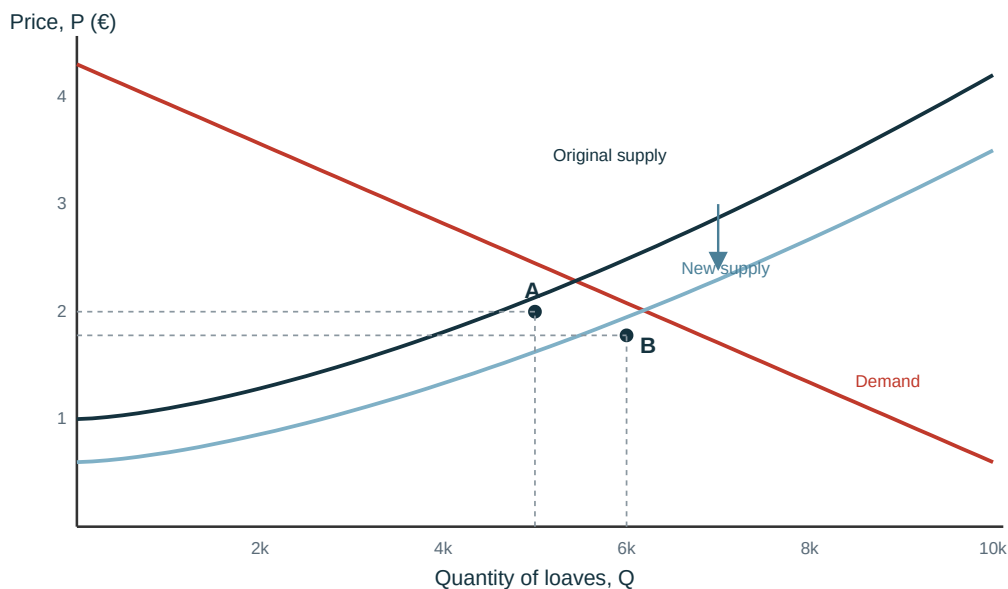


Figure 8. A cost-reducing shock shifts supply out (A → B): equilibrium price falls and quantity rises as the market re-clears along the demand curve.

COMMON MISCONCEPTION

People often expect the market price to fall by **the same amount** as the cost reduction. **Not so** — how much price falls (vs how much quantity rises) depends on the **price elasticities** of demand and supply. This is why manufacturers complain that retailers don't cut retail prices one-for-one with wholesale prices.

Why elasticity decides the price-vs-quantity split

Take the **same** downward supply shift, but with two different demand curves through the original equilibrium A:

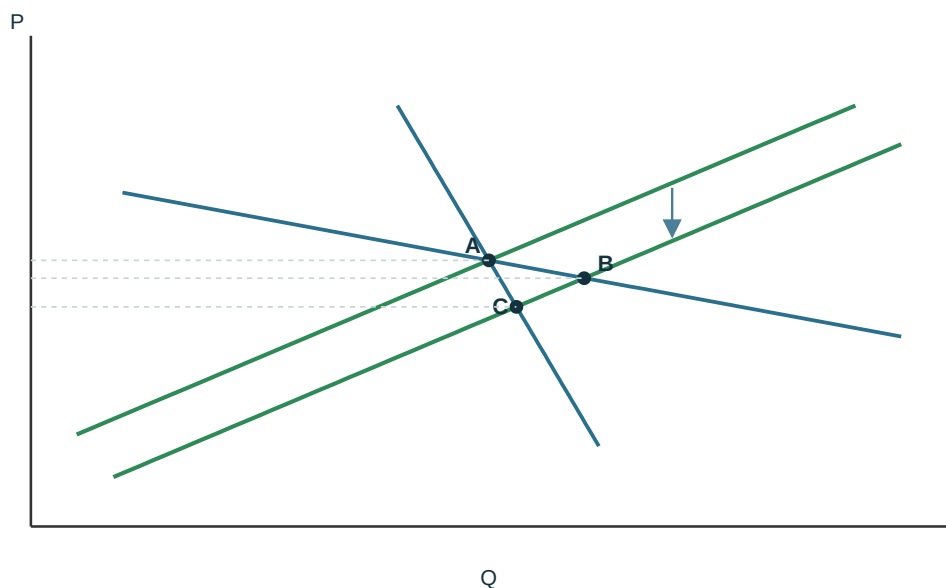


Figure 9. With **elastic** (flat) demand, the new equilibrium is **B**: price drops a little, quantity rises a lot. With **inelastic** (steep) demand, it is **C**: price drops a lot, quantity barely moves. Same cost shock, very different price response — set by demand elasticity.

Other shocks — entry and exit

- The supply curve also shifts through **market entry/exit**. If existing firms earn **economic rents** and the **costs of entry** are not too high, **new firms enter** → supply shifts **outward (down)**.
- **Demand shocks** are analysed the same way; their price impact depends on **supply** elasticity.

Short-run vs long-run equilibrium

In the bakery market, suppose equilibrium A has price €2. At that price an **individual bakery is making a profit** (you can see it because price sits **above its average cost**). That profit attracts entry.

LONG-RUN EQUILIBRIUM CONDITION

New bakeries keep entering (each wanting a share of the rents) → supply **shifts out and flattens** → market price and profits **fall** — until economic rent is **zero**:

$$P = MC = AC$$

At this long-run point (C) firms just break even and entry stops.

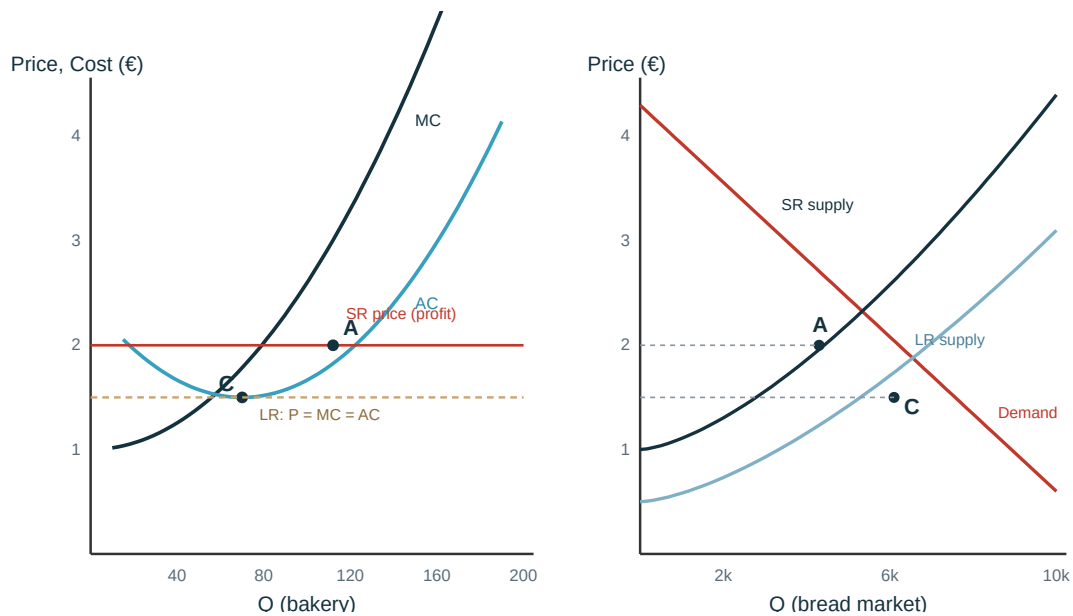


Figure 10. Left: the firm's MC and U-shaped AC. Short-run price (A) sits above minimum AC → economic profit. Right: that profit draws entry, shifting market supply out until price falls to minimum AC (C), where $P = MC = AC$ and rent is zero. The difference between short and long run depends on the production process (a gardener adjusts fast; a chemical plant slowly).

SHORT-RUN VS LONG-RUN ELASTICITY

The number of firms is **exogenous in the short run** but **endogenous in the long run**. Therefore **supply is more elastic in the long run**, as more firms can enter. **Investment warning:** a positive demand shock raises prices, but the rise is **smaller in the long run**; basing an investment on the short-run price spike risks **over-investment**.

SECTION E

Applications — Taxes

Governments tax to **raise revenue** and/or **discourage consumption**. Two types:

TWO KINDS OF TAX

- **Ad valorem tax** — a fixed **percentage of value** (e.g. VAT).
- **Specific (unit) tax** — a fixed **amount per quantity** (e.g. €10 per airline ticket).

A tax can be collected from **either side**, and it drives a **wedge** between three prices:

- **Seller price p_S** — what the seller keeps after paying the tax.
- **Buyer price p_B** — what the buyer actually pays, tax included.
- **Market price p** — the price at which the good trades.

For a specific tax t : if the **seller** remits it, $p_S = p - t$ and $p_B = p$. If the **buyer** remits it, $p_B = p + t$ and $p_S = p$. In both cases: **express demand and supply as functions of the market price, then set supply = demand**.

Worked example — a €10 airline-ticket tax

Demand and supply (thousands of tickets):

$$q_D = 4920 - 5 \cdot p_B \quad q_S = -1746.67 + \frac{10}{3} \cdot p_S$$

No tax: buyer price = seller price = market price → equilibrium **b** at **p = 800, q = 920**.

Tax of €10 collected from sellers ($p_S = p - 10$): supply becomes $q_S = -1746.67 + \frac{10}{3}(p - 10) = -1780 + \frac{10}{3}p$. Setting $q_D = q_S$ gives market price **p = 804** (= the buyer price), quantity **q = 900**, and seller price **$p_S = 804 - 10 = 794$** .

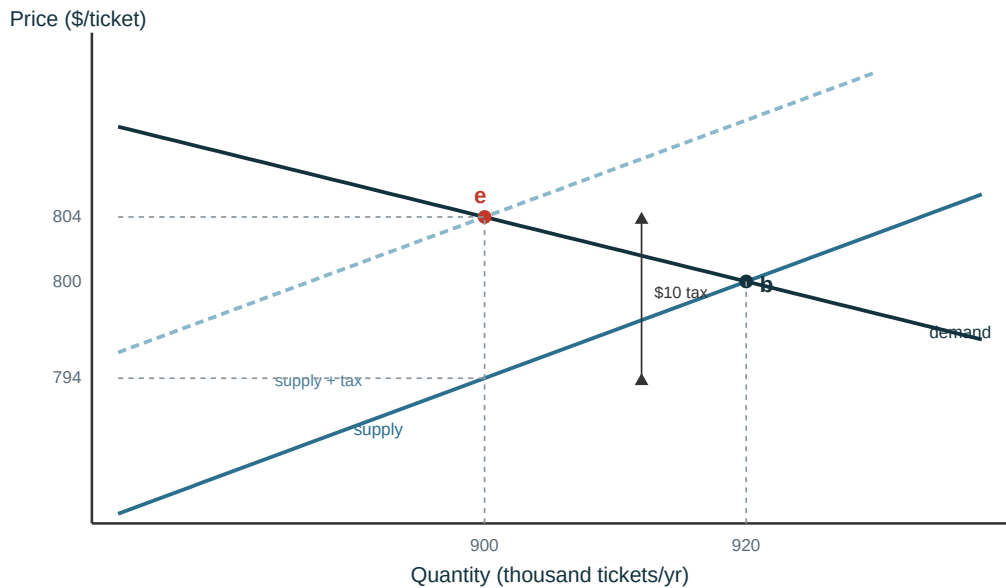


Figure 11. Collecting the tax from sellers shifts the supply curve **up by €10** (sellers now need €10 more per ticket). Equilibrium moves from **b** (800, 920) to **e** (804, 900). The tax drives a **wedge**: buyers pay €804, sellers keep €794.

TAX EQUIVALENCE — WHO "PAYS" DOESN'T MATTER

Collect the **same €10 from buyers instead** ($p_B = p + 10$): demand becomes $q_D = 4920 - 5(p + 10) = 4870 - 5p$. Equating with the untaxed supply gives market price **$p = 794$** , buyer price **$794 + 10 = 804$** , seller price **794**, quantity **900** — **exactly the same outcome**. Whether the buyer or the seller physically remits the tax is irrelevant to prices and quantities. **Subsidies are just negative taxes** and are analysed the same way (shift supply down or demand up).

Surplus, government revenue and deadweight loss

Before the tax, total welfare = consumer surplus + producer surplus. The tax rearranges this:

- **Consumer surplus** — area under demand above the **buyer** price (804).
- **Producer surplus** — area above supply below the **seller** price (794).
- **Government revenue** — the tax $\times$ quantity traded (10×900).
- **Total welfare = CS + PS + government revenue.**

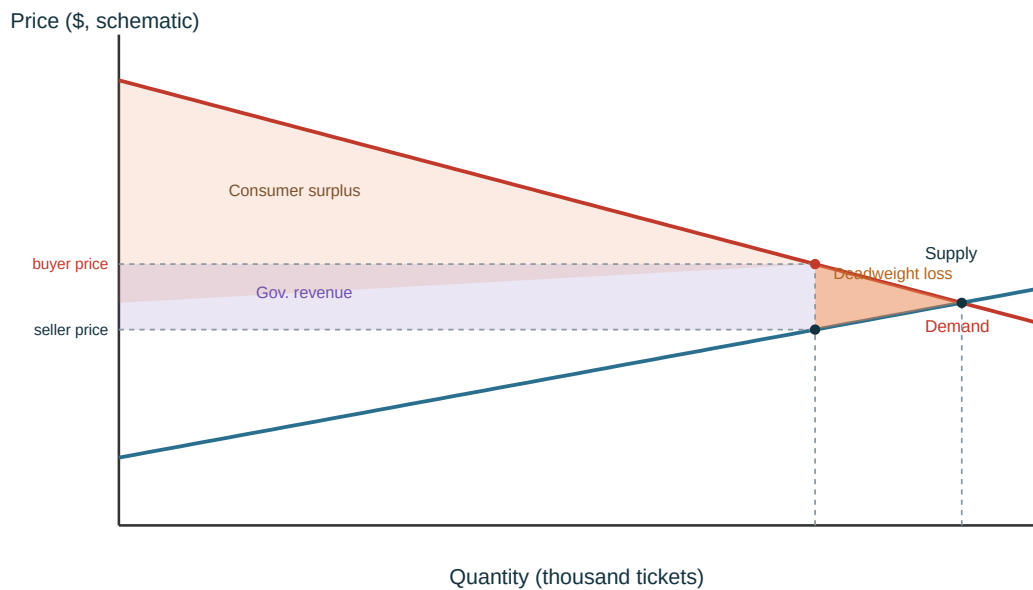


Figure 12. The tax splits the price into buyer (804) and seller (794) prices and cuts quantity from 920 to 900. Government collects the middle rectangle. But the **orange triangle** — trades that were worthwhile ($WTP > MC$) yet no longer happen — is lost to everyone: the **deadweight loss**. Total welfare falls by this triangle.

WHY DEADWEIGHT LOSS ARISES

The tax drives a wedge between the **marginal cost of producing** and the **marginal benefit to consumers**. Some consumers whose willingness to pay **exceeds** the cost of production can no longer buy → total surplus falls and quantity traded is **too low**.

Tax incidence — who really bears the tax?

INCIDENCE & ELASTICITY

The effect of a tax is the **same whether collected from buyers or sellers** (equivalence). With moderate elasticities: the **buyer price rises by less than the tax**, the **seller price falls by less than the tax**, and the **quantity falls**. Crucially, **the side of the market that is less sensitive to price (less elastic) bears the larger share of the tax**.

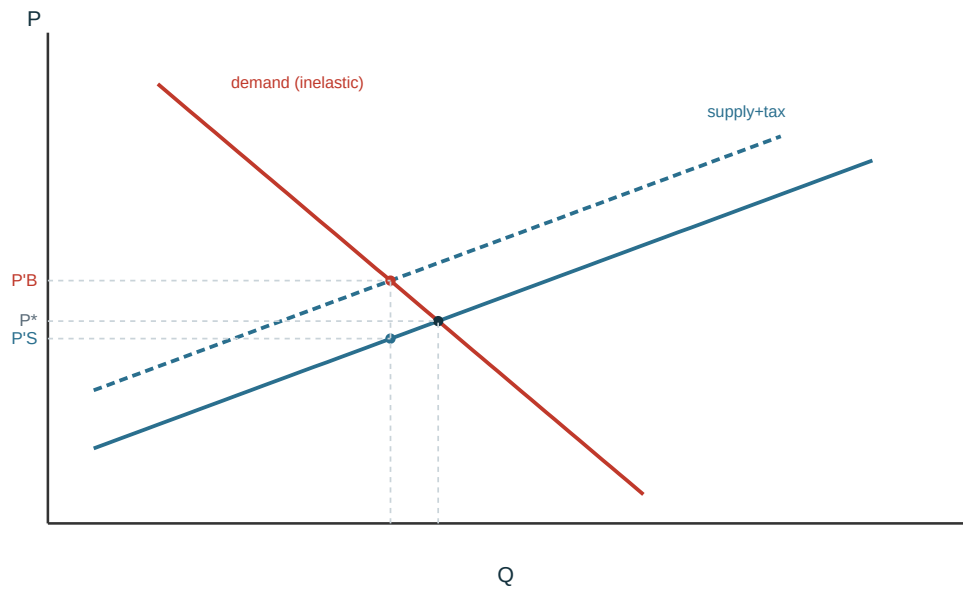


Figure 13. When **demand is more inelastic than supply**, buyers bear more: the buyer price jumps from P^* up to P'_B (a lot), while the seller price only slips to P'_S (a little). In the airline example demand is more elastic than supply, so the split is the other way — the buyer price rises €4 and the seller price falls €6.

Real-world example — Denmark's butter (saturated-fat) tax

In 2011 Denmark introduced a tax per kg on saturated fat, **≈ 22% of the average butter price**. Consumption of butter and related products **fell 15–20%**. But the tax was **scrapped** soon after because of the **administrative burden** of collecting it.

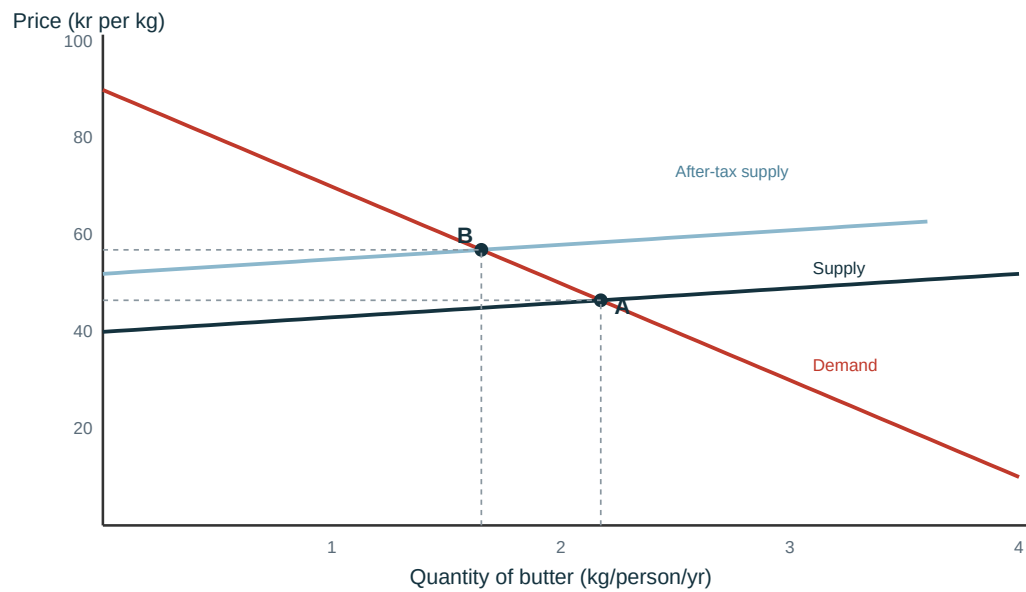


Figure 14. The saturated-fat tax shifts the supply curve up; equilibrium moves from A to B — a higher consumer price and lower quantity. It worked (consumption fell) but the collection costs outweighed the benefits, and it was repealed.

SECTION F

Perfect Competition

The efficiency result of Section C is the pay-off of a specific idealised market structure, **perfect competition**. It pulls together every assumption used along the way.

THE PERFECTLY COMPETITIVE MODEL

A market is **perfectly competitive** when:

- there are **many buyers and sellers**, all **price-takers**;
- products are **homogeneous** (identical);
- there is **free entry and exit** in the long run;
- information is good and there are **no external effects**.

Under these conditions the market reaches $P = MC$ in the short run and $P = MC = AC$ in the long run, and the equilibrium is **Pareto-efficient** — it exhausts all gains from trade with **no deadweight loss**.

LISTEN FOR IN THE LECTURE

Perfect competition is a **benchmark**, not a description of reality ("hard to find price-takers in real life"). Its value is as a yardstick: departures from it — market power (previous unit), taxes (Section E), and externalities (Unit 12) — are exactly the cases where the efficiency result **breaks**, and each creates a deadweight loss.

CHECK YOURSELF

- Which single assumption fails for a firm with market power, and why does that create deadweight loss?
- In long-run equilibrium a competitive firm earns zero economic profit — is it therefore making an accounting loss? (Recall Unit 2: economic vs accounting profit.)

Summary, Formulas & Glossary

The unit in one paragraph

In a competitive market, firms are **price-takers**: each faces a horizontal demand at the market price and produces where $P = MC$ (so its supply curve is its marginal-cost curve above AC). Summing firms gives the upward-sloping **market supply**. The **competitive equilibrium**, where supply = demand, **maximises total surplus** (CS + PS) with **no deadweight loss** — provided price-taking, complete contracts and no externalities hold. Shocks **shift** the curves; how price vs quantity responds depends on **elasticities**, and **entry** drives the long run to $P = MC = AC$ (zero rent). A **tax** drives a wedge between buyer and seller prices, is **independent of who remits it**, transfers surplus to government, and creates a **deadweight loss**; its **incidence** falls more heavily on the **less elastic** side.

Formula & rule sheet

Concept	Rule / expression
Price-taker's output	produce where $P = MC$ (since $MR = P$)
Firm supply curve	the MC curve, for the part above AC
Shut-down / exit rule	produce only if $P \geq AC$ (long run)
Producer surplus	area between price and MC, over units sold
Market supply	horizontal sum of firms' quantities at each price
Elasticity of supply	$\epsilon_S = \% \Delta \text{ quantity supplied} / \% \Delta \text{ price}$
Competitive equilibrium	quantity supplied = quantity demanded ($P = P^*$)
Long-run equilibrium	$P = MC = AC$ (zero economic rent)
Specific tax (seller remits)	$p_S = p - t$, $p_B = p$
Specific tax (buyer remits)	$p_B = p + t$, $p_S = p$
Total welfare with tax	CS + PS + government revenue (– deadweight loss vs no tax)
Tax incidence	less elastic side bears the larger share

Glossary

Term	Meaning
Price-taker	Firm/agent that takes the market price as given; faces horizontal demand.
WTP / WTA	Willingness to pay (demand) / willingness to accept (supply).
Marginal cost / revenue	Cost / revenue of producing one more unit; optimum at $MR = MC$ (= P for a price-taker).
Producer / consumer surplus	Seller / buyer gains from trade (areas relative to price).
Elasticity of supply/demand	$\% \Delta \text{ quantity} \div \% \Delta \text{ price}$; flatter = more elastic.
Competitive equilibrium	Price-taking market where supply = demand.
Pareto efficiency	No one can be made better off without making another worse off; competitive eq. achieves it under the assumptions.
Deadweight loss	Lost total surplus from trades that don't happen (e.g. under a tax).
Short vs long run	Number of firms fixed vs free to enter/exit; long-run supply is more elastic.
Tax wedge / incidence	Gap between buyer and seller price / how the burden is split.
Ad valorem vs specific tax	% of value (VAT) vs fixed amount per unit.

Exam-style questions to practise

1. A firm has $MC = 2 + 0.01Q$. At a market price of €5, how much does it supply, and what is its producer surplus?
2. Demand is $Q = 100 - 2P$, supply $Q = -20 + 2P$. Find the equilibrium price and quantity, and identify excess supply/demand at $P = 35$ and $P = 25$.
3. A €4 specific tax is levied on a good. Explain why the outcome is identical whether buyers or sellers remit it.
4. Using elasticities, explain who bears more of a tax on cigarettes (inelastic demand) versus a tax on a specific brand of soft drink (elastic demand).
5. Draw the short-run and long-run adjustment to a permanent rise in demand for bread, and explain why the long-run price rise is smaller.
6. Why does a competitive equilibrium maximise *total* surplus but not *consumer* surplus? Illustrate the deadweight loss created by a price ceiling.

Study guide prepared as a companion to *Microeconomics — Prosperity and Inequality, Unit 8: Supply and Demand — Price-taking and Competitive Markets* (KU Leuven, 2025–2026). Equilibria, surplus areas and tax wedges are computed from the stated demand/supply functions; the airline and tanker examples reproduce the slides' numbers.

MICROECONOMICS · PROSPERITY AND INEQUALITY

Technological Change, Population & Growth

Unit 2 — Study Guide & Lecture Notes

- › Decision-making, opportunity cost & economic rent
- › Specialization & comparative advantage
- › Technology choice, isocost lines & the Industrial Revolution
- › Economic models & the climate challenge

BEFORE YOU BEGIN

How to use these notes

These notes follow the lecture slides one-to-one, but expand the reasoning so you can read them on their own, annotate them during class, and revise from them before the exam. Every important graph from the slides has been redrawn cleanly so you can see exactly what the lecturer is pointing at.

DEFINITION BOX

The precise definition you are expected to reproduce in the exam.

WORKED EXAMPLE

Numbers carried all the way through, matching the slide examples.

LISTEN FOR IN THE LECTURE

Points the lecturer usually stresses verbally — leave room to add your own notes here.

CHECK YOURSELF

Quick questions to test recall before moving on.

The big question of this unit

For almost all of human history, living standards barely moved. Then, roughly **200–250 years ago**, income per person in a handful of countries began to rise at a rate never seen before — the famous "**hockey-stick**" of growth. This unit builds the economic toolkit needed to answer two linked questions:

1. Why did living standards and population **stagnate** for centuries (the **Malthusian trap**)?
2. Why did they suddenly **take off** with the Industrial Revolution — and why in **Britain** first?

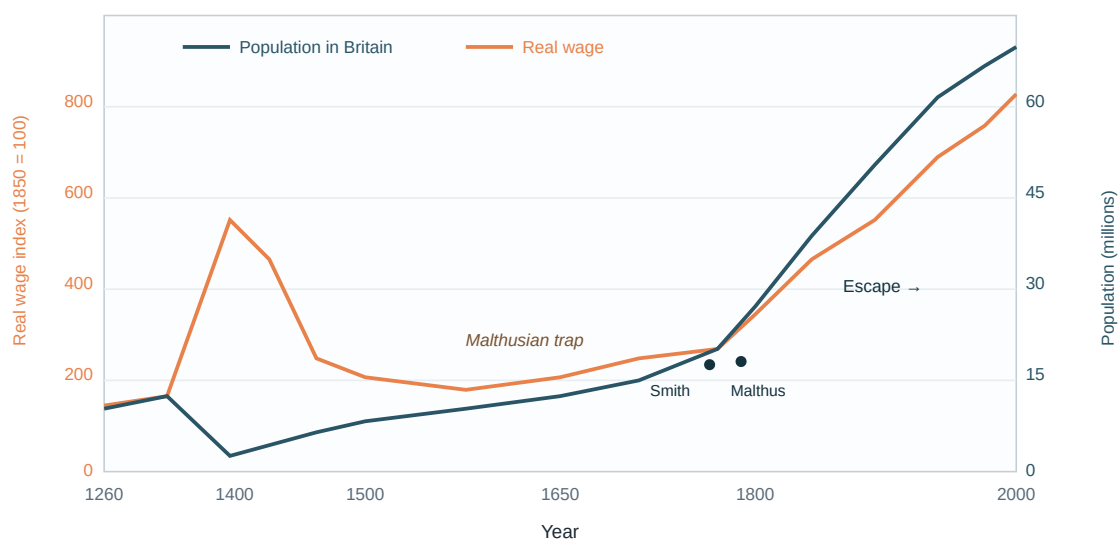


Figure 1. The escape from stagnation. For ~500 years real wages and population were locked in the **Malthusian trap** (flat, with wages falling whenever population rose). From ~1800 both wages *and* population climb together — the Industrial Revolution breaks the trap.

LISTEN FOR IN THE LECTURE

The lecturer uses this chart to frame the whole unit: *autarky vs. trade, Malthus vs. Smith, and why "escape" happens exactly when technology starts improving permanently*. Note the brief dip around 1350 — the Black Death killed a third of the population, so the survivors' real wages jumped. That single event is the classic illustration of the Malthusian mechanism.

SECTION A

Introduction

The rapid, sustained rise in income and living standards we take for granted today is, historically speaking, **brand new**. It is largely the product of **technological progress** — our growing ability to produce more output from the same inputs.

Two puzzles the unit will solve

- **How did the technological revolution start?** What changed so that inventions began to be adopted, one after another, in a self-sustaining wave?
- **Why did it not start earlier?** Humans were clever for millennia — water wheels, windmills and gunpowder all pre-date 1750. So why did take-off wait until the 18th century?

The slide's three images — a woman spinning by hand, the **spinning jenny**, and a mechanised **cotton mill** — trace exactly this transition: from home production, to a single clever machine, to the factory system. Textiles were the first industry to be transformed, and it is the industry we will keep returning to when we model technology choice in Section D.

KEY IDEA OF THE UNIT

We will use **economic models** to explain (i) the centuries of **stagnation** before 1800 and (ii) the **explosive growth** in living standards and population since. The tools introduced along the way — opportunity cost, comparative advantage, production functions and isocost lines — are the core of microeconomics.

LISTEN FOR IN THE LECTURE

The framing question "*Why an island off the coast of Europe, and why then?*" recurs throughout. Keep it in mind: by the end of Section D you should be able to answer it with a diagram, not just a story.

SECTION B

Decision Making

Before we can explain the behaviour of firms and inventors, we need a clean theory of **how a rational agent chooses**. Economics models every decision as a **trade-off between the benefits and the costs** of an action. The subtlety — and the whole point of this section — is that "cost" means more than the money that leaves your wallet.

Direct costs vs. opportunity costs

DEFINITION — OPPORTUNITY COST

The **opportunity cost** of an action is the **net benefit of the next-best alternative action** that you give up in order to take it. It is a cost even though no money changes hands.

- Part of the opportunity cost of **studying** is the **wage you could have earned** working instead.
- The opportunity cost of **starting your own business** is the **salary you could earn as an employee**.

DEFINITION — ECONOMIC COST

$$\text{Economic cost} = \text{Direct (out-of-pocket) cost} + \text{Opportunity cost}$$

A decision is worth taking only if its benefit exceeds this *full* economic cost — not merely the accounting cost.

Worked example — the clothing shop in Brussels

A woman owns a clothing shop on the *Nieuwstraat*. The facts:

- Expected **revenue** this year: €400,000
- **Material expenses**: €300,000
- The **building** is worth €1,000,000. She inherited it 20 years ago and its value is fully written off (so accounting shows no rent cost).
- She could invest money in a fund earning **5% per year**.
- She could earn **€60,000 per year** working elsewhere.

Income Statement (Accounting)

Revenue	400,000
Material expenses	– 300,000
Accounting profit	100,000

Economic Profit

Revenue	400,000
Material expenses	– 300,000
Accounting profit	100,000
Opportunity costs	
Building ($5\% \times \text{€}1\text{m}$)	– 50,000
Forgone wage	– 60,000
Economic profit	– 10,000

The **accounting profit is a healthy €100,000**. But the best alternative course of action — **sell the building, invest the €1m, and take a job** — would yield €50,000 (interest) + €60,000 (wage) = **€110,000**. Because that alternative earns more, keeping the shop open actually **destroys €10,000 of value per year**. The **economic profit (or economic rent)** of staying is **–€10,000**.

WHY THE BUILDING MATTERS EVEN THOUGH IT'S "FREE"

Owning the building outright does *not* make it costless. Every year she holds it, she gives up the €50,000 it could earn if sold and invested. This forgone return is a genuine opportunity cost — a favourite exam trap.

The decision rule

Two equivalent ways to state the same rule:

1. Take the action if **benefit > economic cost** (direct cost + opportunity cost).
2. Take the action if its **net benefit** (benefit – direct cost) exceeds the **net benefit of the next-best alternative** (the opportunity cost).

In the example: net benefit of the shop = **€100,000**; net benefit of the alternative = **€110,000**. Since $100,000 < 110,000$, she should **not** keep the shop.

Economic rent and innovation rent

DEFINITION — ECONOMIC RENT

$$\text{Economic rent} = \text{benefit} - \text{direct costs} - \text{opportunity costs} = \text{net benefit} - \text{opportunity cost}$$

A **positive economic rent** means the action beats every alternative; it is the reward for doing something better than the next-best option.

DEFINITION — INNOVATION RENT

A special case of economic rent. The **innovation rent** is the difference between the profit earned with a **new technology** and the profit you would have earned **sticking with the old one**. It is the carrot that gives firms an **incentive** to innovate — and it is the engine that will drive the whole Industrial-Revolution model in Section D.

CHECK YOURSELF

- If the fund's return rose to 8%, what would the shop's economic profit become?
(Answer: opportunity cost of the building = €80,000, so economic profit = 100,000 – 80,000 – 60,000 = –€40,000 — even worse.)
- Why can a firm show an accounting profit yet still be "failing" economically?

SECTION C

Specialization & Comparative Advantage

Technological progress and rising output are inseparable from **specialization** — people and firms concentrating on a narrow range of tasks and trading for everything else. This section explains *why* specialization raises output, and the surprising result that trade benefits **everyone**, even a producer who is worse at everything.

The gains from specialization

The rise of **firms** lets people specialize in what they are good at. Two mechanisms:

- **Division of labour within firms** — a firm splits one product into many small tasks.
- **Firms focusing on what they do best** — specialization *between* firms.

Specialization raises the **productivity of labour** for three reasons:

THREE SOURCES OF THE GAINS

- **Learning by doing** — repetition makes you faster and better at a task.
- **Differences in ability** — people are naturally better at different things.
- **Economies of scale** — producing many units is more cost-effective per unit than producing few.

Crucially, people can **only** specialize if they have a way to acquire everything else they need. In a capitalist society that mechanism is the **market**.

Adam Smith's pin factory

Adam Smith (*The Wealth of Nations*, 1776) gave the classic illustration. Making a pin involves **18 distinct operations** (drawing the wire, straightening, cutting, pointing, grinding...). Instead of one worker doing all 18:

- A single worker doing everything might make **~20 pins/day**.
- **10 workers**, each specialising in a few operations, produce **48,000 pins/day** — thousands of times more per person.

Why? Time is saved by not switching tasks, skill rises through **learning by doing**, and each person does the operation they are best at. And it does not stop there — the same logic drives **further mechanisation**:

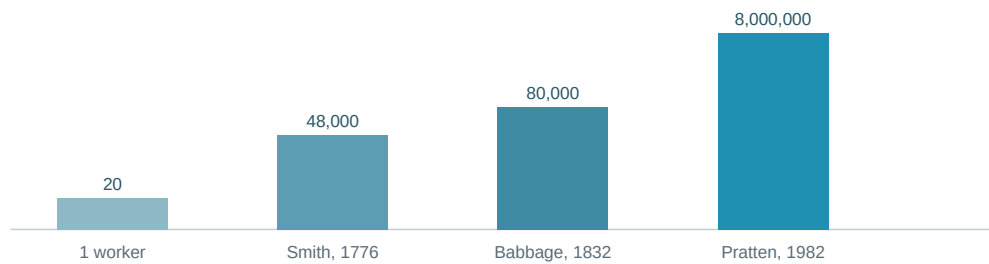


Figure 2. Output per pin-factory day, on a log scale. Specialization (1776) plus two centuries of mechanisation multiplied productivity by a factor of hundreds of thousands. Technology and the division of labour reinforce each other.

Absolute vs. comparative advantage

Now the deep result. When abilities differ, **all** producers gain by specialising — *even if one of them could produce every good more cheaply*. What matters is not who is better in absolute terms, but who has the lower **relative** (opportunity) cost. David Ricardo (1817) proved this with the famous Portugal–England example.

Hours of labour needed	Wine (per unit)	Clothes (per unit)
England	120	100
Portugal	80	90

These are **labour requirements** (the inverse of labour productivity). Portugal needs fewer hours for *both* goods, so Portugal has an **absolute advantage** in everything. It looks like Portugal has nothing to gain from England. Ricardo showed the opposite.

DEFINITION — COMPARATIVE ADVANTAGE

A country (or person) has a **comparative advantage** in a good if its **opportunity cost** of producing that good — how much of the *other* good it must give up — is **lower** than the other country's. A country can be worse at producing both goods (absolute disadvantage) yet still hold a comparative advantage in one of them.

Production Possibilities Frontier (PPF)

Suppose each country has **360,000 working hours**. Devoting them all to one good gives the **maximum production**:

- **England:** 3,000 wine or 3,600 clothes.
- **Portugal:** 4,500 wine or 4,000 clothes.

All in-between mixes lie on a straight line, the **production possibilities frontier (feasible frontier)**. For England the labour constraint is $100 \cdot \text{Clothes} + 120 \cdot \text{Wine} \leq 360,000$, i.e.

$$\text{Clothes} = 3600 - \frac{6}{5} \cdot \text{Wine} \quad (\text{England}) \quad \text{Clothes} = 4000 - \frac{8}{9} \cdot \text{Wine} \quad (\text{Portugal})$$

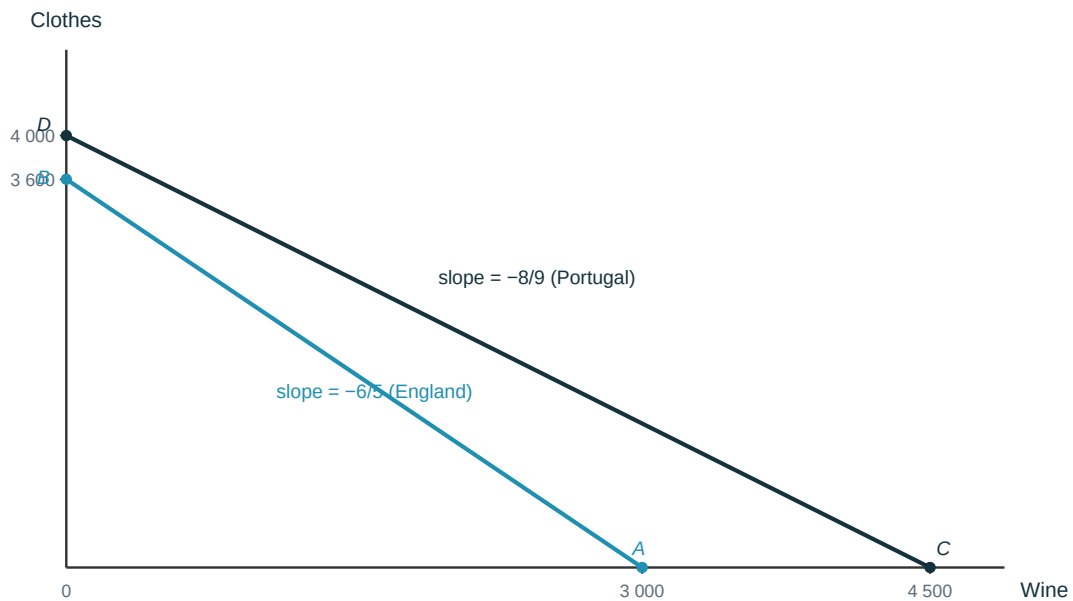


Figure 3. PPFs for the two countries. The **slope of each line is the opportunity cost of wine** (clothes forgone per unit of wine). Portugal's line is flatter ($-8/9$): giving up wine costs it less clothing than it costs England ($-6/5$).

Reading off the comparative advantage

	Opportunity cost of 1 wine	Opportunity cost of 1 clothes
England	$120/100 = 6/5$ clothes	$100/120 = 5/6$ wine
Portugal	$80/90 = 8/9$ clothes	$90/80 = 9/8$ wine

- **Portugal** gives up only **8/9** clothes per wine vs England's **6/5** → **Portugal has the comparative advantage in wine.**
- **England** gives up only **5/6** wine per clothes vs Portugal's **9/8** → **England has the comparative advantage in clothes.**

So the pattern of trade is: **Portugal specializes in wine, England in clothes, and they swap.** Comparative advantages create the *opportunities* for specialization and trade.

The terms of trade and the gains

What price (of wine in terms of clothes) will they trade at? It must sit **between the two opportunity costs**:

$$\frac{8}{9} < \text{price of wine (in clothes)} < \frac{6}{5}$$

- Below $8/9$ → Portugal would rather make its own clothes.
- Above $6/5$ → England would rather make its own wine.

Suppose they agree on the simple rate **1 wine = 1 clothes**. Both can now consume **beyond their own PPF**:

WORKED EXAMPLE — ENGLAND GAINS FROM TRADE

Under **autarky**, England might produce and consume 3,000 clothes and 500 wine (labour used = $3000 \cdot 100 + 500 \cdot 120 = 360,000$ ✓). With trade, England **specialises fully in clothes (3,600)**, keeps 3,000 for itself, and trades **600 clothes for 600 wine** with Portugal. Result: 3,000 clothes **and** 600 wine — more of one good, same of the other. Portugal gains symmetrically.

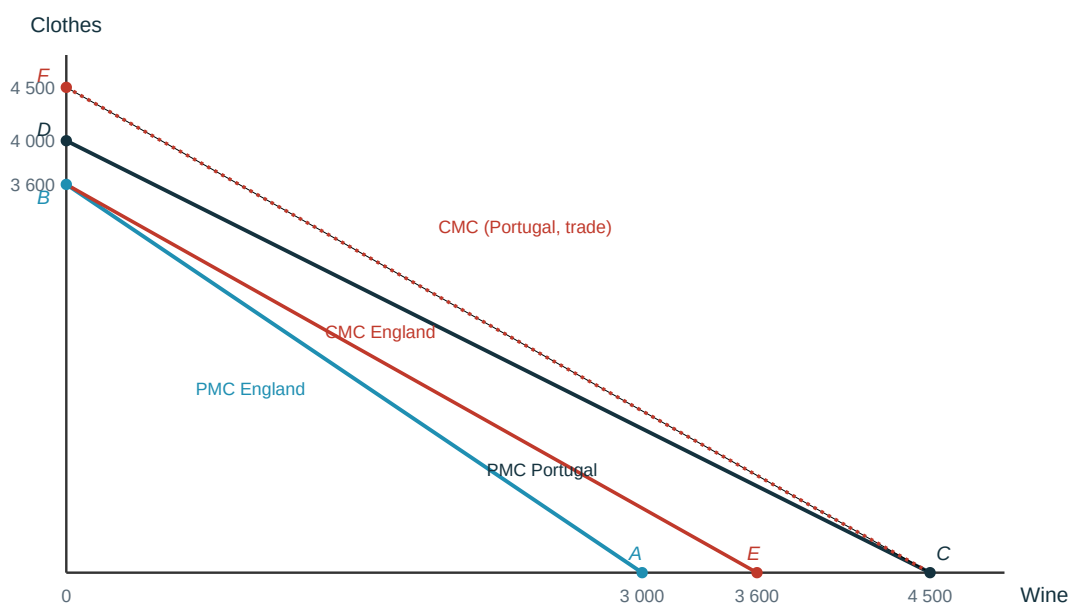


Figure 4. Production (solid) vs consumption possibilities (red) once trade opens at 1:1. Each country's red consumption line lies **outside its own production frontier** — the visual proof of the gains from trade. The same logic applies to **two individuals** inside one country, not just to nations.

CHECK YOURSELF

- Portugal has an absolute advantage in both goods — so why does it still gain by trading with England?
- If the trade price were exactly $6/5$ clothes per wine, which country would capture *all* the gains?
- What does the **slope** of a straight-line PPF represent, and why is it constant here?

SECTION D

Technologies & Innovation

This is the heart of the unit. We build a **model of the firm's technology choice** and use it to answer the framing question: *why did the Industrial Revolution start in 18th-century Britain?* The many possible explanations — high wages, cheap coal, the Enlightenment, culture, colonies — are narrowed to a mechanism we can **draw**: a change in **relative input prices** made energy-hungry machines profitable, and Britain happened to have exactly the right prices.

Firms, technology and the production function

DEFINITIONS

- **Technology** — the method a firm uses to convert inputs into output.
- **Factors of production** — the inputs (labour, machines, energy, land...).
- **Production function** — the relationship between inputs and output, e.g. $Y = f(X)$ for one input, or $Y = f(M, N, E)$ for machines, workers and energy.

Fixed vs. variable proportions

Consider a simple **olive-oil** technology: three machines (milling, mashing, pressing), one worker to operate them, using 80 kWh, producing 50 litres.

Machines M	Workers N	Energy E (kWh)	Output Y (litres)
3	1	80	50
6	2	160	100
9	3	240	150
12	4	320	200

TWO KEY PROPERTIES

- **Fixed-proportions technology**: to raise output the inputs must be scaled up *together* in fixed ratios (1 worker : 3 machines : 80 kWh). You cannot swap one for another.
- **Constant returns to scale**: doubling every input exactly doubles output.

By contrast, the **food/apples** production of the previous unit was **variable-proportions**: you could add labour to a *fixed* plot of land, giving the familiar **concave** production function (diminishing returns to labour). Fixed proportions is a special case where no substitution is possible.

Graphing a two-input technology

Output depends on two inputs, so a full picture is 3-D and awkward. The trick: put the **two inputs on the axes** and mark a point for each output level. With fixed proportions, equal output levels line up on a **straight ray through the origin**.

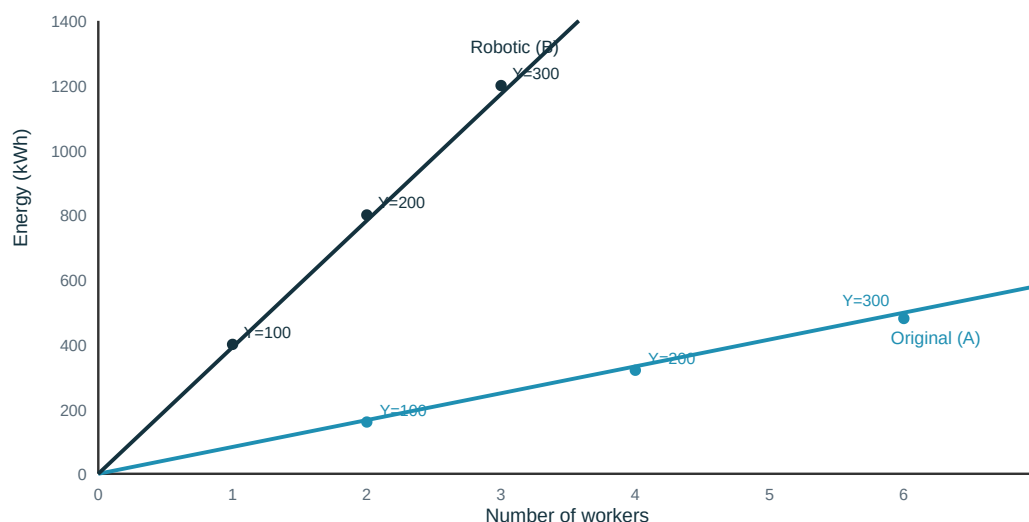


Figure 5. Two **fixed-proportions** technologies for olive oil. The **original (A)** uses more **labour**; the **robotic (B)** uses more **energy** but only 1 worker per 100 litres. Neither is "better" in the abstract — the best choice depends entirely on the **relative prices** of labour and energy.

LISTEN FOR IN THE LECTURE

The point the lecturer is driving at: a technology is just a **bundle of input requirements**. "Which is better?" is **not** answerable until we know input prices. That is the bridge to isocost lines.

Modelling the choice of technology (cloth)

Back to the Industrial Revolution. There are **five ways to produce 100 metres of cloth**, each using labour (workers) and energy (tonnes of coal). Technology **E** is **labour-intensive**; technology **A** is **energy-intensive**.

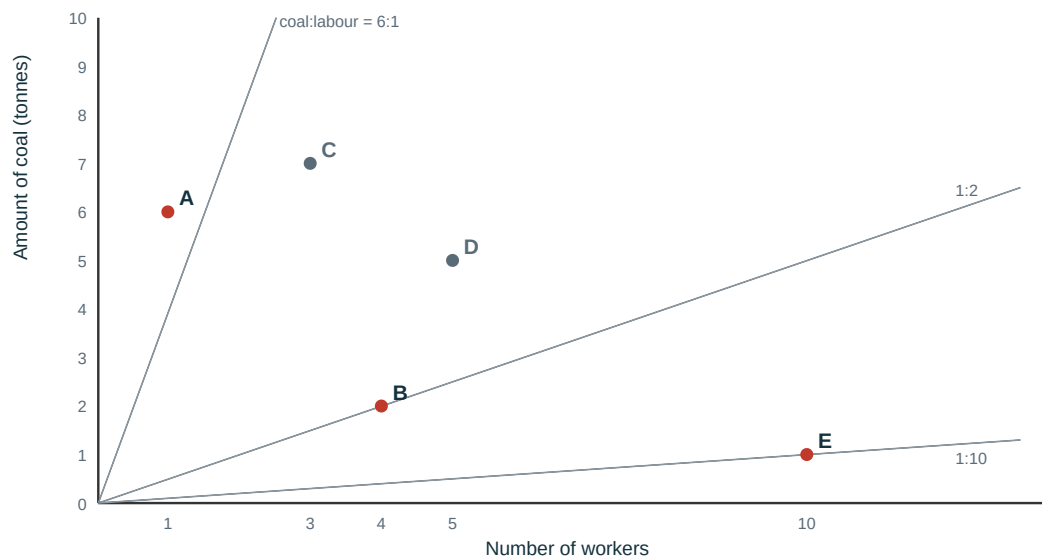


Figure 6. Five technologies (each produces the same 100 m of cloth). Rays from the origin show constant coal-to-labour ratios. Red points (A, B, E) will turn out to be the **sensible** choices; grey points (C, D) are wasteful — the next figure shows why.

Ruling out inferior (dominated) technologies

DEFINITION — DOMINATION

Technology 1 is **dominated** by technology 2 if technology 1 uses **at least as much of every input** (and more of at least one). A dominated technology will **never** be chosen, whatever the input prices are.

Compare the points: **C (3 workers, 7 coal)** needs **both** more workers *and* more coal than **A (1 worker, 6 coal)** → **C is dominated by A**. Likewise **D is dominated by B**. That leaves **A, B and E** — none of which is dominated (each is best on at least one input). To choose among *these*, we finally need **prices**.



Figure 7. Any technology lying **up-and-to-the-right** of A uses more of both inputs and is dominated by A. C sits in that region → eliminated. A, B and E survive.

CHECK YOURSELF

Is technology E dominated by anything? Why can a firm rationally pick the coal-heavy A or the labour-heavy E depending on circumstances? (Hint: nothing sits below-and-left of either.)

Cost Minimisation & Isocost Lines

Firms want to maximise profit, which — for a given amount of cloth — means producing it at **the least possible cost**. The total cost of a technology using L workers and R tonnes of coal is:

$$\text{cost} = (\text{wage} \times \text{workers}) + (\text{coal price} \times \text{tonnes}) = (w \times L) + (p \times R)$$

This is why the choice of technology depends on **economic information** — the **relative prices** of the inputs — and not on engineering alone.

Isocost lines

DEFINITION — ISOCOST LINE

An **isocost line** joins all input combinations that cost the **same total amount**. Rearranging the cost equation:

$$R = \frac{c}{p} - \frac{w}{p} \cdot L$$

Its **slope is $-w/p$** , the **relative price of the inputs**. Points **above** a line cost more; points **below** cost less. **Lower isocost lines = cheaper**. The firm picks the technology sitting on the **lowest attainable isocost line**.

Numerical anchor (wage £10, coal £20/tonne, so slope = $-10/20 = -1/2$):

Point	Workers	Coal (tonnes)	Total cost (£)
P_1	2	3	80
P_2	6	1	80
Q_1	3	6	150
Q_2	5	5	150

P_1 and P_2 lie on the **same £80 line**; Q_1 and Q_2 on the higher **£150 line**. Any point on a higher (further-out) line is more expensive.

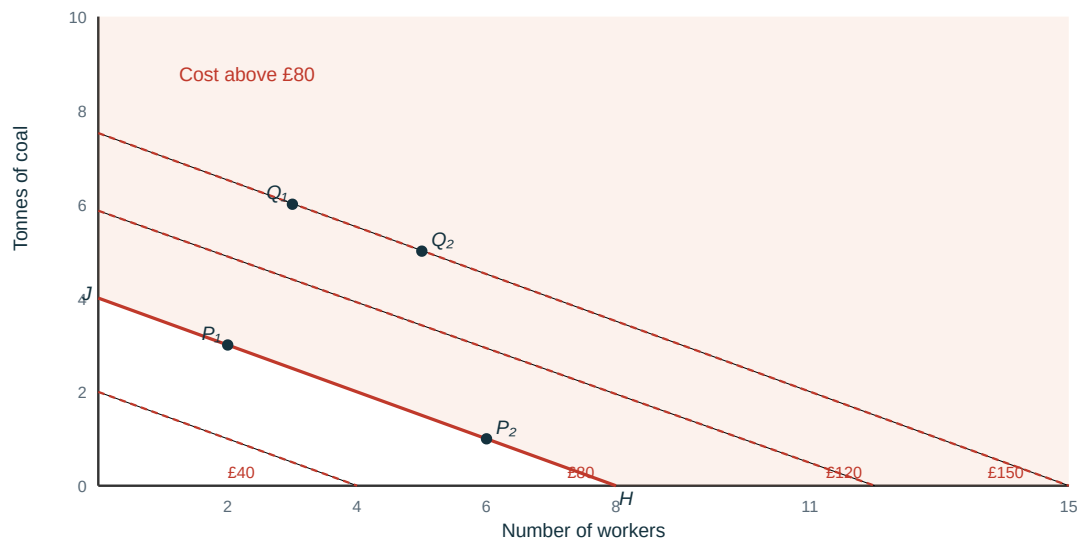


Figure 8. A family of parallel isocost lines (slope $-w/p = -\frac{1}{2}$). P_1 and P_2 share the £80 line; Q_1 , Q_2 the £150 line. The firm wants the **lowest** line that still reaches a feasible technology.

Why Britain? Changing relative prices

Now put technologies and isocost lines **on the same graph**. Recall the surviving technologies for 100 m of cloth: **A = (1 worker, 6 coal)** — energy-intensive — and **B = (4 workers, 2 coal)** — labour-intensive.

Before the Industrial Revolution (coal £20, labour £10)

- Cost with **B**: $(10 \times 4) + (20 \times 2) = \text{£}80 \rightarrow$ B sits on the lowest isocost line.
- Cost with **A**: $(10 \times 1) + (20 \times 6) = \text{£}130 \rightarrow$ A is far more expensive.
- **Firms choose B**. There is **no incentive** to develop the coal-hungry technology A.

After coal gets cheaper (coal £5, labour £10)

The isocost slope becomes $-10/5 = -2$ — the lines get **steeper**. Now:

- Cost with **A**: $(10 \times 1) + (5 \times 6) = \text{£}40$.
- Cost with **B**: $(10 \times 4) + (5 \times 2) = \text{£}50$.
- **A is now on the lowest isocost line** — firms switch to the energy-intensive technology.

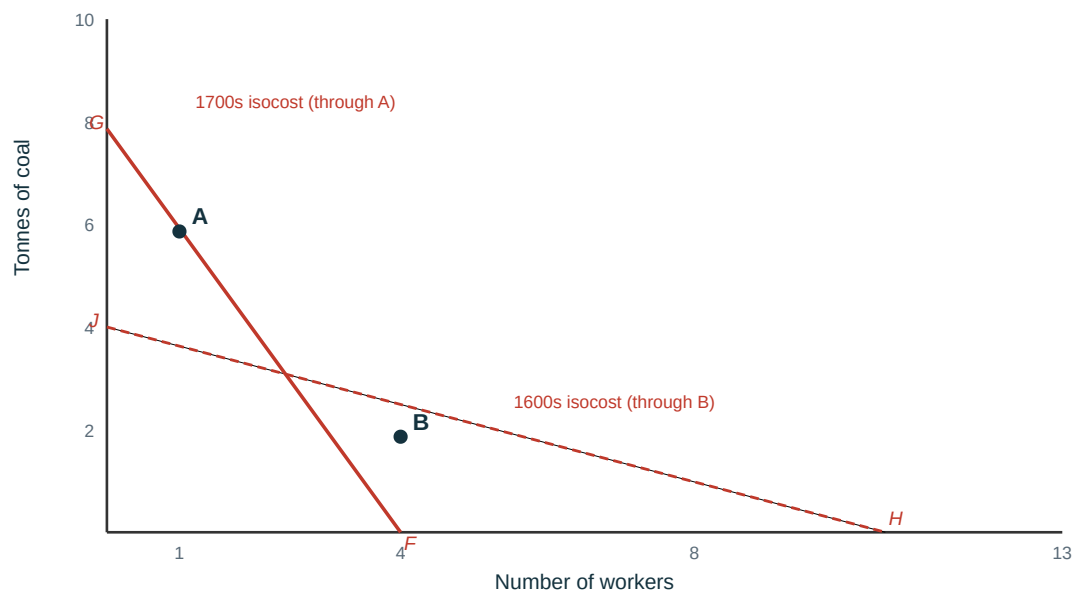


Figure 9. The **same two technologies**, two eras. In the **1600s** the flat isocost (cheap-ish labour relative to coal) makes **B** cheapest — A lies *outside* line HJ, so no one develops it. By the **1700s** labour is dear and coal cheap: the steep isocost makes **A** cheapest (B now lies outside FG). Britain's price signal is what triggered the switch.

THE BENEFIT OF INNOVATING — INNOVATION RENT

A firm that switches first to the newly-cheapest technology gains an advantage over rivals. Since $\text{profit} = \text{revenue} - \text{costs}$, the extra profit equals the **fall in cost**:

- Cost with old technology B: $(4 \times 10) + (2 \times 5) = \text{£}50$
- Cost with new technology A: $(1 \times 10) + (6 \times 5) = \text{£}40$

Innovation rent = £10 per 100 m of cloth. This rent is the incentive that pulls firms toward better technology.

Why Britain, and why then?

The model's answer is now sharp: the Industrial Revolution needed **(i) the capacity to innovate** and **(ii) relative prices that made energy-intensive technology cheapest**. Britain had both.

Uniquely, **English wages were high** (relative to the cost of capital/energy) **and coal was cheap** — precisely the configuration that rewards labour-saving, coal-using machines.

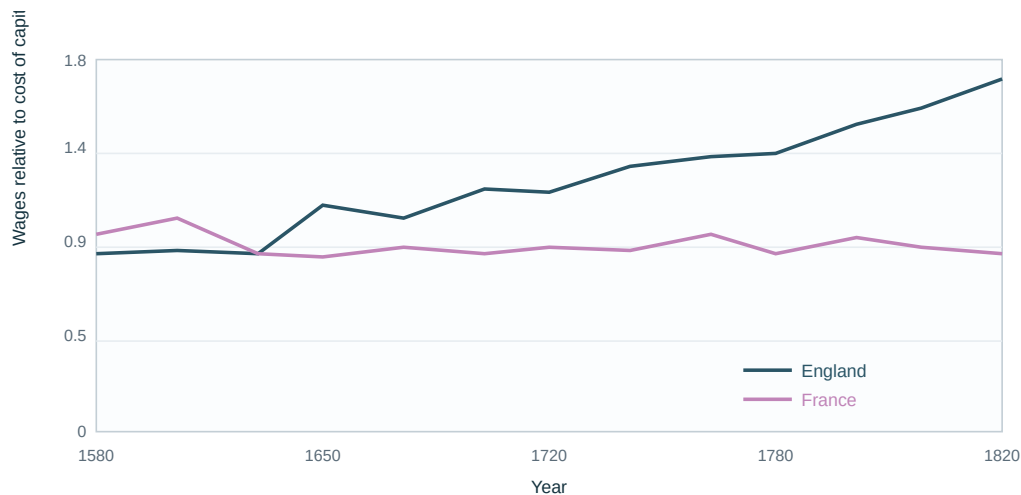


Figure 10. Wages relative to the cost of capital, 1580–1820. England's ratio climbs steadily (labour became expensive relative to machines/energy); France's stays flat and low. Only in England did the price signal steeply reward capital- and energy-intensive innovation — the answer to "why Britain first?"

Why did other countries eventually adopt it too?

- **Technological progress:** continued innovation reduced the inputs needed per unit (the technology moved from A toward an even leaner A'), lowering costs for everyone.
- **Falling energy costs and rising wages** spread internationally, so the coal-intensive technology eventually became cheapest elsewhere as well.

Entrepreneurs and creative destruction

DEFINITIONS — SCHUMPETER'S ENGINE

- **Entrepreneur** — the **first adopter**, the firm willing to try new technologies and start new businesses.
- **Schumpeterian (innovation) rents** — the temporary extra profits the first adopters enjoy.
- **Creative destruction** — the process by which old technologies and the firms that fail to adapt are **swept away** by the new, because they can no longer compete in the market.

The reward (innovation rent) and the threat (creative destruction) together make capitalism a **permanent innovation machine** — the mechanism absent from all the stagnant economies before 1800.

CHECK YOURSELF · EXAM MCQ

Two technologies A and B produce a car, with **machines on the Y-axis** and **workers on the X-axis**. Which statement is correct?

- (a) "A **dominates** B if, at current prices, a car is cheaper with A." — **False**. Domination means using *less of every input*; it is independent of prices. What (a) describes is A being *cheaper*, not dominating.
- (b) "If A is on an isocost line **above** B's, A is cheaper." — **False**. Higher (further-out) isocost lines cost *more*, so A would be dearer.
- (c) "Isocost lines become **flatter** if the price of machines goes up." — **True — this is the answer**. With machines on Y, the cost line is $M = c/p_m - (w/p_m) \cdot L$, so slope = $-w/p_m$. A higher machine price p_m shrinks the slope's magnitude → the lines get **flatter**.
- (d) "None of the above." — **False**, because (c) is correct.

Take-away: keep three ideas separate — **domination** (input quantities, price-free), **cost** (which isocost line you sit on), and **slope** (the input *price ratio*, read off the axis convention).

SECTION E

Economic Models

Everything we did in Sections C and D was **model-building**. This section steps back to ask: what is an economic model, why do we deliberately leave things out, and what makes one good?

Why do we need models?

An economy is the actions and interactions of **millions** of people. No one can hold all of that in mind at once. A **model** is a deliberate simplification that lets us see the **big picture**. Building one means separating:

- the **essential features relevant to the question** — kept in the model, and
- **unimportant details** — safely ignored.

KEY MINDSET

A model **omitting** detail is a **feature, not a bug**. The omissions are what make the important elements visible.

The metro-map analogy. A metro map is "wrong" about almost everything — distances, street layout, the curve of the lines. But for the one question a passenger actually asks — "*which line do I take, and where do I change?*" — it is far more useful than a geographically accurate map. A good economic model is exactly like this: it distorts the irrelevant to make the relevant unmistakable. (The Barcelona and Brussels metro maps on the slides make the same point — both are schematic, both are useful.)

Building a model — five steps

1. **Begin with a well-defined question.**
2. **Capture the elements** of the economy that matter for that question.
3. **Describe how agents act** and interact with each other and with the model's elements.
4. **Determine the outcomes** of these actions — an **equilibrium**.
5. **Study what happens when conditions change** (comparative statics).

Key ideas inside a model

VOCABULARY YOU MUST KNOW

- **Equilibrium** — a **self-perpetuating** situation: nothing of interest changes unless an **external force** alters the model's set-up.
- **Endogenous variable** — its value is **determined by the model**.
- **Exogenous variable** — its value is **determined outside the model** (taken as given).
- **Ceteris paribus** — "all else equal": study the relation between two variables while **holding everything else constant**.
- **Mathematics** — economic models are usually expressed with **equations and graphs**.

WORKED DISTINCTION — THE MALTHUS MODEL

In the Malthusian model, **population and living standards are endogenous** (the model determines them), while **technological progress is exogenous** (it arrives from outside). The whole story of the Industrial Revolution is what happens when that exogenous variable stops being a one-off shock and becomes **permanent**.

What makes a good model?

Four tests — a good model is:

- **Clear** — it helps us understand something important.
- **Predicts accurately** — its predictions are consistent with the evidence.
- **Improves communication** — it makes explicit what we agree and disagree about.
- **Useful** — we can use it to find ways to improve how the economy works.

CHECK YOURSELF

In the firm's technology-choice model of Section D, which variables are **exogenous** (given from outside) and which are **endogenous** (chosen by the firm)? (*Hint: input prices and the available technologies are exogenous; the technology actually chosen and the cost are endogenous.*)

SECTION F

Economic Growth & Climate Change

The same forces that produced the hockey-stick of prosperity also produced a hockey-stick of **carbon emissions**. This closing section connects the model to the defining challenge of the century.

Capitalism + carbon = growth + climate change

The Industrial Revolution was, at its core, an **energy transition**:

THE CENTRAL TRANSITION

From an economy where **photosynthesis** was the main energy source — so that **land was the binding constraint** on growth (the Malthusian world) — to an **energy-rich economy based on fossil fuels**, where that constraint was lifted.

- It produced **unprecedented increases in per-capita income**.
- It also produced **unprecedented increases in the Earth's surface temperature**.
- The ongoing reduction in global poverty **cannot** be achieved by the same "**coal-plus-capitalism**" recipe — the emissions would be catastrophic.

An unsustainable path

Across countries, richer means dirtier: CO₂ emissions per person rise strongly with GDP per person. Continuing to lift billions out of poverty along this line would push emissions far beyond safe limits.

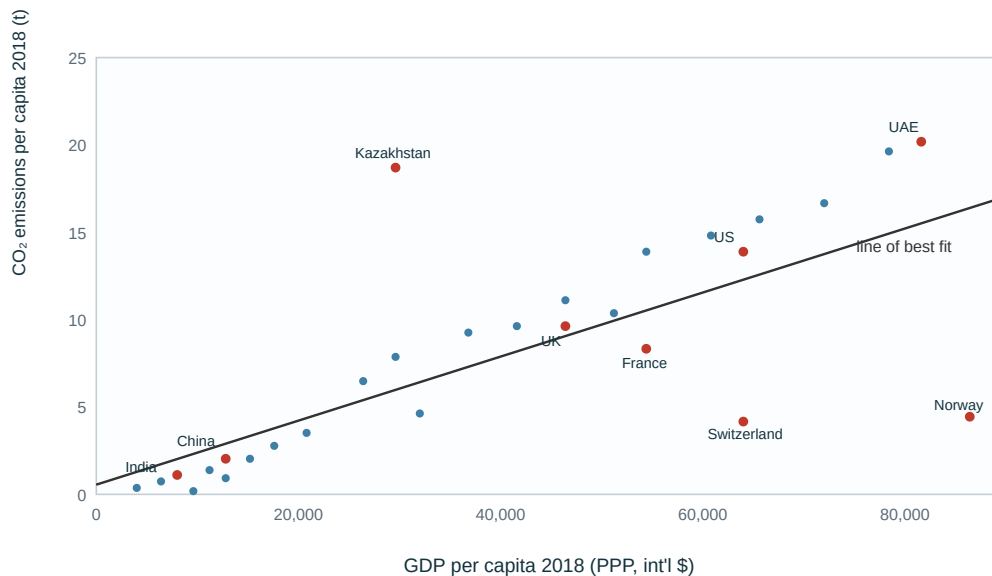


Figure 11. Richer countries emit more CO₂ per person (positive best-fit line). But the **scatter around the line matters**: France, Switzerland, Norway and Sweden are rich yet **low-carbon**, proving prosperity and low emissions *can* coexist — a matter of technology and energy mix, not income alone.

Some hope? Cheap clean energy and decoupling

The same innovation logic from Section D now works **for** the climate. As cumulative solar capacity grew, the price of solar modules collapsed along a steep **experience curve** — every doubling of installed capacity cut costs by a roughly constant percentage. Renewables have gone from luxury to **cheapest new electricity**.

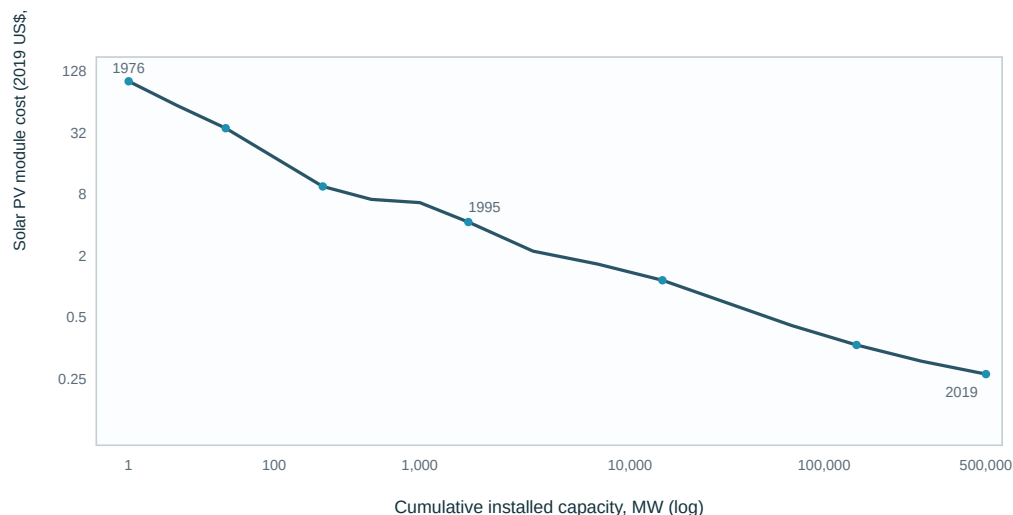


Figure 12. The solar-PV **experience curve** (log–log). Costs fell from ~\$100/W (1976) to well under \$0.50/W (2019) — a >99% decline — driven by cumulative deployment. This is **creative destruction working for decarbonisation**: the same innovation-rent mechanism, now pointed at clean energy.

There is also early evidence of **decoupling**: in several advanced economies, GDP per capita keeps rising while energy use per capita **falls** — even after adjusting for imported ("trade-adjusted")

energy, so it is not merely offshoring emissions.

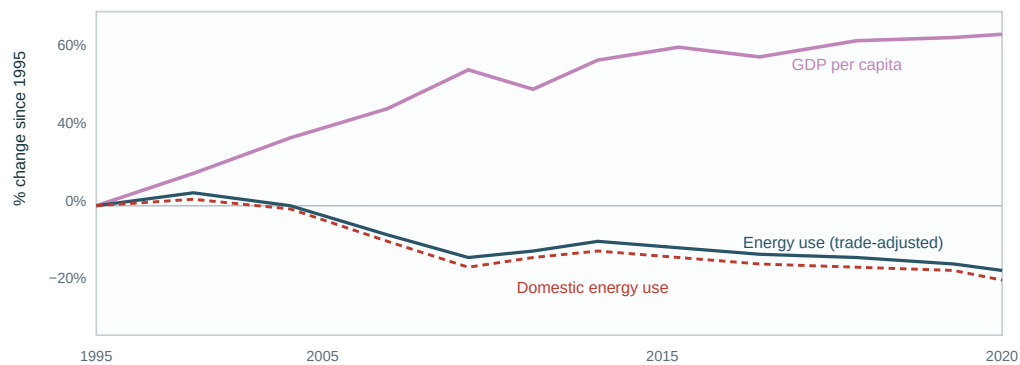


Figure 13. Relative decoupling: GDP per capita up ~54% since 1995 while energy use per capita is down ~10%. Because the fall holds even on a **trade-adjusted (consumption-based)** measure, it is genuine efficiency — not just outsourcing industry abroad.

LISTEN FOR IN THE LECTURE

The optimistic and pessimistic readings sit side by side. Pessimistic: the historical growth engine *runs on carbon* (Fig. 11). Optimistic: the *same* innovation mechanism can now make clean energy cheapest (Fig. 12) and break the income–energy link (Fig. 13). The exam question is usually about the **mechanism**, not which mood you pick.

Summary, Formulas & Glossary

The unit in one paragraph

Rational agents weigh benefits against the **full economic cost** (direct + opportunity), and the surplus over the best alternative is **economic rent**. **Specialization** — driven by **comparative advantage** — lets everyone consume beyond their own production frontier through **trade**. Firms choose **technologies** to minimise cost given input prices, represented by **isocost lines**; a change in **relative prices** (dear labour, cheap coal) made energy-intensive technology cheapest in **Britain**, and the **innovation rent** plus **creative destruction** turned one-off invention into permanent progress. **Models** are the deliberate simplifications that let us see all this. The same engine produced the climate crisis — and may yet produce its solution.

Formula sheet

Concept	Formula
Economic cost	direct cost + opportunity cost
Economic rent	benefit – direct cost – opportunity cost = net benefit – opportunity cost
Opportunity cost of a good	(units of other good given up) – the slope of the PPF
Comparative advantage in good X	lower opportunity cost of X than the other party
PPF (England)	$\text{Clothes} = 3600 - (6/5) \cdot \text{Wine}$
PPF (Portugal)	$\text{Clothes} = 4000 - (8/9) \cdot \text{Wine}$
Firm's cost	$\text{cost} = (w \times L) + (p \times R)$
Isocost line	$R = c/p - (w/p) \cdot L$ (slope = $-w/p$)
Innovation rent	cost(old technology) – cost(new technology)
Profit	profit = revenue – costs

Glossary of terms to memorise

Term	Meaning
Opportunity cost	Net benefit of the next-best forgone alternative.
Economic vs accounting profit	Economic profit subtracts opportunity costs too; can be negative when accounting profit is positive.
Economic / innovation rent	Surplus over the best alternative; innovation rent = gain from a new technology vs the old.
Division of labour	Splitting production into specialised tasks (Smith's pin factory).
Learning by doing / economies of scale	Two engines of rising productivity from specialization.
Absolute advantage	Producing a good with fewer inputs than another party.
Comparative advantage	Producing a good at lower <i>opportunity</i> cost than another party.
Production Possibilities Frontier	All maximal output combinations for given resources; slope = opportunity cost.
Terms of trade	The price at which goods are exchanged; lies between the two opportunity costs.
Production function	Relation between inputs and output, $Y = f(\cdot)$.
Fixed vs variable proportions	Inputs must scale together vs. can be substituted.
Constant returns to scale	Doubling all inputs doubles output.
Dominated technology	Uses at least as much of every input as another → never chosen.
Isocost line	Equal-cost input combinations; slope = $-\text{relative input price}$.
Entrepreneur / creative destruction	First adopter earning Schumpeterian rents; laggard firms are swept away.
Model / equilibrium	Deliberate simplification; a self-perpetuating outcome.
Endogenous / exogenous	Determined inside / outside the model.
Ceteris paribus	"All else equal."

Exam-style questions to practise

1. A café owner reports €80,000 accounting profit but could earn €50,000 elsewhere and €40,000 by renting out her premises. Compute her economic profit and advise her.
2. Country X needs 10 hrs/car and 5 hrs/laptop; Country Y needs 8 hrs/car and 2 hrs/laptop. Who has the comparative advantage in cars? Show the opportunity costs.

3. Two technologies produce steel: A = (2 workers, 8 coal), B = (6 workers, 3 coal). At wage 20 and coal 10, which is cheaper? At wage 20 and coal 30? Explain using isocost slopes.
4. Explain, with a diagram, why the Industrial Revolution's energy-intensive technologies were adopted in Britain before France.
5. Define an equilibrium and give the endogenous and exogenous variables of the Malthusian model.
6. "Prosperity requires high emissions." Use Figures 11–13 to argue both for and against.

Study guide prepared as a companion to *Microeconomics — Prosperity and Inequality, Unit 2: Technological Change, Population, and Growth* (KU Leuven, 2025–2026). All diagrams are clean redrawings of the lecture figures for study purposes. Cross-references such as "extension 2.4 in the handbook" point to the CORE / course handbook.

Scarcity, Work & Choice

Unit 3 — Study Guide & Lecture Notes

- › Preferences, utility & indifference curves
- › The feasible frontier, MRS and MRT
- › The utility-maximising choice: $MRS = MRT$
- › Income & substitution effects and working hours

BEFORE YOU BEGIN

How to use these notes & where this unit fits

This guide follows the Unit 3 slides section-by-section and expands the reasoning so you can read it on its own or annotate it in class. Every diagram — indifference curves, the feasible frontier, the tangency condition, and the income/substitution decomposition — is redrawn cleanly, with the convex curves computed rather than sketched so the geometry is exactly right.

DEFINITION BOX

The precise definition to reproduce in the exam.

WORKED EXAMPLE

Numbers carried through, matching the slide examples.

LISTEN FOR IN THE LECTURE

Points usually stressed verbally — leave room for your own notes.

CHECK YOURSELF

Quick questions to test recall.

The thread from Units 1–2

- **Unit 1:** labour is **work**.
- **Unit 2:** labour is an **input** in production, and new technologies raise the **productivity of labour**.
- **Unit 3 (this unit):** if technology makes an hour of work more productive (higher wage), **how does an individual respond?** Do they take the gain as more **consumption**, more **free time**, or both?

THE QUESTION THIS UNIT ANSWERS

Build a **model of individual choice** that explains the differences in **work hours** across countries and over time — and, in particular, why rising wages have historically been taken partly as **more leisure**.

Outline

1. **A.** Introduction — the puzzle of hours and income
2. **B.** Scarcity and choice: preferences, indifference curves, MRS, the feasible frontier and MRT
3. **C.** Decision-making under scarcity: the optimal choice, $MRS = MRT$
4. **D.** Income and substitution effects
5. **E.** Application to technological change

SECTION A

Introduction

Since 1870, living standards have risen enormously in most countries. But the gains have **not** all been taken as extra income: there are large disparities in **free time** and **income** across countries. Some rich countries work long hours; others have converted much of their productivity gains into shorter working weeks and longer holidays.

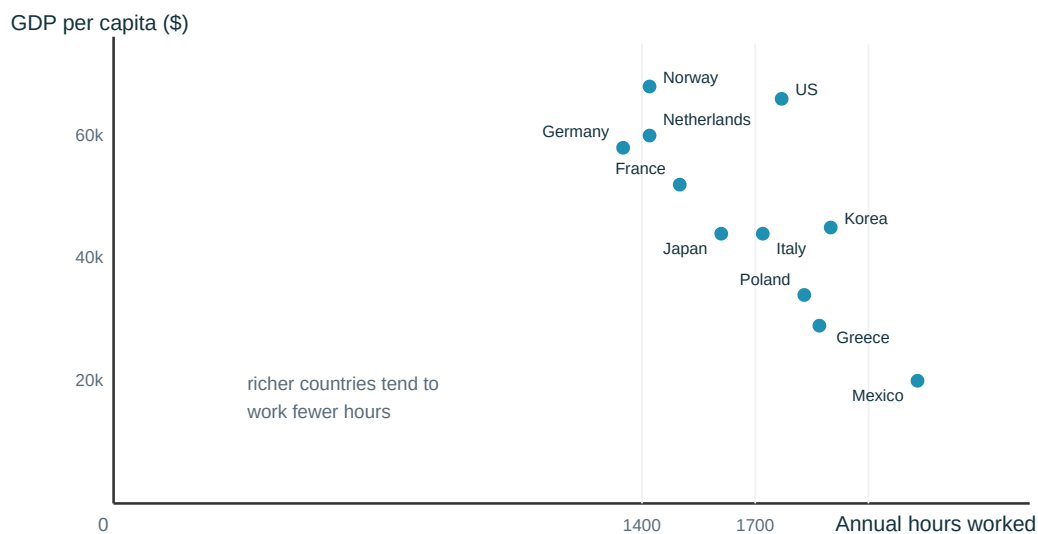


Figure 1. Across richer countries, higher income per person tends to come with **fewer** annual hours worked (e.g. Germany, the Netherlands, France) — while some equally rich countries (e.g. the US, Korea) work much more. Income alone does not pin down hours; **preferences** (and culture, and politics) do. This unit builds the model that makes sense of the scatter.

LISTEN FOR IN THE LECTURE

The framing question for the whole unit: *a new technology raises the productivity of labour* — how does that affect (i) living standards and (ii) the free time and working hours people choose? Keep both halves in mind; the answer turns out to hinge on two opposing forces (income and substitution effects).

SECTION B

Scarcity & Choice: Key Concepts

Individuals want to consume as much as possible **and** have as much free time as possible. But a day has only a limited number of hours — this is **scarcity**. Increasing consumption means **sacrificing free time** (you must work to earn). So we need a model of how a person picks the best combination of the two.

TWO INGREDIENTS OF THE MODEL OF CONSTRAINED CHOICE

- **Preferences** — how the individual ranks combinations of consumption and free time.
- **The trade-off (constraint)** — the feasible combinations of consumption and free time, i.e. how much consumption an hour of free time costs.

Simplifying assumptions

- Individuals **don't care about the future**, so there are **no savings**.
- Individuals **cannot borrow**.
- Therefore **income = consumption** (whatever you earn, you spend now).

WHY MAKE "UNREALISTIC" ASSUMPTIONS?

Recall from Unit 2: a model deliberately omits detail to expose the essential mechanism. Ruling out saving/borrowing lets us focus cleanly on the one trade-off that matters here — **free time vs consumption**.

Preferences and utility

People like **both** free time and consumption, and can **rank** the many possible combinations.

DEFINITION — UTILITY

Utility is a numerical indicator of the value a person places on an outcome. Higher-utility combinations are preferred to (chosen over) lower-utility ones.

Indifference curves

DEFINITION — INDIFFERENCE CURVE

An **indifference curve** shows all combinations of two goods that give the **same utility** (the person is "indifferent" between them).

Four properties you must know and be able to justify:

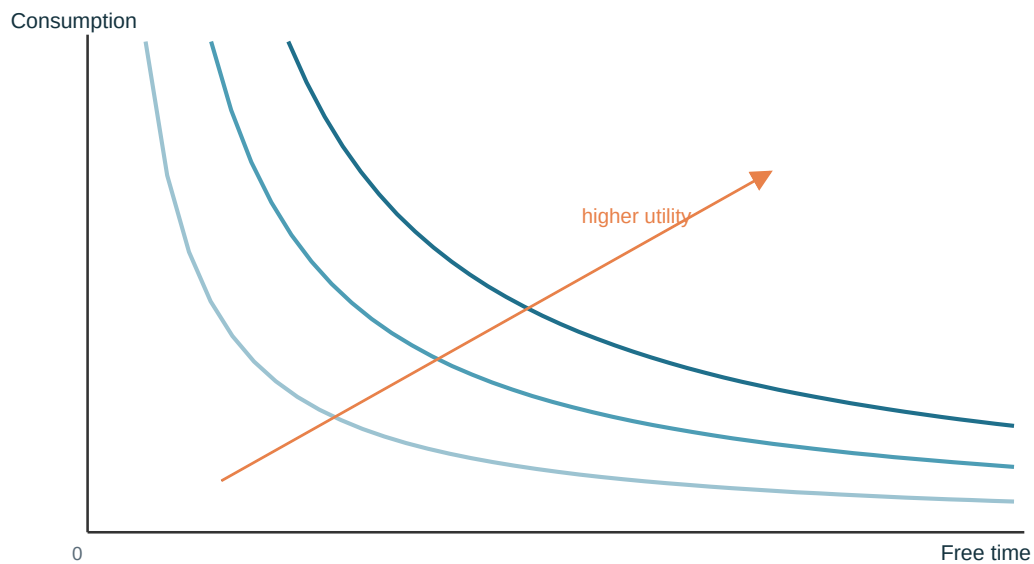


Figure 2. A family of indifference curves for free time and consumption. **Property 1:** they slope **downward** (more is better, so to stay equally happy you give up one good to get more of the other).

Property 2: curves **further from the origin** mean **higher utility** (more of both goods).

THE FOUR PROPERTIES

1. **Downward-sloping** — because more of a good is better.
2. **Higher curves = higher utility** — they contain bundles with more of both goods.
3. **They never cross** (see below).
4. **They get flatter to the right** — a diminishing marginal rate of substitution (convex to the origin).

Property 3 — why indifference curves cannot cross

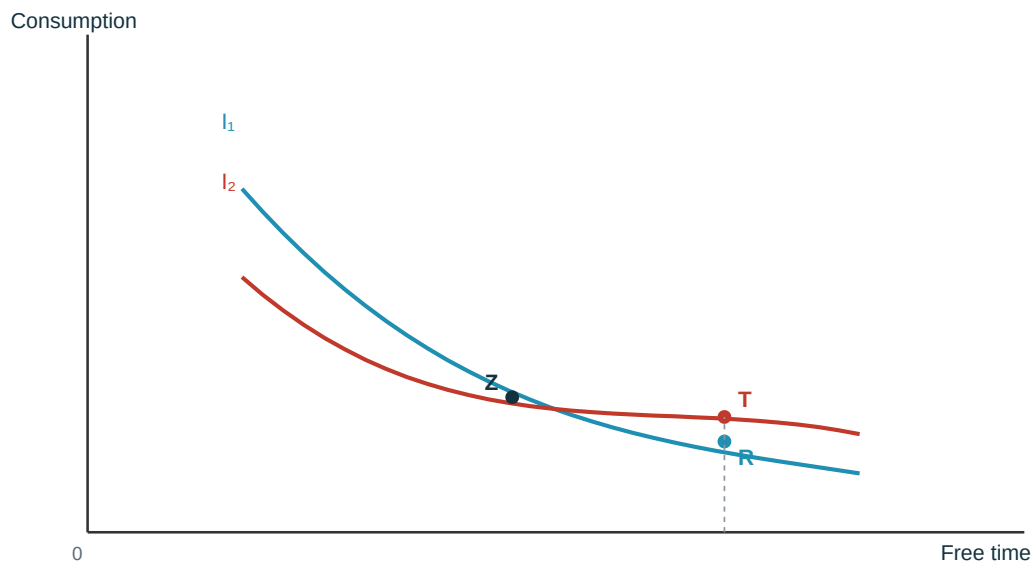


Figure 3. Suppose two indifference curves crossed at Z . Take points R (on I_1) and T (on I_2) directly above one another. Being on the same curve as Z , R gives the same utility as Z , and T also gives the same utility as Z — so R and T must be equally good. But T has **more consumption than R for the same free time**, so T is strictly better. Contradiction — therefore curves **cannot cross**.

Property 4 — the marginal rate of substitution (MRS)

DEFINITION — MARGINAL RATE OF SUBSTITUTION

The **MRS** is the amount of one good a person is **willing to give up** to get one more unit of the other, holding utility constant. Graphically it is the **(magnitude of the) slope of the indifference curve** at a point. It **diminishes** as you move right: the more you already have of a good, the less of the other you'll sacrifice for still more of it.

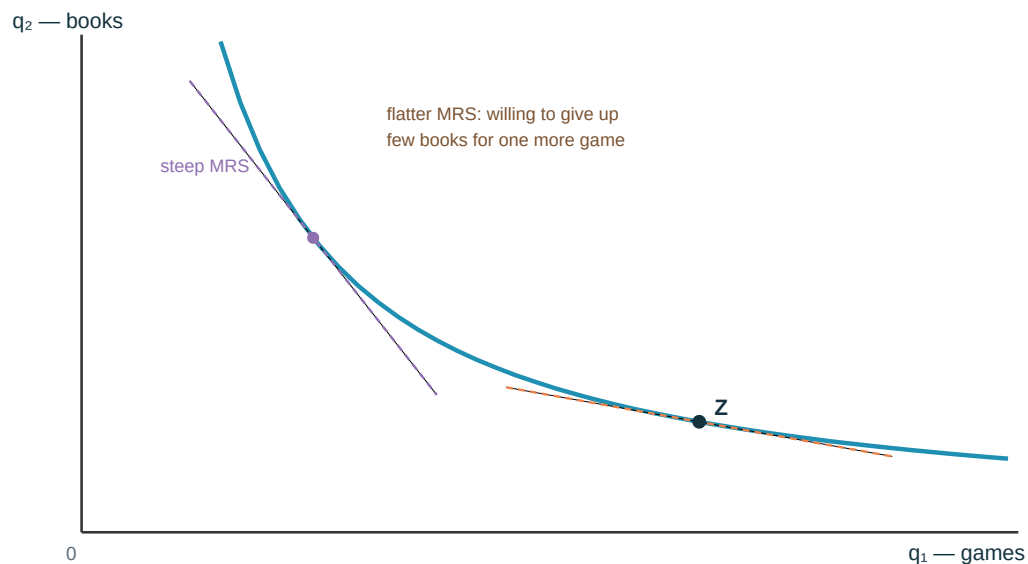


Figure 4. A consumer choosing between games (q_1) and books (q_2). At bundle **Z** she already has **many games and few books**, so she will give up only a **few books** for one more game — the curve is **flat** (small MRS). Far to the left, with few games, she'd sacrifice **many** books for a game (steep, large MRS). The MRS is the slope of the tangent.

LISTEN FOR IN THE LECTURE

The MRS is a **subjective willingness to trade**, read off preferences. Later it will be set equal to the MRT (an **objective** trade-off read off the constraint) to find the optimum. Keep the two straight: **MRS = preferences, MRT = constraint**.

Putting a Price on the Priceless

The MRS is powerful because it lets us value things that have **no market price** — marriage, children, health. There's no market, so no observed monetary value, and asking "how much would you pay for...?" directly is unreliable. The trick is **indirect measurement via happiness**.

- Ask people "**How satisfied are you with your life?**" and relate the answer to observable characteristics (income, marital status, health, employment...).
- The **MRS between, say, marriage and income** then reveals the willingness to pay for marriage — how much extra income would compensate a single person for being unmarried, at constant happiness.

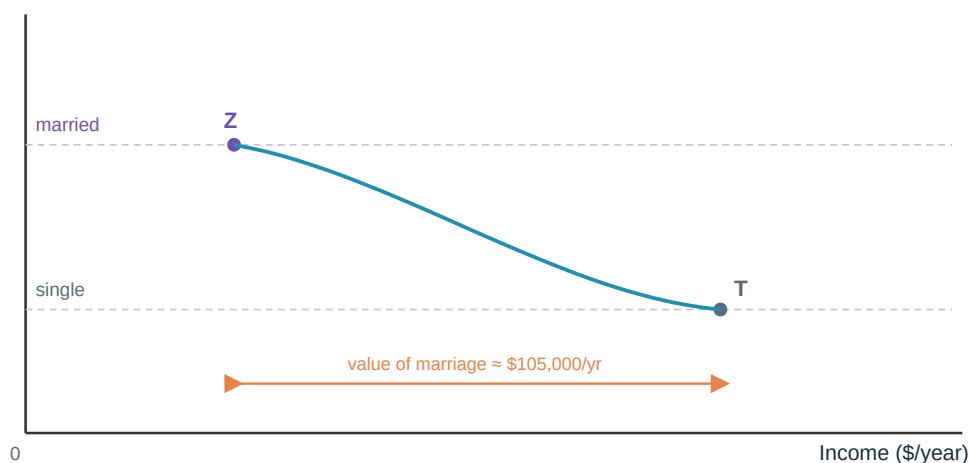


Figure 5. Same-happiness (indifference) reasoning applied to marriage. A married person at income Z is exactly as happy as a single person at the higher income T. The horizontal gap — the extra income needed to keep a single person as happy — is the **monetary value of marriage**, estimated at roughly **\$105,000 per year**.

What?	In comparison to?	Value (per year)
Excellent health	Very bad health	+\$463,170
Married	Single	+\$105,000
Divorce	No divorce	−\$34,000
Unemployed	Full-time job	−\$114,248
Death of partner	No death of partner	−\$308,780

Source: estimates via life-satisfaction regressions (Blanchflower & Oswald-style; slide cites bakadesuyo.com). Signs show whether the event raises or lowers well-being.

But do we get used to everything? Hedonic adaptation

A caveat to these numbers: over time people tend to **return to a long-run "set-point"** of happiness after major life events. The initial jump (or drop) in satisfaction fades.

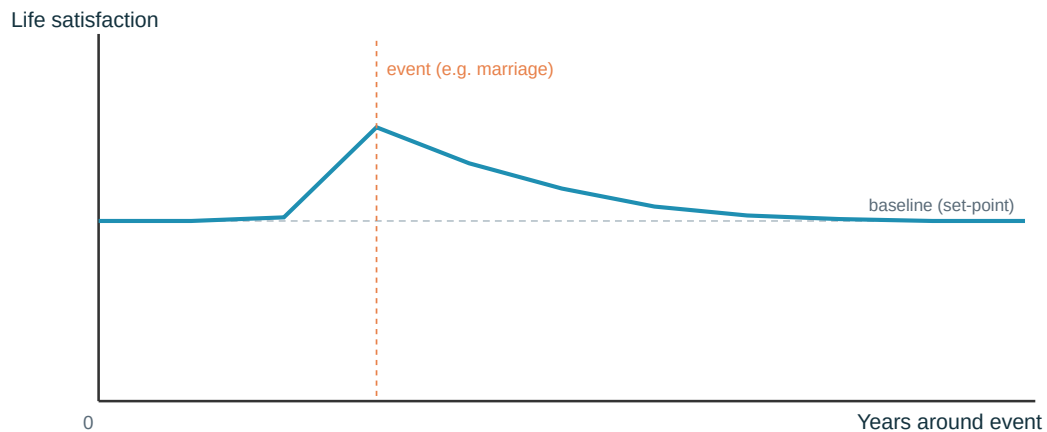


Figure 6. Life satisfaction around a major event (e.g. marriage). There is a spike, but within a few years people drift back toward their **baseline**. (Source: Clark, Diener, Georgellis & Lucas, 2008.) This **adaptation** means the money-values above should be read as short-to-medium-run, not permanent.

The Feasible Frontier & the MRT

Preferences tell us what the person **wants**; now we need what is **possible**. The trade-off between consumption and free time is drawn as the **feasible frontier**.

DEFINITIONS

- **Feasible frontier** — the combinations of free time and consumption that can just be **reached** (work all available hours → max consumption, no free time; work none → max free time, no consumption).
- **Feasible set** — the frontier **plus everything inside it** (all attainable combinations).

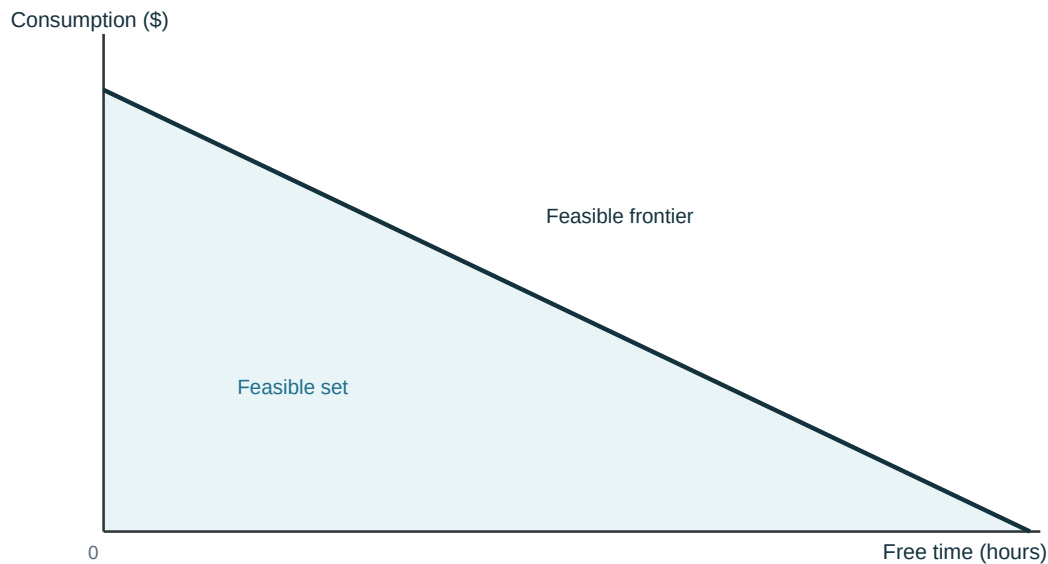


Figure 7. The feasible frontier and the feasible set. Points on the frontier use all available time; points inside waste some potential consumption or time; points outside are unattainable.

DEFINITION — MARGINAL RATE OF TRANSFORMATION

The **slope of the feasible frontier** is the **opportunity cost of free time** — the consumption you forgo for one more hour off. This is the **marginal rate of transformation (MRT)**: the rate at which free time can be **transformed** into consumption (i.e. the wage).

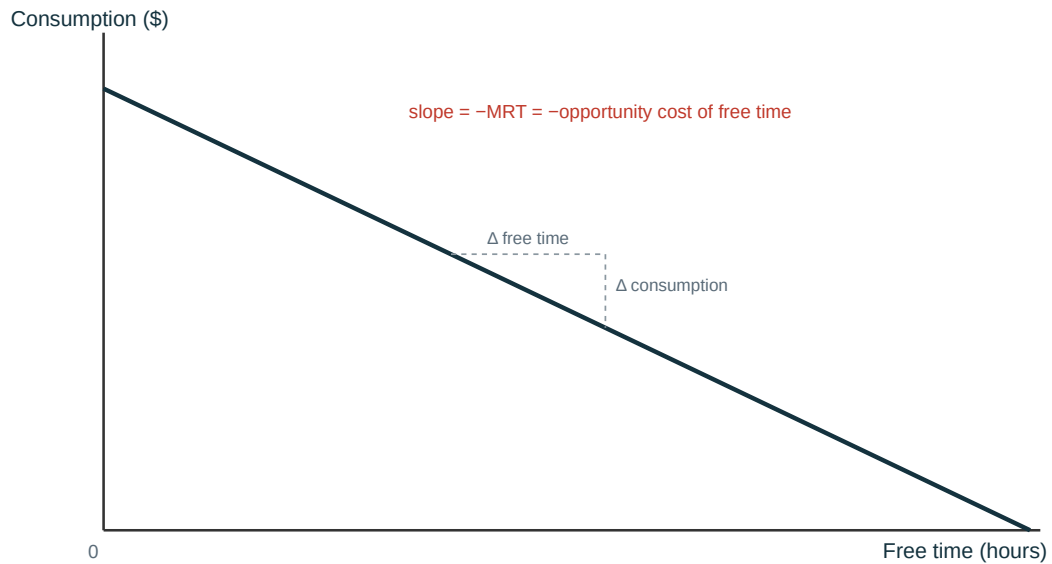


Figure 8. The MRT is the frontier's slope. If an hour of free time costs $\$w$ of consumption (the wage), the frontier has slope $-w$ and the $MRT = w$. A **higher wage** → **steeper frontier** → **higher opportunity cost of free time**.

CHECK YOURSELF

- Why must indifference curves be convex (flatten to the right)? What would a *concave* one imply about willingness to trade?
- The MRS and MRT are both "slopes" — whose slope is each, and what does each measure (willingness vs ability to trade)?

SECTION C

Decision-Making Under Scarcity

Now combine the two halves. Which feasible combination will the person choose? This is a **constrained choice problem**: pursue an **objective** subject to a **constraint**.

THE CONSTRAINED OPTIMISATION

- **Objective:** maximise **utility**.
- **Constraint:** the **feasible frontier**.

The optimal choice is the combination lying on an indifference curve **as far from the origin as possible** while still being feasible.

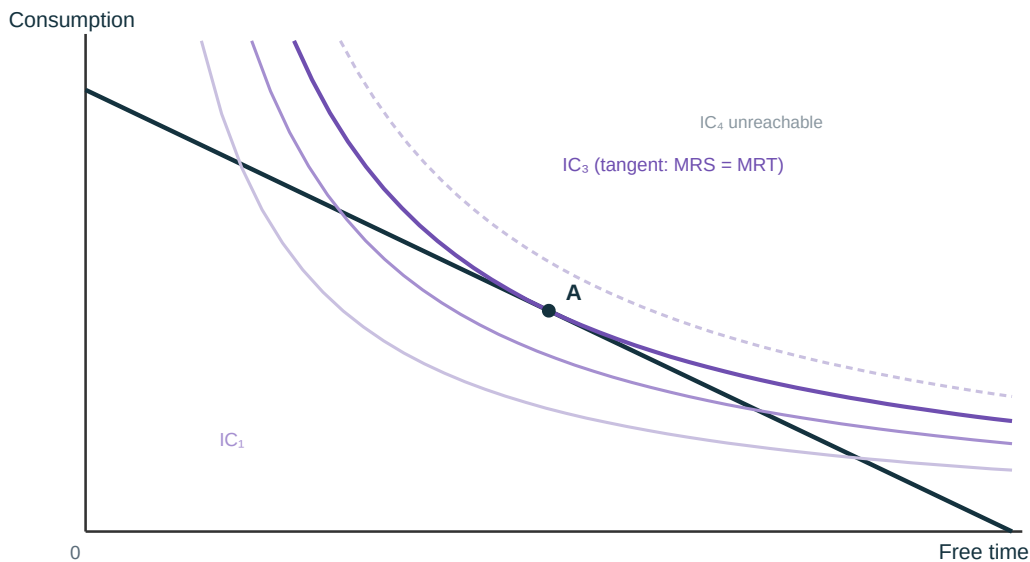


Figure 9. Overlay the indifference curves on the feasible frontier. IC_4 would be nicer but is **unreachable**; IC_1 is feasible but wasteful (you can do better). The best attainable curve just **touches** the frontier — the tangency point **A**.

OPTIMALITY CONDITION — $MRS = MRT$

The utility-maximising choice is where the **indifference curve is tangent to the feasible frontier**. At that point the rate at which the person is **willing** to trade the goods (MRS) equals the rate at which they **can** trade them (MRT):

$$MRS = MRT$$

Why the tangency is best — the intuition

At a non-tangency point the two rates differ, and you can always improve. Consider a point **D** where the indifference curve is **steeper** than the frontier ($MRS > MRT$):

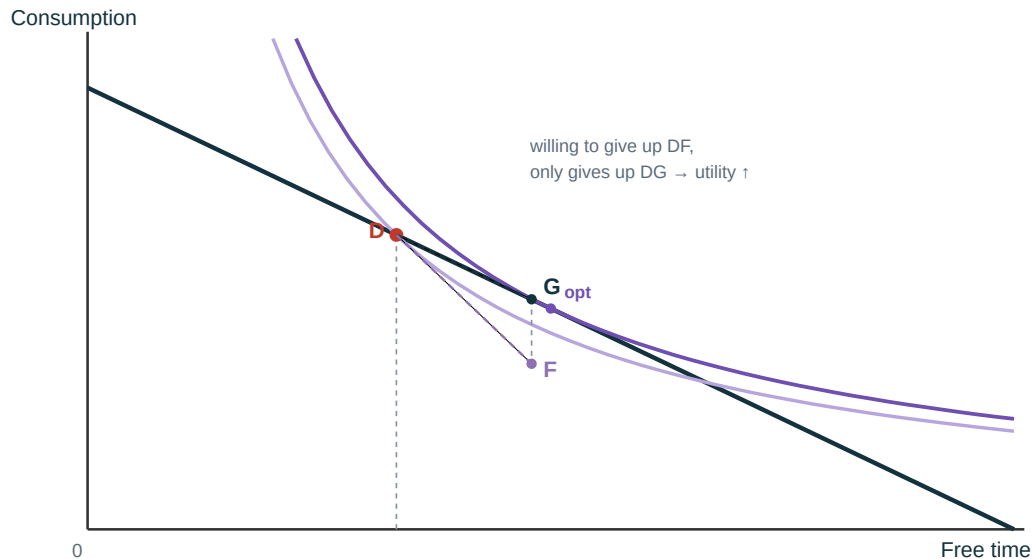


Figure 10. At D , to gain one more hour of free time the consumer is **willing** to give up DF of consumption (her steep MRS), but the frontier only **makes** her give up DG (the flatter MRT). Since she sacrifices **less than she was willing to** ($DG < DF$), taking more free time raises her utility — she moves to a higher indifference curve. She keeps going until $MRS = MRT$ (the optimum).

LISTEN FOR IN THE LECTURE

The logic is completely general: whenever $MRS \neq MRT$ there is an unexploited gain, so only the tangency can be optimal. If $MRS > MRT$, take more free time; if $MRS < MRT$, work more. This is the same "equate the margins" idea you'll meet again and again in microeconomics.

SECTION D

Income & Substitution Effects

Set-up — the summer-job example

You are planning your **10-week summer break**. You can work at **\$90 per day**, but you also value time for friends, holiday and study. How many days should you work?

Your feasible frontier has slope **-\$90/day** (the wage = the MRT = the opportunity cost of a day off). The optimum is the tangency **A**.

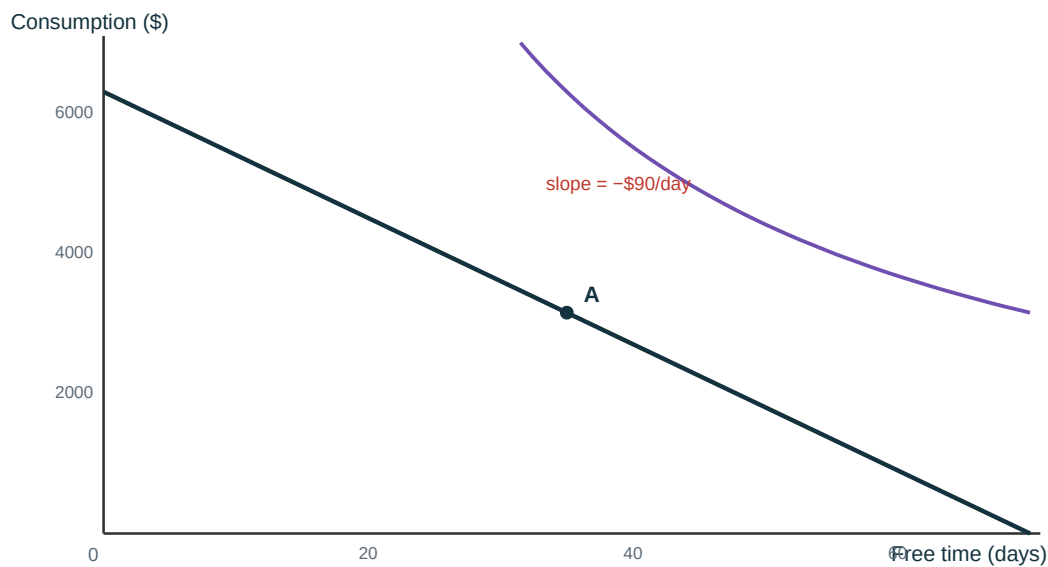


Figure 11. The summer-job feasible frontier: work all days → maximum consumption, no free time; take all days off → maximum free time, no consumption. Slope = $-\$90/\text{day}$. The chosen point A is where $\text{MRS} = \text{MRT}$.

A pure increase in income (a gift)

Your mother gives you **\$1000**, no strings attached. This does **not** change the wage, so the frontier shifts **up in parallel by \$1000** — the opportunity cost of free time is unchanged. You re-optimize at a new tangency.

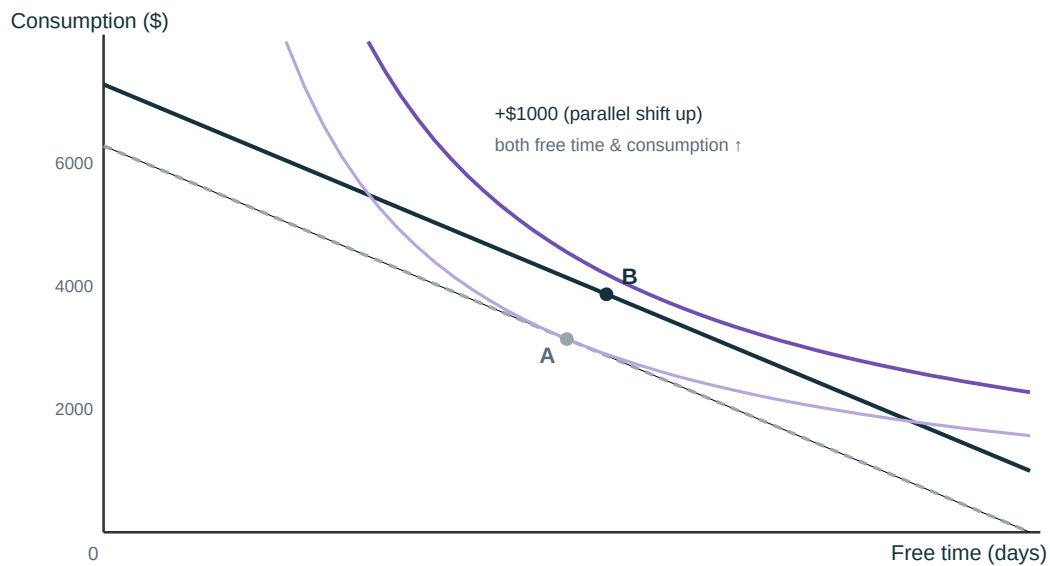


Figure 12. A \$1000 gift shifts the frontier up in **parallel** (same slope). The new optimum B typically has **both more consumption and more free time** — free time is a **normal good**. (With particular preferences free time might stay the same, but it should never **fall**.)

KEY TAKEAWAY

A rise in **unearned** income (no change in the wage) has only an **income effect**: more of everything you value, including free time. We expect free time to **rise or stay equal** — **never fall**.

A Wage Rise: Two Opposing Effects

Now a genuinely different job pays **\$130 per day**. Unlike the gift, a higher **wage** does two things at once, and they pull free time in **opposite** directions:

THE TWO EFFECTS OF A WAGE INCREASE

- **Income effect** — at your old hours you now earn more, so you can afford more of everything, including **more free time** (free time is normal). **Pushes free time up.**
- **Substitution effect** — free time is now more **expensive** (each hour off costs \$130 rather than \$90 of consumption), so you substitute away from it. **Pushes free time down.**

We separate them with a **hypothetical** frontier: a line with the **old slope** ($-\$90$) drawn tangent to the **new, higher** indifference curve.

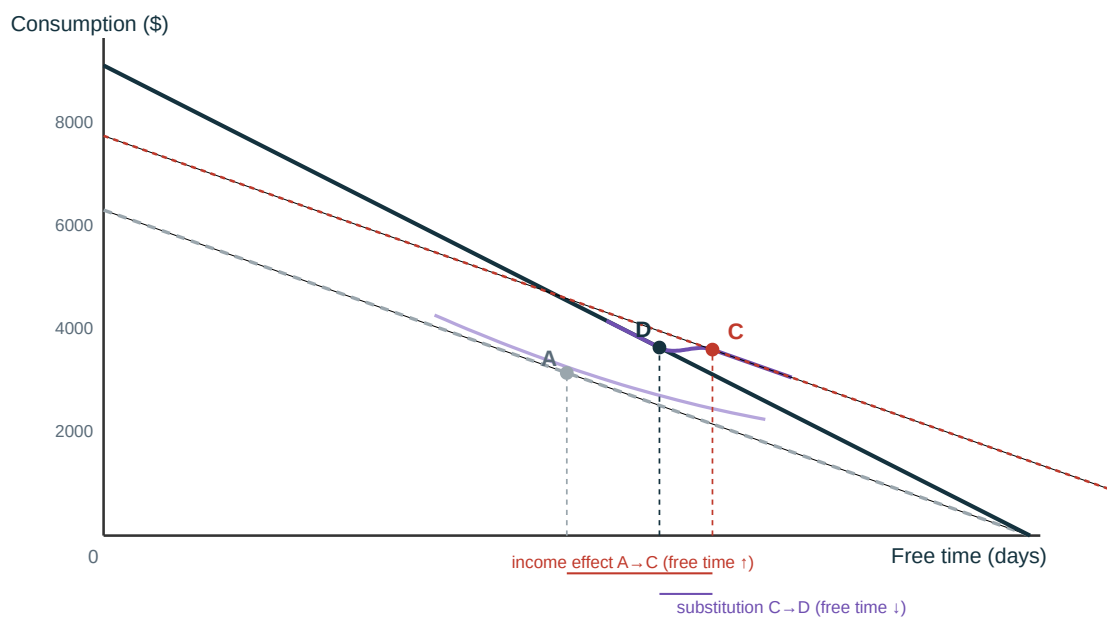


Figure 13. Decomposing a wage rise. **A → C (income effect):** a parallel move to the new higher indifference curve at the **old** relative price — free time rises. **C → D (substitution effect):** along the **same** indifference curve, the frontier steepens to the true new wage — free time falls. The **net** move A → D here shows free time **rising**, so the income effect dominates.

THE OVERALL EFFECT

$$\text{Overall effect} = \text{Income effect} + \text{Substitution effect}$$

Because the two effects have **opposite signs** for free time, the net direction depends on **which is larger** — and that depends on **individual preferences**. Free time (and hence hours worked) can rise **or** fall when the wage rises.

LISTEN FOR IN THE LECTURE

The construction is fiddly the first time. Anchor it on three points: **A** (old optimum), **D** (new optimum, on the true new frontier), and **C** (the hypothetical point — **same slope as old, same indifference curve as new**). Income effect = A → C, substitution = C → D. The consumption axis reveals the puzzle's resolution; the free-time axis reveals its sign.

Application to Technological Change

Differences over time

Now use the model to explain the historical fall in working hours. Take a US worker in **1900** at point A. Over the century, technological progress raised the wage, **steepening** the budget constraint. Decompose exactly as in Figure 13:

- **Income effect:** higher wage → afford more → **free time and consumption both rise** (A → C).
- **Substitution effect:** free time dearer → **free time falls**, consumption rises (C → D).
- **Net:** historically the **income effect dominated**, so **free time increased** — the working week and working year got shorter as countries got richer.

THIS IS EXACTLY FIGURE 13

The "technological change over time" diagram **is** the wage-rise decomposition, with the income effect winning. That is the model's explanation for the long-run decline in hours worked in rich economies.

Cross-country differences

Why do equally rich countries choose such different hours (Figure 1)? The model's baseline answer is **differences in preferences**. But preferences are shaped by more than taste:

- **Culture / norms** — differing attitudes to work and leisure.
- **Politics** — legal limits on working hours, mandated holidays.
- **Social preferences** — e.g. "keeping up with the Joneses": if status depends on relative consumption, people over-work.

Is this a good model?

WEAKNESSES

Not literally realistic: people don't compute MRS/MRT, and **most cannot freely choose their hours**.

WHY IT STILL WORKS

A good **approximation**: over time people **learn** the mix that suits them; hours shift through culture and politics (**indirect** choice); and people choose **which jobs to apply for**. It **explains real patterns** — hours across countries and over time — via preferences and income/substitution effects.

Summary, Formulas & Glossary

The unit in one paragraph

A person chooses between **consumption** and **free time** under scarcity. **Preferences** are drawn as convex, non-crossing **indifference curves**, whose slope is the **MRS** (willingness to trade). The **feasible frontier** is the constraint, whose slope is the **MRT** = the wage = the opportunity cost of free time. The best feasible bundle is the **tangency** where **MRS** = **MRT**. A pure income gain shifts the frontier out in parallel (only an **income effect**: free time rises or stays equal). A **wage rise** rotates the frontier and triggers **two opposing forces** — an income effect (free time $\uparrow$) and a substitution effect (free time $\downarrow$) — whose net depends on preferences. Historically the income effect has dominated, so richer societies work **fewer** hours.

Formula & concept sheet

Concept	Meaning / expression
Utility	numerical ranking of bundles; higher = preferred
Indifference curve	bundles with equal utility; convex, downward, never cross
MRS (marginal rate of substitution)	slope of indifference curve = willingness to trade
Feasible frontier	consumption = wage $\times$ (hours available - free time)
MRT (marginal rate of transformation)	slope of frontier = wage = opp. cost of free time
Optimal choice	MRS = MRT (tangency)
Income effect (wage $\uparrow$)	free time $\uparrow$ (normal good)
Substitution effect (wage $\uparrow$)	free time $\downarrow$ (leisure dearer)
Overall effect	income effect + substitution effect

Glossary

Term	Meaning
Scarcity	Limited time (and resources) forces trade-offs.
Constrained choice	Maximising an objective (utility) subject to a constraint (the frontier).
Utility	Numerical index of preference satisfaction.
Indifference curve	Equal-utility combinations of two goods.
MRS	Willingness to give up one good for one more unit of the other; slope of the IC; diminishing.
Feasible frontier / set	Attainable boundary / all attainable bundles.
MRT	Rate free time converts into consumption; slope of the frontier; the wage.
Income effect	Response to being richer at unchanged relative prices.
Substitution effect	Response to a changed relative price at constant utility.
Normal good	Demand rises with income (free time is one).
Hedonic adaptation	Tendency to return to a happiness set-point after events.

Exam-style questions to practise

1. Draw a feasible frontier for a wage of \$15/hour over a 16-hour day and mark the optimum. What is the MRT?
2. Explain, using MRS and MRT, why a point inside the feasible set can never be optimal.
3. A worker's wage doubles but she works *the same* hours. What does this tell you about the relative size of her income and substitution effects?
4. Use an income/substitution diagram to explain why average annual hours fell across the 20th century in rich countries.
5. Two countries have identical technology and wages but very different working hours. Give three model-consistent explanations.
6. Why can indifference curves not cross? Prove it with a diagram.

and tangencies are geometrically accurate; data figures are clean redrawings of the lecture charts for study purposes.

Social Interactions

Unit 4 — Social Interactions & Game Theory

- › Games, best responses & the Nash equilibrium
- › Dominant strategies & the prisoner's dilemma
- › Resolving social dilemmas: altruism, repetition, rules
- › Sequential games & coordination

BEFORE YOU BEGIN

How to use these notes & where this unit fits

This guide follows the Unit 4 slides section-by-section and expands the reasoning so you can read it alone or annotate it in class. Every payoff matrix is redrawn in the KU Leuven bi-matrix style (row player's payoff bottom-left, column player's top-right), Nash cells outlined in green, so you can read the strategic logic at a glance.

DEFINITION BOX

The precise definition to reproduce in the exam.

WORKED EXAMPLE

Numbers carried through, matching the slide examples.

LISTEN FOR IN THE LECTURE

Points usually stressed verbally — leave room for your own notes.

CHECK YOURSELF

Quick questions to test recall.

Where this unit sits

Previous units modelled choices that **did not depend on other people's decisions**. Adam Smith argued that self-interested individuals, guided by the "**invisible hand**", can produce outcomes good for society (entrepreneurship, innovation). But self-interest can also be **harmful**. This unit uses **game theory** to model interactions where my payoff depends on your action, and asks: why do these problems arise, and what can we do about them?

DEFINITION — SOCIAL DILEMMA

A **social dilemma** is a situation in which actions taken **independently by self-interested individuals** result in a **socially sub-optimal outcome** — e.g. traffic jams, climate change, COVID-19. Two classic forms:

- **Tragedy of the commons** — common resources are **over-exploited** (Lloyd 1833, overgrazing; Hardin 1968). E.g. fishing, climate, a shared kitchen.
- **Free riding** — benefiting from others' contributions to a cooperative project **without contributing yourself**.

The unit asks how **altruism**, **repetition**, and **government policy** can resolve them.

Outline

1. **A.** Introduction — self-interest, social dilemmas
2. **B.** Game theory: key concepts — games, Nash equilibrium, dominant strategies, prisoner's dilemma
3. **C.** Evaluating outcomes — the Pareto criterion
4. **D.** Resolving social dilemmas — social preferences, repeated games, changing the rules
5. **E.** Coordination — multiple equilibria

SECTION B

Game Theory: Key Concepts

DEFINITIONS

- **Social interaction** — a situation involving more than one party, where one's actions affect **both their own and others'** outcomes.
- **Strategic interaction** — a social interaction where people are **aware** of how their actions affect others.
- **Strategy** — an action a player can take in a social interaction.

Games are studied with **game theory** (used in social science *and* biology).

DEFINITION — A GAME

A **game** describes a social interaction through five elements:

1. **Players** — who is involved.
2. **Feasible strategies** — the actions each player can take.
3. **Order of play** — what each player knows when deciding.
4. **Information** — what each knows when choosing an action.
5. **Payoffs** — outcomes for every combination of actions.

Worked example — crop choice

Two farmers, **Anil** and **Bala**, each choose which crop to specialise in — **Rice** or **Cassava**. They interact only once (a **one-shot game**) and choose **simultaneously** (neither knows the other's choice). Payoffs depend on market prices and land quality.

		Bala	
		Rice	Cassava
Anil	Rice	4, 4	6, 3
	Cassava	6, 6	5, 2

Payoffs: **bottom-left** = Anil (row), **top-right** = Bala (column).

Best response & Nash equilibrium

DEFINITIONS

- **Best response** — the strategy that yields the highest payoff, **given the other player's strategy**.
- **Nash equilibrium** — a set of strategies (one per player) such that **each player's strategy is a best response to everyone else's**. No player can gain by **deviating unilaterally**. (After John Nash, 1928–2015, Nobel 1994 — see *A Beautiful Mind*.)

Finding the Nash equilibrium of the crop game by checking best responses:

- If Bala plays Rice, Anil's best response is **Cassava** ($6 > 4$).
- If Bala plays Cassava, Anil's best response is **Rice** ($6 > 5$).
- If Anil plays Rice, Bala's best response is **Rice** ($4 > 3$).
- If Anil plays Cassava, Bala's best response is **Rice** ($6 > 5$).

		Bala	
		Rice	Cassava
Anil	Rice	4 / 4	3 / 6
	Cassava	6 / 6	2 / 5

The unique Nash equilibrium (green): **Anil grows Cassava, Bala grows Rice** → payoffs (6, 6). Here it is also Pareto-efficient — but that need not be true in general.

LISTEN FOR IN THE LECTURE

Reading a bi-matrix: to find **Anil's** best response, fix a Bala **column** and compare the **bottom-left** numbers; to find **Bala's**, fix an Anil **row** and compare the **top-right** numbers. A cell is a Nash equilibrium when it is best for *both* at once. **A game can have more than one Nash equilibrium** — or none in pure strategies.

Finding Nash Equilibria

A 3×3 example

Player 1 chooses a row (T/M/B); Player 2 a column (L/C/R). Find every best response:

		Player 2		
		L	C	R
Player 1	T	1 2	2 0	3 0
	M	1 1	1 1	0 1
	B	1 0	0 2	2 2

Bottom-left = Player 1, top-right = Player 2. Unique Nash equilibrium: **(B, R)** → (2, 2).

- If Player 2 plays **L**, Player 1's best is **T** ($2 > 1 > 0$).
- If Player 2 plays **C**, Player 1's best is **B** ($2 > 1 > 0$).
- If Player 2 plays **R**, Player 1's best is **B** ($2 > 1 > 0$).
- Likewise for Player 2. **Only in cell (B, R)** are both strategies best responses to each other — no player can improve by unilaterally changing. That is the **Nash equilibrium**.

Dominant strategies & the prisoner's dilemma

DEFINITIONS

- **Dominant strategy** — a strategy that is the best response **regardless of what the other player does** (does not always exist).
- **Dominant strategy equilibrium** — a Nash equilibrium in which **every** player plays a dominant strategy.

Back to Anil and Bala, now deciding how to deal with pest insects: **Toxic Tide** (kills every insect but pollutes) vs **Integrated Pest Control / IPC** (less effective, no pollution).

		Bala	
		IPC	Toxic Tide
Anil	IPC	3 / 3	1 / 4
	Toxic Tide	4 / 1	2 / 2

No matter what the other does, **Toxic Tide** beats IPC for each player (a dominant strategy) → equilibrium (Toxic, Toxic) = (2, 2).

DEFINITION — PRISONER'S DILEMMA

A game with a **dominant-strategy equilibrium** in which playing the dominant strategy yields **lower individual AND total payoffs** than another (cooperative) outcome. Here both would be better off at (IPC, IPC) = (3, 3), but self-interest drives them to (Toxic, Toxic) = (2, 2). (Originally told as two burglars caught by police.)

CHECK YOURSELF · EXAM MCQ

Which is correct?

- (a) "An equilibrium in dominant strategies is always a Nash equilibrium." — **True**. If each plays a best response regardless of the other, it is certainly a best response to what the other actually does.
- (b) "The prisoner's dilemma outcome is never a Nash equilibrium, as all players can improve by changing strategies." — **False**. It is a Nash equilibrium; no one can improve by changing *unilaterally* (both must change together to reach the better outcome).
- (c) "A strategy combination that maximises the sum of payoffs is always a Nash equilibrium." — **False**. The efficient (IPC, IPC) maximises the total but is *not* a Nash equilibrium.

Answer: (a).

SECTION C

Evaluating Outcomes — the Pareto Criterion

DEFINITIONS — PARETO

- Allocation A **Pareto-dominates** B if **at least one party is better off under A and nobody is worse off**.
- An allocation is **Pareto-efficient** if **nobody can be made better off without making somebody worse off**.

Often **many** outcomes are Pareto-efficient — e.g. every way of splitting €1000 between two people is efficient as long as no money is left on the table.

Plot the four pest-control outcomes in **payoff space** (Anil's payoff on the x-axis, Bala's on the y-axis):

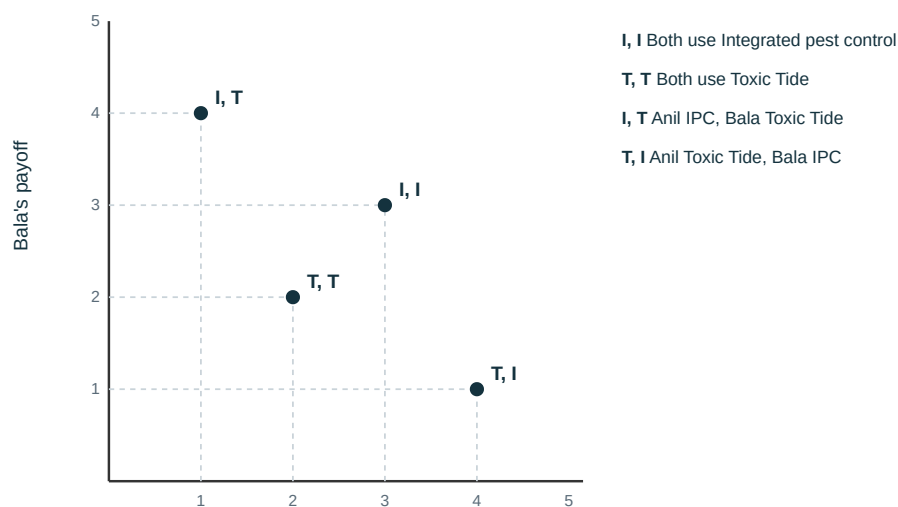


Figure 1. The four outcomes of the pest game. $(I, I) = (3, 3)$ Pareto-dominates $(T, T) = (2, 2)$ — both do better. But (I, I) does **not** dominate (I, T) or (T, I) , where one player does better and the other worse. (I, T) , (I, I) and (T, I) are all Pareto-efficient.

Caveats of the Pareto criterion

TWO IMPORTANT LIMITS

- **It often can't choose.** With many efficient allocations, the Pareto criterion is silent about which to pick. It may not even rank an efficient against an inefficient one — e.g. (T, I) is efficient but does **not** Pareto-dominate the inefficient (T, T).
- **It says nothing about fairness.** Many **unfair** allocations are Pareto-efficient — e.g. keeping €99.99 of a found €100 and giving your friend 1 cent is efficient (no money wasted) yet hardly fair.

LISTEN FOR IN THE LECTURE

Pareto efficiency is a **low bar**: "no waste", not "good" or "fair". The Nash equilibrium of a prisoner's dilemma is **Pareto-inefficient** (both could do better) — which is exactly why we look for ways to escape it in Section D.

SECTION D

Resolving Social Dilemmas

The prisoner's-dilemma outcome arose for three reasons. Each suggests a route out:

THREE ROUTES OUT OF A SOCIAL DILEMMA

1. Players **only cared about their own payoffs** → introduce **social preferences / altruism**.
2. Nobody could make players **pay for the consequences** of their actions on others → introduce **repeated games, social norms, peer punishment**.
3. Players **could not coordinate** beforehand → **change the rules of the game** (institutions and policies).

D.i — Social preferences: altruism

Social/prisoner's dilemmas arise when players care **only** about their own payoffs. But in experiments many players show **altruism** — they choose the dominated (cooperative) strategy. If altruism is **strong enough**, the socially desirable outcome can be reached (in *Golden Balls*, (split, split) is common; likewise in public-good games).

How altruistic are we? The public-good game

PUBLIC-GOOD GAME (PRISONER'S DILEMMA FOR N PLAYERS)

Each of **n** players gets **T** tokens and chooses how many to keep (z_i) vs contribute to a common pool ($g_i = T - z_i$). The payoff is:

$$\text{payoff}_i = z_i + \frac{\beta}{n}(g_1 + g_2 + \dots + g_n)$$

- If $1 > \beta/n$, a purely self-interested player **keeps all tokens** (private return on a contributed token, β/n , is below 1).
- If $\beta > 1$, it is **socially optimal to contribute everything** (each token multiplied by $\beta > 1$ across the group).
- For $1 > \beta/n$ and $\beta > 1$ together, the game is a **social/prisoner's dilemma**.

Experimental finding: in public-good games worldwide ($\beta = 1.6$, $n = 4$, 20 tokens each), people start by contributing a **lot** (≈ 8 – 14 of 20), far above the self-interested prediction of zero — clear evidence of **altruism/cooperation**. But without punishment, contributions **decline period by period** as cooperators get discouraged by free-riders.

Modelling altruistic preferences

Represent preferences with **indifference curves** over the two players' payoffs and combine with the feasible set (as in Unit 3). Example: Zoë wins **£200** in a lottery — will she keep it all, or share with her flatmate Yvonne?

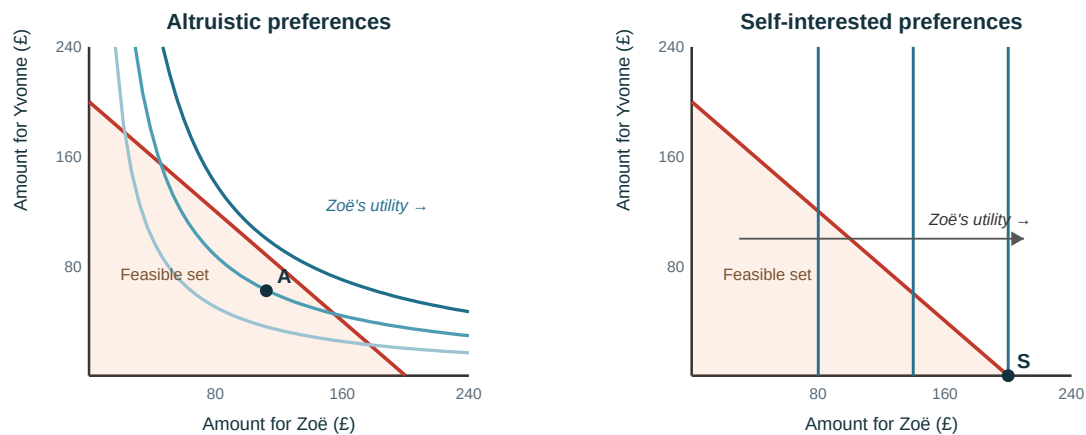


Figure 2. Altruistic Zoë (left) has **convex indifference curves** over money-for-Zoë and money-for-Yvonne, so her optimum **A** shares some money (interior point on the £200 budget line). **Self-interested Zoë** (right) has **vertical** indifference curves — she only values her own money — so her optimum **S** keeps all £200 (corner).

Repeated Games & Punishment

So far, **one-shot** games. Better outcomes can arise in **repeated interactions** through **social norms**, **reciprocity**, and **peer punishment**: behaving selfishly today has **consequences in future periods**, so it may **no longer be a dominant strategy**.

Worked example — the cartel's dilemma (two petrol stations)

Jupiter and Saturn each choose to **Maintain** a high price or **Cut** it. In a **one-shot** game, cutting is a dominant strategy for both → both cut (the **cartel's dilemma**).

		Saturn	
		Maintain	Cut
Jupiter	Maintain	1000 / 1000	700 / 1300
	Cut	1300 / 700	800 / 800

Profits (bottom-left = Jupiter, top-right = Saturn). One-shot Nash: **(Cut, Cut) = (800, 800)** — worse for both than (Maintain, Maintain) = (1000, 1000).

TRIGGER STRATEGY IN THE REPEATED GAME

When the game repeats, a firm can follow a "**trigger**" strategy: start with the **high** price, and keep it as long as the rival does too; if the rival **cuts**, retaliate with a low price for a while, then return to the high price. This creates a **trade-off**: high profit **today** from cutting vs high profits in the **future** from maintaining. Cooperation (Maintain, Maintain) can be an **equilibrium** if players value future money highly enough — because **punishment becomes possible**.

Punishment sustains cooperation — the evidence

In the public-good game, contributions normally **decline** over the rounds. But when players are allowed to **punish** free-riders (at a cost to themselves), contributions become **much higher and stay high** — peer punishment sustains cooperation. The willingness to punish, even at personal cost, is itself a social preference.

LISTEN FOR IN THE LECTURE

The key insight: repetition turns a one-shot prisoner's dilemma into a game where **defection can be punished later**, so the dominant strategy to defect disappears. This is how **social norms, reputation and reciprocity** support cooperation in real life without any government.

Rules of the Game — Sequential Games

The third route: **change the rules**. One important rule is the **order of play**. When players move in **sequence** (observing earlier choices), the game is drawn in **extensive form** (a **game tree**).

DEFINITIONS

- **Extensive form (game tree)** — **nodes** are points where a player picks a move; **branches** are the possible choices.
- **Backward induction** — solve by first finding the optimal choice at the **final** nodes, then reasoning **backward** to the first node.

Worked example — two TV stations

Channels **Z** and **Delta** choose a time slot (7:30 or 8:00 pm) for the evening news. **Z chooses first**, and Delta observes Z's choice before deciding.

Green = backward-induction path → outcome (Z:4, D:3)

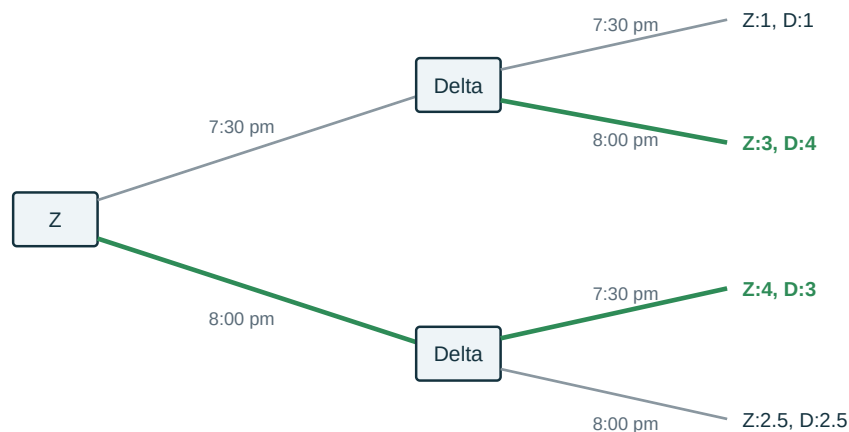


Figure 3. Solve by **backward induction**. If Z picks 7:30, Delta prefers 8:00 ($D:4 > D:1$) → (Z:3, D:4). If Z picks 8:00, Delta prefers 7:30 ($D:3 > D:2.5$) → (Z:4, D:3). Anticipating this, Z compares 3 vs 4 and picks **8:00**. Outcome: **(Z:4, D:3)**.

The ultimatum game

DEFINITION — ULTIMATUM GAME

A **sequential game** for dividing an economic rent (e.g. a cash prize). The **Proposer** offers a split; the **Responder** observes it and **accepts** (both get the proposed amounts) or **rejects** (both get **nothing**).

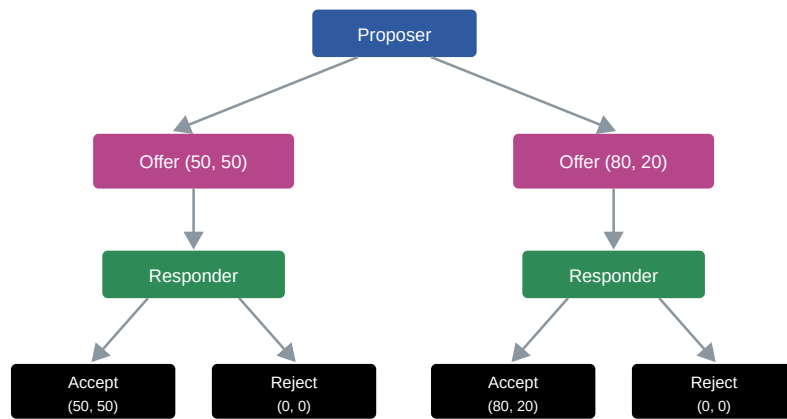


Figure 4. The ultimatum game in extensive form. If players cared **only about their own payoffs**, the Nash equilibrium (by backward induction) is that the Proposer offers **almost nothing** and the Responder **accepts** (since anything > 0 beats 0). In reality, offers are often near **50–50**.

WHY REAL OFFERS ARE GENEROUS

The Proposer's offer may be motivated by **altruism, fairness (50–50), inequality aversion, social norms, or reciprocity**. Responders routinely **reject** low offers — punishing unfairness at a cost to themselves. **Evidence (Kenyan farmers vs US students):** offers are consistent with social preferences *but also* with expected-payoff maximisation (proposers offer more where low offers are more likely to be rejected). And **the rules matter**: introducing **competition between responders** (two responders vying to accept) pushes outcomes back toward the **self-interested** prediction.

SECTION E

Coordination — Multiple Equilibria

Not every social dilemma is a prisoner's dilemma. Some games have **several** Nash equilibria, and the problem is **coordinating** on one of them.

Driving on the left or the right

Two drivers approach each other; each chooses to drive on the Left or the Right. If they choose the same side, all is well (payoff 0); if they mismatch, they crash (−10 each).

		Driver 2	
		Left	Right
Driver 1	Left	0 / 0	−10 / −10
	Right	−10 / −10	0 / 0

Two Nash equilibria: (Left, Left) and (Right, Right). Coordination is possible — and highly important.

COORDINATION GAMES & INSTITUTIONS

With **multiple equilibria**, no player has an incentive to deviate from either, so the outcome is **indeterminate** without coordination. **Institutions and conventions** select the equilibrium: a traffic law (drive on the right), or a coordinated switch. Sweden's "**Dagen H**" (**3 Sept 1967**) — the whole country switched from left to right overnight — is a real example of institutions moving everyone from one equilibrium to another.

Climate change as a game

For purely self-interested players, "**business as usual**" (**BAU**) is a dominant strategy (a prisoner's dilemma), and an emissions treaty would deliver the socially optimal outcome. But if players have **inequality aversion and reciprocity**, the game can become a **coordination game** with **two Nash equilibria**, one of which is the **social optimum** (both restrict).

		US	
		Restrict	BAU
China	Restrict	BEST BEST	GOOD WORST
	BAU	WORST GOOD	BAD BAD

With social preferences, (Restrict, Restrict) and (BAU, BAU) are both Nash equilibria — so coordination (a treaty) can move both countries to the **BEST** outcome.

LISTEN FOR IN THE LECTURE

The type of game depends on **preferences**: pure self-interest → prisoner's dilemma (one bad equilibrium); social preferences → coordination game (a good equilibrium becomes available). The lecturer's point: **changing preferences or rules changes the game itself**, and institutions exist to steer us toward the good equilibrium (cf. the Dvorak keyboard, another coordination trap).

Summary, Formulas & Glossary

The unit in one paragraph

Social interactions are modelled as **games** (players, strategies, order, information, payoffs). Players choose **best responses**, and a **Nash equilibrium** is a profile where everyone is best-responding — nobody gains by deviating alone. A **dominant strategy** is best whatever others do; when a dominant-strategy equilibrium leaves everyone **worse off** than a cooperative outcome, it is a **prisoner's dilemma** — a Pareto-inefficient Nash equilibrium. The **Pareto criterion** only flags "no waste" (not fairness) and often can't rank outcomes. Social dilemmas can be escaped through **social preferences/altruism**, **repetition with punishment** (trigger strategies, reputation), or **changing the rules** (sequential order, institutions). Some games instead have **multiple equilibria**, where the challenge is **coordination**, and institutions select the good equilibrium.

Concept sheet

Concept	Meaning
Game	players, strategies, order, information, payoffs
Best response	highest-payoff strategy given others' strategies
Nash equilibrium	each plays a best response; no profitable unilateral deviation
Dominant strategy	best response regardless of others' choices
Prisoner's dilemma	dominant-strategy eq. with lower individual & total payoff than cooperation
Pareto dominance / efficiency	A better for someone, worse for none / no waste
Public-good game	$\text{payoff} = z_i + (\beta/n) \cdot \Sigma g$; dilemma when $1 > \beta/n$ and $\beta > 1$
Repeated game / trigger	future punishment can sustain cooperation
Backward induction	solve sequential game from the final nodes back
Coordination game	multiple Nash equilibria; institutions select one

Glossary

Term	Meaning
Social / strategic interaction	Actions affect others / players are aware they do.
Social dilemma	Independent self-interest → socially sub-optimal outcome.
Tragedy of the commons / free riding	Over-exploiting shared resources / benefiting without contributing.
Best response / Nash equilibrium	Optimal reply / mutual best responses.
Dominant strategy (equilibrium)	Best whatever others do (all players do so).
Prisoner's dilemma	Dominant-strategy eq. worse for all than cooperating.
Altruism / social preferences	Caring about others' payoffs, fairness, reciprocity.
Repeated game / peer punishment	Interaction over many periods / costly punishment of defectors.
Extensive form / backward induction	Game tree / solving from the end.
Ultimatum game	Proposer splits a rent; responder accepts or rejects.
Coordination game	Multiple equilibria; the problem is agreeing which.

Exam-style questions to practise

1. Define a Nash equilibrium. In the crop game, verify that (Cassava, Rice) is Nash by checking both players' best responses.
2. Why is the prisoner's-dilemma outcome a Nash equilibrium *and* Pareto-inefficient? Reconcile these with a 2×2 matrix.
3. In a public-good game with $\beta = 1.6$ and $n = 4$, show that a self-interested player contributes zero but full contribution is socially optimal.
4. Explain how a trigger strategy can sustain the (Maintain, Maintain) outcome in the petrol-station game. What condition on the value of future profits is needed?
5. Solve the two-TV-station game by backward induction. Why does moving first benefit Channel Z?
6. Give the two Nash equilibria of the driving game. Why can't the Pareto criterion pick between them, and what selects the outcome in practice?

the crop, pest-control, cartel, climate and driving games reproduce the slides' payoffs, and the sequential and ultimatum trees the slides' extensive forms.

Market Successes & Failures

Unit 10 — Study Guide & Lecture Notes

- › Externalities: private vs social cost, and inefficiency
- › Coase bargaining, Pigouvian taxes & compensation
- › Public goods & asymmetric information
- › The limits of markets

BEFORE YOU BEGIN

How to use these notes & where this unit fits

This guide follows the Unit 10 slides section-by-section and expands the reasoning so you can read it alone or annotate it in class. The banana-and-fishery externality diagrams are redrawn cleanly, with the private, external and social cost curves and the efficient quantity computed so the geometry is exact.

DEFINITION BOX

The precise definition to reproduce in the exam.

WORKED EXAMPLE

Numbers carried through, matching the slide examples.

LISTEN FOR IN THE LECTURE

Points usually stressed verbally — leave room for your own notes.

CHECK YOURSELF

Quick questions to test recall.

Where this unit sits

The previous units showed that a **perfectly competitive market generates an efficient (Pareto-optimal) outcome**. But that result relies on strong conditions: **property rights must be well defined**, institutions must be able to **enforce** them, and they must be written into **complete, enforceable contracts**. Often these conditions **fail** — and the market delivers an inefficient outcome. This unit catalogues the main **market failures** and the tools to fix them.

THE FOUR SOURCES OF MARKET FAILURE (ROADMAP)

- **External effects** (externalities) — Section B.
- **Public goods** — Section C.
- **Asymmetric information** — Section C.
- **Market power** ($P > MC$) — Section C.

Section D then asks a deeper question: even where markets *work*, **should** they allocate everything?

Outline

1. **A.** Introduction — when the efficiency result breaks
2. **B.** External effects — inefficiency, bargaining, government intervention
3. **C.** Other types of market failure — public goods, asymmetric information, market power
4. **D.** Limits to markets — should everything have a price?

SECTION A

Introduction

In perfectly competitive markets the market generates an **efficient outcome** — but only if:

- **property rights are well defined**,
- institutions can **enforce** those property rights, and
- they can be written into **complete and enforceable contracts**.

Often these conditions are **not** satisfied — most importantly because of **external effects (externalities)** and **asymmetric information**.

Motivating example — the pesticide chlordane

The **banana weevil** caused large losses on banana plantations. The pesticide **chlordane** was highly effective against it — but **hazardous to humans and the ecosystem** (known since the 1970s). It was used heavily on Caribbean plantations in Guadeloupe and Martinique until 1993:

- The chemical washed off the land into rivers, **contaminating freshwater prawn farms and lobster fisheries**.
- By 2012, Martiniquean men had **the highest rate of prostate cancer in the world**; neurological damage in children was documented.

THE CORE IDEA

The banana plant owners **did not take into account the costs they imposed on fishermen** (and on health). A cost (or benefit) that one party's decision imposes on another, but that the decision-maker ignores, is an **externality** — and it is a source of market failure.

SECTION B

External Effects

DEFINITION — EXTERNALITY

An **external effect (externality)** arises when one party **directly conveys a cost or benefit to another party** that is not reflected in market prices.

- a **cost** imposed on others → **negative externality** (e.g. factory emissions);
- a **benefit** conferred on others → **positive externality** (e.g. a store's advertising helping neighbours).

WHY EXTERNALITIES BREAK EFFICIENCY

Economic efficiency requires **marginal social benefit = marginal social cost**. But an individual firm or consumer only considers its **own (private)** costs and benefits. With externalities, **marginal social cost/benefit differs from marginal private cost/benefit** — so the private optimum is **not** the efficient one. **Maximum efficiency is not achieved.**

Negative externality — bananas and fisheries

Abstracting from the health issues, growing bananas imposes a cost on fisheries (lower catch). Three cost concepts:

THREE MARGINAL COSTS

- **Marginal private cost (MPC)** — the plantation's own marginal cost of growing bananas.
- **Marginal external cost (MEC)** — the extra cost imposed on fishermen per ton of bananas.
- **Marginal social cost (MSC = MPC + MEC)** — the full cost to society.

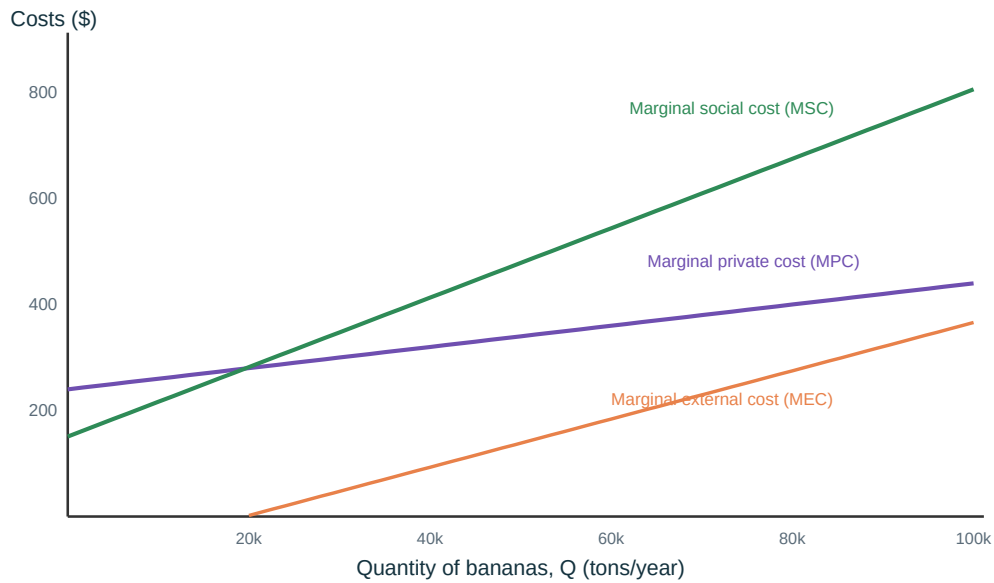


Figure 1. The marginal social cost (green) is the vertical sum of the plantation's marginal private cost (purple) and the marginal external cost imposed on fishermen (orange). The gap between MSC and MPC is exactly the externality.

The private outcome is not efficient

The plantation produces where its **marginal benefit (= the price of bananas)** equals its **marginal PRIVATE cost** — point **A** at **80,000 tons**. Society, however, should produce where marginal benefit equals marginal **SOCIAL** cost — point **B**.

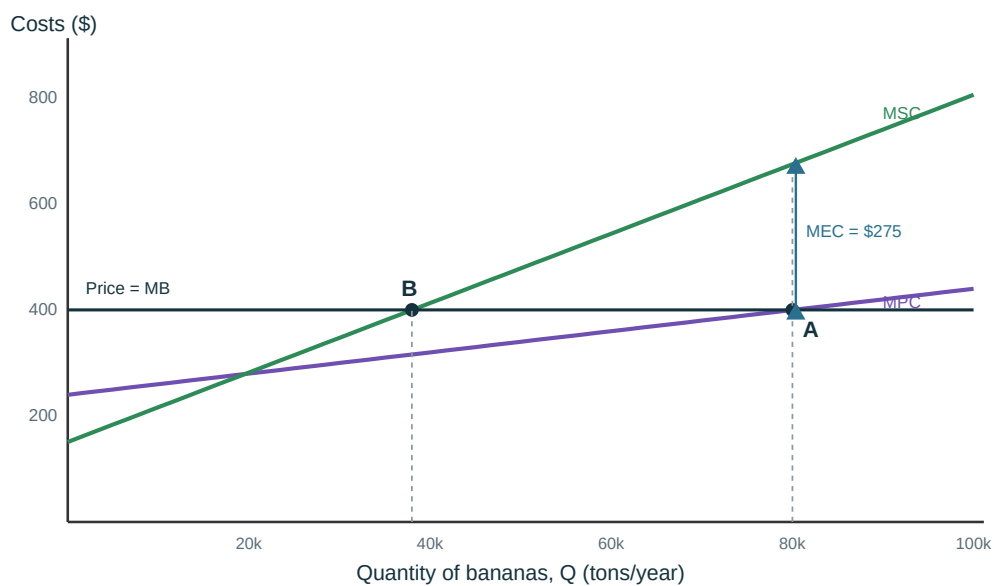


Figure 2. The plantation chooses A (80,000 tons), where price = MPC. But at 80,000 tons the marginal external cost is **\$275** — the gap between MSC and MPC. Since price (marginal benefit) is **below** the marginal social cost there, the last tons are socially wasteful. The outcome is **not Pareto-efficient**.

WORKED EXAMPLE — A PARETO IMPROVEMENT EXISTS

At 80,000 tons the marginal external cost is **\$275**. Fishermen would pay up to **\$275** to cut banana output by one ton. Cutting production by one ton lowers the plantation's revenue by **\$400** (the price) but also lowers its cost by nearly \$400 ($MPC \approx \text{price at A}$), so the plantation would accept **any compensation above roughly zero** to cut back. Since fishermen will pay up to \$275 and the plantation needs almost nothing, **both can be made better off** — the hallmark of an inefficient starting point.

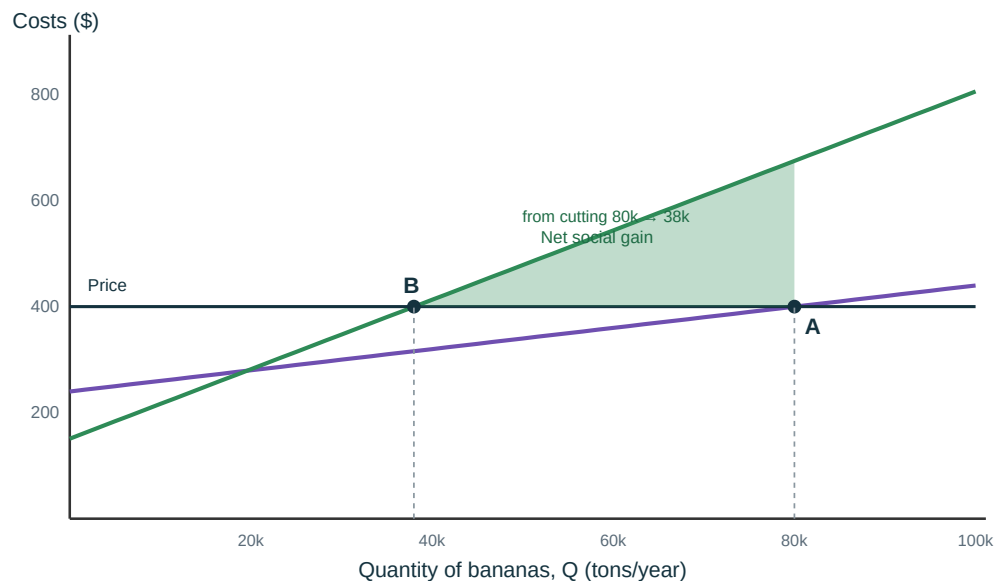


Figure 3. Pareto improvements are possible as long as the price (marginal benefit) is **below** the marginal social cost. The efficient outcome is where **marginal benefit = MSC** — a production rate of about **38,000 tons**. The green triangle is the **net social gain** from cutting production from 80,000 to 38,000.

Positive external effects

Actions can also confer **benefits** on others: a **positive externality** where the **marginal social benefit exceeds the marginal private benefit**. Example: a large store (e.g. FNAC) opening in a shopping street also benefits **other shops selling non-competing products**. Positive externalities lead to **under-provision** (too little of the good activity), the mirror image of the over-production under negative externalities.

LISTEN FOR IN THE LECTURE

The whole section rests on one move: replace the **private** cost/benefit the decision-maker sees with the **social** one. Negative externality → private cost too low → **too much** produced. Positive externality → private benefit too low → **too little**. Every policy below is a way to make the decision-maker face the social figure.

Fixing Externalities

Solution #1 — Bargaining (the Coase idea)

COASE THEOREM

Legally **assign property rights** over the externality (e.g. the right to pollute, or the right to clean water). Then **private bargaining** between the parties will reach a **Pareto-efficient allocation regardless of which party holds the rights** — in the absence of transaction costs. Private bargaining can be **more effective than government intervention** because the parties have more of the relevant **information**.

But **transaction costs** — the costs of acquiring information, enforcing the contract, or organising collective action — can be a major obstacle in reality.

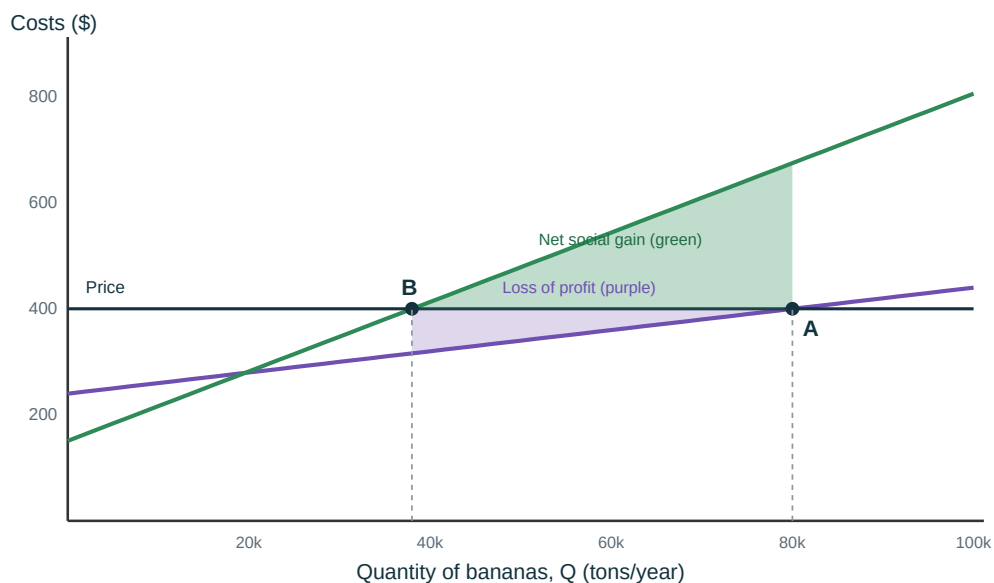


Figure 4. The size of the deal. The **plantation's minimum acceptable offer** = its **lost profit** from cutting output (purple area between price and MPC over 38k–80k). The **fishermen's reservation option** (maximum they'd pay) = the **whole external cost avoided** = green + purple. Because the **net social gain** is positive, a deal exists; where in the range the price lands depends on **relative bargaining power**.

PRACTICAL LIMITS OF BARGAINING

- **Collective action** — finding a representative and agreeing how to split the gains within each side.
- **Missing information** — calculating each fisherman's exact loss and each plantation's contribution to pollution.
- **Enforcement** — a court may struggle to verify whether plantations complied.
- **Limited funds** — fishermen may not have the money to pay the required compensation.

Solution #2 — Government intervention

Three instruments, each with **different distributional consequences** (and each with limits: missing information / measuring social costs, and lobbying).

(a) Regulation

The government **caps** banana production at the Pareto-efficient level (38,000 tons). Plantation profits fall (lower volume), but there is no tax bill.

(b) Pigouvian taxes / subsidies

DEFINITION — PIGOUVIAN TAX/SUBSIDY

A tax on firms generating a **negative** externality (or a subsidy for a **positive** one), set **equal to the marginal external cost at the Pareto-efficient level**. It **forces producers to face the full social cost** of their decisions, so their private optimum coincides with the social optimum.

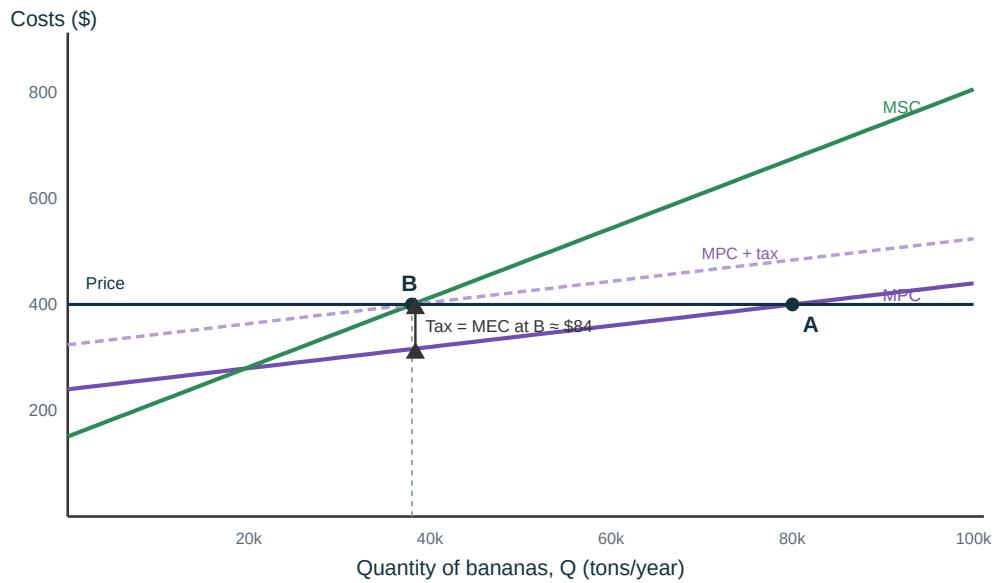


Figure 5. A Pigouvian tax ($\approx$ the marginal external cost at B) shifts the plantation's effective marginal cost up to **MPC + tax** (dashed). Its new private optimum is now at **38,000 tons** — the efficient level. Note this is **worse for plantations than regulation**: they cut volume *and* pay the tax.

(c) Compensation

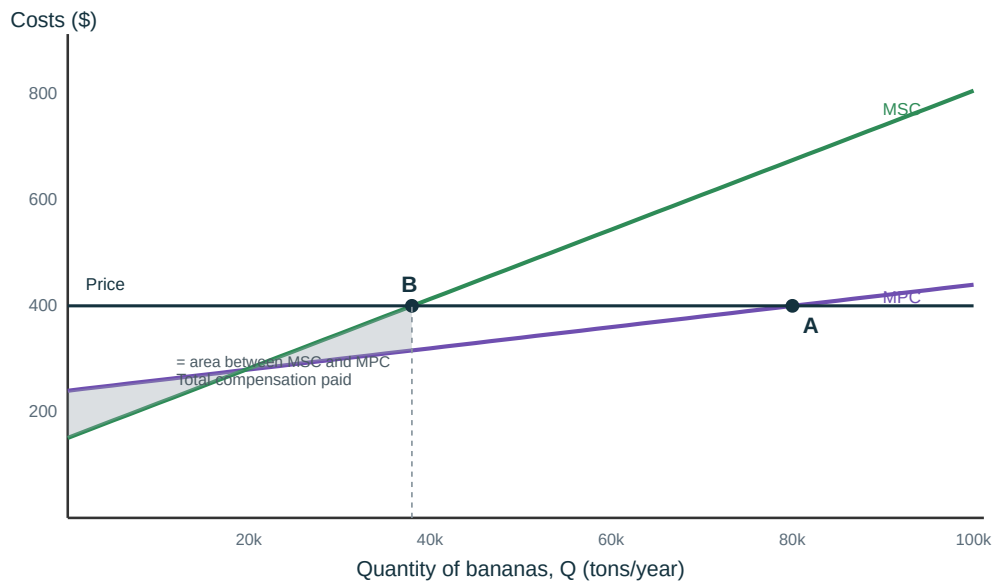


Figure 6. Plantations are required to **compensate** fishermen by the difference between marginal social and marginal private cost (= the marginal external cost) for each ton produced. This drives them to the efficient 38,000 tons. Compared with regulation, plantation profits are **lower** but **fishermen are better off** (they receive the compensation).

Why do external costs occur? Incomplete contracts

ROOT CAUSE

External costs cause market failure because of **incomplete contracts** — contracts that do not specify, in an **enforceable** way, every aspect of the exchange affecting all parties. Contracts that would internalise external costs are unenforceable because the relevant information is **not verifiable** or is **asymmetric** (not known to the decision-maker). So in reality it is **impossible** to use contracts or property rights to fold all social costs into the decision.

REAL-WORLD TWIST — DISNEY'S SECRET LAND PURCHASES

Disney bought the land **around** its Anaheim park **secretly** (and in Orlando via multiple shell companies / agents; in Hong Kong via an **option** on adjoining real estate). Why secretly? Because a park generates a huge **positive externality** on surrounding land values. If sellers knew Disney was buying, they would hold out for the higher post-park price — so Disney bought quietly to **internalise the externality** (capture the land-value gain) itself.

SECTION C

Other Types of Market Failure

Besides externalities, markets fail to reach a Pareto-efficient outcome for three more reasons: **public goods**, **asymmetric information**, and **market power**.

Public goods

Goods are classified along two dimensions — whether they are **rival** (my use reduces yours) and whether they are **excludable** (I can stop non-payers using them).

DEFINITIONS

- **Non-rival** — use by one person **does not reduce** its availability to others (a flood-control dam, street lighting, national defence, a TV broadcast).
- **Non-excludable** — it is **impossible to exclude** anyone from access (fish in the open ocean, flood protection).
- A **public good** is a **non-rival** good (which may or may not be excludable).

	Rival	Non-rival
Excludable	Private goods (food, clothes, houses)	Artificially scarce goods (subscription TV, uncongested toll roads, patented knowledge)
Non-excludable	Common-pool resources (ocean fish stocks, common grazing land)	Public goods & "bads" (a lunar eclipse view, public broadcasts, national defence, noise & air pollution)

After CORE Figure 12.8 — private goods and public goods.

WHY MARKETS FAIL FOR PUBLIC GOODS

Markets allocate **private** goods well, but tend to fail for the other three types:

- **Non-rival goods** have a **marginal cost of zero**, so pricing at $P = MC$ is impossible unless provision is **subsidised** or there is **market power**.
- **Non-excludable goods** can't be priced at all — the provider **cannot exclude non-payers** (the **free-rider** problem). Example: if one farmer builds a community irrigation project, others enjoy the **external benefit** for free, so it is hard to make them pay or to write a contract guaranteeing the efficient level.

Asymmetric information

DEFINITION

Information is **asymmetric** when **one party knows something relevant to the transaction that the other does not**. Two forms:

- **Hidden action** → a **moral hazard** problem (an action one party takes that the other can't observe). *Example: insured people take more risks.*
- **Hidden attributes** → an **adverse selection** problem (a characteristic one party can't observe). *Example: used-car buyers don't know the car's quality, but sellers do.*

ADVERSE SELECTION — HEALTH INSURANCE

Insurance is voluntary and the insurer **can't observe** buyers' health; buyers **know** theirs, and the less healthy are keener to buy. To break even the insurer must charge a **high price**, so **only the less healthy buy** — most buyers already have a problem. The result is a **missing market**: many healthier people who would like insurance stay uninsured.

MORAL HAZARD — CAR INSURANCE

Any insurance also has a **hidden-action** problem: once covered, the buyer takes **more risk** (full damage cover → more careless driving). Insurers can write **some** contractual limits but **cannot enforce** others (e.g. your driving speed), so the behaviour can't be fully controlled.

Market power — price above marginal cost

Firms with **market power** set **price above marginal cost**. This is a **market failure** because the allocation is **not Pareto-efficient** (some consumers who value the good above its marginal cost don't buy it). Policies try to **limit market power** — but with two cautions:

- **Incentives for innovation** — some markup can be the reward that funds R&D.
- Market power arising **solely from product differentiation** may be fine — people **value having different choices**.

SECTION D

Limits to Markets

Everything so far asked *when markets fail to be efficient*. A different, deeper question: even where a market **would** function, **should** it allocate the good at all?

ARGUMENTS AGAINST USING MARKETS FOR EVERYTHING

- **Repugnant markets** — creating a market for some goods/services would **violate ethical or social norms** (e.g. a market in people — slavery).
- **Other institutions may be more effective** — governments, families.
- **Crowding out** — market mechanisms can **crowd out** norms and social preferences (paying for something can destroy the intrinsic motivation to do it).
- **Merit goods** — goods that **should be available to everyone regardless of ability to pay** (e.g. education, basic health care).

LISTEN FOR IN THE LECTURE

This section is **normative**, not just positive: it's about values, not only efficiency. A market can be perfectly efficient *and* still be the wrong way to allocate something. Keep the two questions separate in the exam — "is it efficient?" vs "should it be a market at all?"

Summary, Formulas & Glossary

The unit in one paragraph

Competitive markets are efficient only when property rights are complete and enforceable. They fail through **externalities** (decisions impose uncoded effects on others — $MSC \neq MPC$, so the private optimum over- or under-produces), **public goods** (non-rival goods have zero marginal cost; non-excludable goods invite free-riding), **asymmetric information** (hidden actions → moral hazard; hidden attributes → adverse selection and missing markets), and **market power** ($P > MC$). Externalities can be tackled by **bargaining** (Coase — efficient regardless of who holds the rights, absent transaction costs), or by government **regulation**, **Pigouvian taxes/subsidies**, or **compensation** — instruments that reach the same efficient quantity but differ in **who bears the cost**. Finally, some goods arguably **should not** be allocated by markets at all (repugnant markets, crowding out, merit goods).

Concept sheet

Concept	Meaning / rule
Efficiency condition	marginal social benefit = marginal social cost
MSC	$MSC = MPC + MEC$ (private + external marginal cost)
Private optimum (neg. externality)	price = MPC → over-production
Efficient quantity	price = MSC
Coase theorem	clear property rights + zero transaction costs → efficient bargain, whoever holds the rights
Pigouvian tax	tax = MEC at the efficient quantity
Public good	non-rival ($\pm$ excludable); $MC \approx 0$
Moral hazard	hidden action (unobservable behaviour)
Adverse selection	hidden attributes (unobservable type) → missing market
Market power failure	$P > MC$

Glossary

Term	Meaning
Externality	Cost/benefit a decision imposes on others, ignored by the decision-maker.
MPC / MEC / MSC	Marginal private / external / social cost.
Pareto efficiency	No one can be made better off without making another worse off.
Coase theorem	With clear rights and no transaction costs, bargaining reaches efficiency regardless of who holds the rights.
Transaction costs	Costs of information, enforcement, collective action.
Pigouvian tax/subsidy	Tax/subsidy set to the marginal externality at the efficient level.
Incomplete contract	Contract that doesn't enforceably cover all affected interests.
Rival / excludable	My use reduces yours / non-payers can be shut out.
Public good / common-pool resource	Non-rival ($\pm$ excludable) / rival but non-excludable.
Free-rider problem	Enjoying a non-excludable good without paying.
Moral hazard / adverse selection	Hidden action / hidden attributes under asymmetric information.
Missing market	A market that fails to exist because of adverse selection.
Merit good / repugnant market	Should be available to all / a market that violates social norms.

Exam-style questions to practise

1. A factory's $MPC = 10 + 0.5Q$ and it imposes $MEC = 0.5Q$ on neighbours; the price is 40. Find the private and the socially efficient output, and the size of the deadweight loss.
2. Explain the Coase theorem. Why does the efficient outcome not depend on whether the polluter or the victim holds the property right? When does it break down?
3. Compare regulation, a Pigouvian tax and compensation for the banana example: which reaches the efficient quantity, and how do they differ in distribution?
4. Classify each as rival/excludable: a lighthouse, a congested motorway, a patented drug, an ocean fishery. Which are public goods, and why might markets under-provide them?
5. Distinguish moral hazard from adverse selection with an insurance example of each. How does adverse selection produce a "missing market"?
6. Give two goods you think should *not* be allocated by markets, and justify using the "limits to markets" arguments.

Study guide prepared as a companion to *Microeconomics — Prosperity and Inequality, Unit 10: Market Successes and Failures* (KU Leuven, 2025–2026). The banana/fishery curves are computed so the private optimum (80,000 tons, MEC = \$275) and efficient quantity ($\approx 38,000$ tons) match the lecture figures; the public-goods matrix follows CORE Figure 12.8.